

Municipal Waste Management in Israel

From Cleanup to System Design

A Civic Handbook



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Municipal Waste Management in Israel: A Civic Handbook

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Introduction

From Cleanup to System Design

Most discussions of waste begin with a question about behaviour: why don't people sort their rubbish, why do they dump things by the bin, why don't they compost. This book begins with a different question: why is the system built in a way that makes the right behaviour inconvenient, expensive, or simply impossible – and what can be done about that.

Why this book, and why now

Israel entered 2026 with an officially confirmed shortage of landfill capacity. According to the State Comptroller, remaining capacity will not be sufficient even for the waste volume expected by the end of 2026. In parallel, two legislative processes are under way – a broad Ministry of Finance reform of sector regulation, and a narrower Ministry of Environmental Protection memorandum amending the Packaging Law – either of which could reset the system's basic parameters for the next fifteen to twenty years. And according to the Ministry's own 2025 national scan, only around 6 percent of household packaging waste reaches a designated collection container at all.

That is a rare combination: a physical infrastructure crisis, an unfinished institutional reform, and a documented capture failure, all visible at once, all before any of it has been settled. That is precisely why this book is being written now rather than a few years from now, once the outcome is already decided. The moment before decisions are made is the moment civic knowledge is worth the most.

This book was written in Haifa, drawing on the experience of a Russian-speaking environmental activist community (the Re-Haifa Lab group) working at the intersection of practice – early rescue of discarded items, circular-economy pilots, organics projects – and analysis: how the system is structured, who runs it, and why it produces the results it produces. This is not an academic treatise or a manifesto. It is an attempt to translate a complex, fragmented system into language that supports meaningful participation in it.

The central argument of this book is that municipal waste management should begin before waste appears – before a useful object, edible food, or organic material becomes part of the waste stream at all.

Israel has a missing layer between everyday household life and the formal collection system: local prevention infrastructure. Neighbourhood circularity hubs can bring together repair, reuse, second-hand exchange, food-sharing, composting, bulky-item transport, creative upcycling, and practical education. Such hubs don't replace municipal waste services; they complement them at the point where the standard system usually starts too late.

In this view, sorting material into fractions remains important, but it is no longer the first or central act. It is the last gate – what happens after prevention, repair, reuse, food redistribution, and local organics work have not been enough. A good waste-management system should not only process waste more efficiently. It should create the conditions under which fewer things become waste in the first place.

Who this book is for

The book addresses four audiences at once.

The activist and resident who want to understand why their efforts to sort waste achieve less than they should – and what can be changed through concrete action at the neighbourhood or city level. Part VII (Chapters 26–28) is written directly for you: data tools, an audit methodology, a seven-day action plan.

NGOs and environmental organisations working professionally on waste issues, or looking for an entry point to partner with a municipality. The book describes not only how the system works, but where its built-in conflicts of interest sit – knowing that makes negotiation more precise.

Municipal officials and local council representatives who make practical decisions every day about contracts, bins, and complaints. The chapters on financial structure (Part V) and on how incentives are built into contractor agreements are written for a reader who already knows, from the inside, what a tonne of waste costs and why the sanitation budget never quite adds up.

Academic readers and reviewers assessing the evidence base. This book applies an explicit evidentiary scale, introduced in Chapter 8, under which every figure in every section on material streams is marked at one of four levels – confirmed, probable, reconstructed, or unknown – with its source and its denominator stated alongside it. No figure is used without saying where it came from and what, precisely, it measures. Given the level of uncertainty still present in Israel's waste sector, that is the only responsible way to write about it.

How the book is organised

The book now runs to twenty-eight chapters, organised into seven parts of progressively more detailed analysis. Each part assumes the one before it, but each can also be read on its own.

Part I. Why the Policy Is Changing (Chapters 1–3)

How to read a waste statistic – the gates a material passes through, and why every percentage needs a denominator, a boundary, and a year before it means anything (Ch.1); the map of institutional actors and where responsibility for a given stream actually sits (Ch.2); and the landfill-capacity crisis, waste-to-energy, and the twenty-five-year infrastructure commitments now being locked in under pressure (Ch.3).

Part II. How Waste Moves Through the City (Chapters 4–7)

The physical route from the kitchen, the yard, and the bin site (Ch.4) through trucks, transfer stations, and sorting facilities – the invisible middle where household behaviour becomes contractor data, or stops being data at all (Ch.5) – to landfill, RDF, and energy recovery (Ch.6), and finally to the question of who lives beside each of those destinations, and why (Ch.7).

Part III. What Actually Happens to Recyclables (Chapters 8–13)

The evidentiary scale as a working tool for separating documented fact from optimistic declaration (Ch.8), then six material streams examined in turn – from paper, cardboard, and deposit PET, where a relatively complete downstream chain exists (Ch.9), through the orange bin's own anatomy (Ch.10), rigid and flexible plastics (Ch.11–12), to glass, metal, and beverage cartons (Ch.13).

Part IV. Labour, Packaging Policy, and Extended Producer Responsibility (Chapters 14–19)

Who physically performs the recovery this book has been describing, and why sorting remains stubbornly resistant to full automation (Ch.14); how TAMIR's extended producer responsibility system was designed and where it stopped measuring what it needed to measure (Ch.15); why the deposit system and TAMIR were never institutionally integrated, despite handling overlapping materials (Ch.16); the 2026 reform now under discussion, and what it would need to make governable that isn't governable today (Ch.17); packaging design as the point where recyclability is decided before a product is ever filled (Ch.18); and what, if anything, Israel can usefully borrow from the EU's PPWR framework (Ch.19).

Part V. Money, Contracts, and Incentives (Chapters 20–22)

The financial architecture behind municipal waste services: gate fees, landfill levies, hidden subsidies, and the gap between who is legally responsible and who actually pays. This part explains why contracts often reward tonnes collected rather than residual waste reduced – and how KPIs, pay-as-you-throw principles, data requirements, and contract design could shift incentives from managing volume to reducing it.

Part VI. Waste Prevention and Local Circularity Infrastructure (Chapters 23–25)

Why prevention can't be reduced to individual behaviour or occasional volunteer effort. Organics, food, clothing, furniture, toys, and household items appear here not as “waste streams” but as materials that can be intercepted before they enter the system at all – organised, where possible, around the neighbourhood circularity hub as a missing layer of municipal infrastructure.

Part VII. Law, Oversight, and Civic Action (Chapters 26–28)

The practical civic toolkit: what residents, NGOs, and local groups have the right to ask; how to audit a waste stream without being an expert; and a one-week framework for working with municipalities, schools, and NGOs – taking part in the reform process not as unpaid janitors, but as informed civic actors who can engage with data, contracts, and policy choices on their own terms.

Three principles that run through the book

First: data matters more than declarations

Israel's official recycling figures sound convincing until their denominator is checked. TAMIR reported 84 percent recycling for 2021 – against a declared base covering roughly half the real packaging stream; its more recent 2022 figure, 97 percent, is measured against the same narrow base, not a wider one. The Ministry promoted separate collection for years before its own 2025 scan showed that only about 6 percent of household packaging reaches the right container. A 2024 inter-ministerial review, forced to revise the Ministry's own estimate of total packaging generation sharply upward, now puts Israel's honest recycling rate somewhere in a range of roughly 28 to 55 percent – a wide range precisely because the underlying uncertainty is real (Ch.16).

This book applies its evidentiary scale to every claim, official and independent alike. A fact with a confirmed source is an argument. A claim with no stated source, or the wrong denominator attached to it, is a declaration. The difference is not a technicality – it is the difference this entire book is built to preserve.

Second: incentives explain behaviour

When a municipality doesn't expand separate collection, that isn't always apathy or negligence. It's often a rational response to a genuine cost structure: sorting infrastructure costs money to run even when it recovers only a modest share of what passes through it, and a hidden municipal subsidy already covers part of the gap between what producers pay and what collection actually costs (Ch.15). When TAMIR favours the commercial stream over the household stream, that isn't arbitrary – it's a documented price signal, roughly a fivefold difference in treatment tariffs between the two (Ch.10).

Understanding the incentives built into the system is not an excuse for its results. It's a tool for engaging with it: knowing why an actor does what it does lets you propose changing the structure, not just appealing to its motivation.

Third: systemic change starts from below and above at once

This book doesn't ask readers to choose between neighbourhood-level civic action and participation in the national legislative process. Both matter, and they reinforce each other. A circularity hub that collects its own data and publishes an annual report becomes an argument in a conversation with the municipality. A municipality with that argument becomes stronger in its dialogue with the Union of Local Authorities. The Union, representing an aggregated position, has genuine influence on the reform itself.

This isn't naive optimism – it's a description of a mechanism that already works in countries with more developed waste-management systems. In 2026, Israel is at the point where that mechanism is only beginning to take shape.

A note on the evidence base, and on this edition

This book was written and is being revised during a period of active change. The State Comptroller's report (July 2025), the Ministry's national scan (August 2025), the Packaging Law memorandum (November 2025), and the Ministry of Finance's reform decision (December 2025) were all published within roughly the twelve months before this edition. That has two consequences. First, the book describes the system at a moment of maximum relevance, drawing on the most current data available at each point of revision. Second, some of what it describes – above all the 2026 reform (Ch.17) – remains genuinely unresolved. Where an outcome is uncertain, this book says so directly and describes the live possibilities rather than predicting a result.

This edition reflects a full methodological revision of Parts I through IV (Chapters 1–19), each fact independently re-checked against primary sources where possible, with a single running bibliography recording what was confirmed, what was updated from the previous draft, and what remains an open question. Parts V through VII are still undergoing the same process at the time of this printing; readers should treat their current chapter numbers and specific figures as provisional until that revision is complete, while the book's overall structure and argument – including where those parts sit in the sequence – are already settled.

For a reader picking this book up a year or two from now

Data on landfill capacity, tariffs, and the legislative status of the reform can change quickly, and some of it already has since earlier drafts of this book. The core analytical frameworks are more durable: the evidentiary scale, the principle that incentives explain behaviour, and the seven-step civic audit cycle all work regardless of whether any specific reform proposal is ultimately adopted in the form described here.

What a reader can take away from this book, in practice

- An understanding of the legal architecture: who is legally responsible for what, and why both sides of almost any waste dispute are often right at the same time (Ch.2).
- A map of the actors and their built-in conflicts of interest – a practical tool for knowing who to address a specific question to (Ch.2, Ch.15–16).
- The evidentiary scale itself – a way of telling a documented fact apart from a likely assumption and from a genuine unknown (Ch.8).
- An understanding of why the system's financial incentives systematically fail to align with its stated environmental goals (Ch.15–16, Part V).
- A first look at who actually performs the physical labour this book describes, and why that labour resists full automation more than the word “automation” usually implies (Ch.14).
- A civic audit methodology – what to request, how to observe, how to turn an observation into an argument (Part VII).
- A concrete action plan at every level of engagement, from an individual resident to a municipal official.

The book begins with the simplest possible question – what – actually counts as waste, and where does the statistical system catch it – and ends with a question about what one person, one neighbourhood, one organisation can do inside a system that looks too large and too slow to change. The answer isn't simple. But it exists, and it is concrete.

Chapter 1

What Counts as Waste – and Where Statistics Catches It

A plastic bottle bought at a corner shop is, for a moment, simply a product. Within a week it might have become four entirely different things: rubbish in a kitchen bin, a coin waiting at a deposit machine, a shred of PET on a sorting line, or a bale of raw material bound for a furnace. The object hasn't changed at all. What has changed is its status – and that quiet shift, easy to miss, is what this chapter is about.

Much of the public argument over how well Israel manages its waste is really an argument over percentages: recycling rates, landfill shares, collection targets, sorting quality. But a percentage means nothing until its denominator is known. So before trusting any figure, it is worth asking one recurring question: 100 percent of what?

That question matters more in Israel than in most places, because the system is legally fragmented, operationally uneven, and statistically inconsistent from one stream to the next. Some materials are regulated by dedicated laws; others fall to municipalities; others still are managed by producer-responsibility bodies, collected by private operators, or exported and lost from view. Some of this is visible in official reports. Much of it disappears into estimates, private contracts, or foreign markets.

This chapter looks at waste three ways – as a legal category, a material category, and a statistical object – to set out a working method for reading everything that follows: every claim about waste needs to be tied to a specific system, a specific point of measurement, and a specific year.

1. Waste as a legal category

In everyday language, waste is simply something a person no longer wants. Law requires more precision than that. Before anyone can say who is responsible for a discarded item, who should pay for its treatment, or which data system ought to record it, that item has to be placed inside a regulatory category.

Israel's basic framework rests on the Maintenance of Cleanliness Law of 1984, supplemented over time by separate laws for packaging, beverage containers, electronic equipment and batteries, tyres, construction waste, and other streams. There is no single, comprehensive "waste act" – only a patchwork of laws, each written in response to a particular problem as it arose.

For municipal waste policy, a handful of categories matter most.

Mixed municipal waste

The ordinary residual stream collected by or on behalf of local authorities – household waste and similar waste from businesses, institutions, and public spaces. The OECD describes it as household and similar waste collected by municipalities, including bulky items but excluding construction and sewage waste.^[64]

Construction waste

A large stream in its own right, generated by building, renovation, and demolition. It often surfaces in public space as illegal dumping, but statistically it is not the same object as household waste.

Bulky waste

Furniture, mattresses, appliances – usually entering the municipal system through special collection days, curbside placement, or informal rescue by passers-by. Legally and operationally it is distinct from ordinary bin waste, a distinction this chapter returns to later (§9).

Packaging waste

Governed by the Packaging Treatment Regulation Law of 2011, which places responsibility for collecting and recycling packaging on manufacturers and importers.^[65] In practice, this law is the legal basis for the TAMIR system and the orange-bin stream.

Deposit beverage containers

A separate economic and legal system under the Deposit Law, adopted in 1999 and extended in December 2021 to containers between 1.5 and 5 litres.^[24]

Textiles

A different position altogether. Unlike packaging, deposit containers, or electronic waste, textile waste in Israel has no dedicated producer-responsibility regime. A dossier compiled for this book – drawing on municipal collection contracts, NGO reporting, and available Ministry of Environmental Protection commentary – describes the sector as fragmented and weakly regulated, with collection left largely to municipalities, NGOs, and private operators rather than governed by any unified national framework.

The point worth holding onto is that legal categories are not material categories. A PET bottle, a shampoo bottle, a cardboard box, a Tetra Pak carton, and a cotton shirt can all become "waste" the moment they're discarded – but they belong to different legal systems, different collection channels, and different statistical worlds.

2. Waste as a material category

Waste policy speaks mostly in the language of legal streams – packaging, deposit containers, construction waste, electronic waste. Recycling facilities speak a different language: paper, cardboard, PET, HDPE, PP, metal, glass, textiles, organics, composites. The two languages overlap, but they don't map onto each other cleanly.

A PET beverage bottle is plastic, but it may also be a deposit container. A plastic yoghurt cup is plastic packaging. A plastic bucket is plastic but not packaging at all. A cardboard delivery box is both paper and packaging; a newspaper is paper but not packaging. A metal can is packaging; a metal pan is not. A shirt is textile, but nothing in Israeli law treats it as packaging or brings it under a producer-responsibility scheme.

The distinction isn't academic – it changes how a material gets counted. A bottle returned through the deposit system is tallied differently from one thrown in the orange bin. A cardboard box collected from a business may enter a commercial recycling stream, while an identical box from a household is counted through the municipal or blue-bin channel instead. A shirt dropped into a textile bin might be resold locally, exported, downcycled, landfilled, or burned abroad – and each of those outcomes belongs to a different ledger.

Nor does recyclability guarantee recycling. A material can be technically recyclable and still be economically unattractive, contaminated, mixed with other materials, or caught in a stream that makes recovery difficult. An item can enter a "recycling" bin and still be rejected, exported, downcycled, or landfilled further down the line.

Which is why "is it recyclable?" is the wrong first question. The next three sections follow three ordinary objects – a bottle, a carton, a shirt – to show what a better question looks like in practice.

3. The deposit bottle

Few objects illustrate the point as cleanly as a deposit bottle.

Returned through the deposit system, a bottle enters a relatively clean, incentive-driven collection channel – post-consumer material belonging to waste and circularity policy, but no longer part of the mixed municipal stream.^[24] Thrown into a green bin instead, the same bottle becomes ordinary municipal waste. Left on the street, it's litter. Placed in an orange bin because it falls outside the deposit system, it becomes packaging.

Nothing about the bottle itself has changed. What changed is the channel it entered – and with it, the system that will eventually report on its fate. That's why deposit containers belong in any serious discussion of waste policy, but shouldn't simply be folded into the totals for ordinary municipal waste. They sit within the broader system of post-consumer materials, tracked by their own legal and statistical apparatus.

4. Beverage cartons and Tetra Pak

A second object makes the same point from a different angle: cartons look simple to residents and turn out to be complicated for the system behind them.

A milk or juice carton resembles cardboard, but it's usually a multilayer package of paper fibre, plastic, and sometimes aluminium – which is why it doesn't belong in the ordinary paper stream. In Israel it's folded instead into the packaging-collection logic and placed in the orange bin.^[65]

The orange bin, though, doesn't mean "recycled locally." TAMIR describes the orange-bin chain as a sequence: packaging collected separately, transported to sorting stations, sorted by material, baled, and sent to recycling plants, mostly within Israel.^[79] But TAMIR's own process page notes an exception – beverage cartons made of cardboard, plastic, and aluminium are sent abroad after sorting, where they're broken down into their components and recycled separately.^[79]

The lesson carries through the rest of this book: the orange bin means entry into the packaging-collection system, not automatic domestic recycling. A carton can be placed correctly, collected correctly, sorted correctly – and still depend on a specialised facility overseas. "Collected for recycling" has to be read with both a system and a geography attached to it, a point the bottle made for legal channels and the carton now makes for physical destination.

5. Textile donation bins

A third object completes the pattern, and here the trail runs thinner than for either the bottle or the carton – which is itself part of the lesson.

Textile bins invite residents to believe that clothing has been "donated" or "saved" the moment it's dropped in. Collection is only the first step. The clearest evidence of where that step actually leads comes from a 2024 Greenpeace Israel investigation, reported by The Times of Israel: activists sewed GPS tags into twenty-four used garments in good, resellable condition and dropped them into municipal bins run by two contractors, Rosnir and Infinia, in towns from Safed and Rosh Pina in the north to Beersheba in the south, then tracked them for ten months.^[76]

Not one of the twenty-four garments was resold in Israel or recycled into new clothing. Eleven of the items placed in Rosnir bins ended up abroad, in Togo, Ghana, and Burkina Faso; others reached an Arab town and a West Bank city, and one reached Gaza; contact was lost with several after they reached sorting centres. All four tracked Infinia bins were in Beersheba: two garments reached the West Bank, one reached the Hiriya waste centre near Tel Aviv, and contact with the fourth was lost.

Beersheba's own account, given to The Times of Israel, adds a second layer. The city chose Infinia in 2022 to handle its waste, including textiles, which are sorted in Modi'in; before the war that began in October 2023, sorted textiles were sold on through a company called Southern Textiles to buyers in Gaza, a commercial link the municipality says the war severed. The city also supplied a precise scale marker: textiles made up less than 0.04 percent of Beersheba's total recyclable waste in 2023.^[76]

None of this is unique to Israel. A parallel 2023 investigation by the UK-based Changing Markets Foundation tracked twenty-one garments through the take-back bins of ten international fashion brands and found more than three-quarters destroyed, dumped, turned into rags, or exported to Africa; only about a quarter were recycled or reached second-hand shops.^[19] Ghana – one of the destinations named in the Israeli investigation – is widely described as the world's largest importer of used clothing, receiving an estimated fifteen million garments a week, of

which a share often put around forty percent is judged unsellable on arrival and ends up in landfills, rivers, or open fires.

Asked about this by Greenpeace Israel under a freedom-of-information request, the Ministry of Environmental Protection said it had no formal relationship with Rosnir, despite the company's use of the ministry's logo on its bins – which it has since ordered the company to stop using. The ministry's public position is that the real solution lies upstream, in reduced consumption, while defending existing collection as still preferable to landfilling despite its flaws.

None of this makes clothing donation pointless. Local reuse can be genuinely valuable, and Beersheba's pre-war textile-to-Gaza channel was itself a working regional reuse route, not merely an export write-off. But a textile bin proves only that an item entered a collection channel – one whose downstream reporting is far less standardised than TAMIR's, and where, as it turns out, even the basic scale of the stream is hard to pin down.

Unlike packaging, where TAMIR publishes national tonnages because the law requires it, Israel has no publicly available national estimate of how much textile waste the country generates each year. The Central Bureau of Statistics' annual survey of local authorities collects textile as one reported material category, alongside plastic, paper, metal, and glass – but no consolidated national figure emerges from it.^[18] The absence is worth noting on its own terms: for this stream, the statistical system has no visible national gate at all, not even the first one, long before the more familiar losses described later in this chapter begin.

For textiles, then, the question isn't only whether an item was collected, but where it went afterward – reused locally, exported, downcycled, landfilled, burned – and, one step earlier than for the bottle or the carton, whether there's even a number to check against. Every "rescued" garment has a geography. For the bottle, that geography is a domestic reprocessing line; for the carton, it may be a plant overseas; for the shirt, GPS tracking now suggests it is very often Ghana, Togo, Burkina Faso, or a town a few kilometres from the bin – and the total scale of that geography remains, at the national level, unmeasured.

6. Where statistics catches waste

Waste statistics don't count objects in the abstract; they count objects at specific gates. A given tonne of material may pass through several such gates before – or instead of – becoming genuinely recycled product:

Statistical gate	What it means
Generated	The material entered the economy and eventually became unwanted.
Discarded	A household, business or institution gave it up.
Placed in container	It was put into a green, orange, blue, purple, deposit, textile or other channel.
Collected	It was physically removed by a municipality, contractor, producer-responsibility system or private operator.
Accepted	The operator or facility accepted it as part of a valid stream.
Sorted	It was separated into a material fraction.
Sent to recycling	It was transferred to a recycler or processor.
Actually recycled	It became secondary raw material or a new product.
Reported as recycled	It was included in an official or corporate reporting system.

These gates aren't interchangeable. A tonne "collected" isn't necessarily a tonne "recycled." A tonne "sent to recycling" isn't necessarily a tonne that became secondary material. A tonne "reported as recycled" may depend on

the reporting conventions of a particular scheme. And a tonne exported from Israel can vanish from national view

Figure	Denominator (100% =)	What it means	What it does NOT mean	Source / year
25.6% ^[74]	All municipal waste collected by local authorities (2023)	Officially sent to recycling: 1.599 million tonnes out of 6.256 million tonnes.	Not the share of household packaging collected separately.	2023 national municipal waste statistics
75.8% ^[74]	All municipal waste generated (2022)	Landfilled: 4.7 million tonnes out of 6.2 million tonnes.	Not proof that all of this material was technically unrecyclable.	2022 national landfill statistic
25% by weight / 47% by volume ^[60]	Contents of household green bins	Share of packaging still found inside mixed household waste.	Not the composition of all municipal waste in Israel.	Household waste composition survey
~6% ^[61]	Household packaging waste	Captured in designated packaging containers.	Not Israel's overall recycling rate.	Capture into dedicated packaging collection
6.2% ^[79]	Packaging collected by TAMIR (2022)	Share coming from the orange-bin household stream. Most of TAMIR's collected packaging in 2022 came from businesses (77.3%) and dedicated municipal streams (16.5%).	Not the share of orange-bin contents that TAMIR recycled.	TAMIR collection structure, 2022
<0.04% ^[76]	Beersheba's total recyclable municipal waste (2023)	The share reported as textiles by the city's waste contractor – the only publicly sourced textile-scale figure this book could locate for any Israeli locality.	Not a national textile generation estimate: no consolidated national tonnage for textile waste is publicly published in Israel.	Beersheba Municipality, via Times of Israel, Nov. 2024

even while its environmental life continues elsewhere.

Collection statistics can therefore create illusions: they show that material entered a channel, not that it completed the journey through it.

7. Every percentage has a denominator

This is where the question posed at the outset – 100 percent of what? – needs a concrete answer, in real Israeli figures. Consider six numbers that circulate regularly in public debate about Israeli waste policy. Each is correct. Each answers a different question.

These figures don't contradict each other; they answer different questions, and the trouble starts only when they're set side by side as though they measured the same thing. Each has its own denominator, its own boundary, its own operator, its own stage in the chain, and its own year – and, as the TAMIR row shows, a headline number (6.2 percent) can sit beside a much larger figure easily missed altogether: businesses and dedicated municipal streams together account for over ninety-three percent of what TAMIR collected in 2022.

The textile row belongs here for a different reason, and it's included deliberately rather than rounded off to something tidier. No consolidated national figure exists for Israeli textile waste, so the only number available at any resolution is one municipality's own reporting. A single city's 0.04 percent can't be read as a national recycling rate the way 25.6 percent can – it simply isn't that kind of number. Set beside five figures that do have a national denominator, it makes the point this chapter has been building toward: for some streams, the system hasn't yet built the gate at all.

Figure 1 (suggested placement here): an infographic version of the table above, for readers who want the comparison at a glance rather than in prose.

The rule worth carrying into the rest of this book: a percentage without a clear denominator, boundary, stage, and year is a number that misleads.

8. Collection is not recycling

"Recycling bin" is a convenient phrase, but a misleading one. A bin recycles nothing by itself; it only opens a possible chain:

- The resident places the item in a container.
- The container is collected.
- The stream is accepted by the operator or sorting facility.
- The material is sorted into a usable fraction.
- The fraction is transferred to a recycler.
- The recycler turns it into secondary raw material or a product.
- The result is reported in a statistical system.

Losses can occur at any link. A container gets contaminated. A driver skips it. A stream mixes with residual waste. A facility runs out of capacity. Sorted material finds no buyer. Material gets exported, or downcycled, or counted as "sent to recycling" without ever becoming genuinely high-quality secondary material.

None of this means source separation doesn't matter – clean separation usually makes recovery both more valuable and more technically possible, and TAMIR itself notes that packaging separated at source sorts more cleanly and recycles more easily.^[79] But separation is not recycling. It's one necessary gate in a longer chain.

9. Prevention comes before accounting

There's a further reason definitions matter, and it's the reason this chapter exists at all.

Start the story only when something enters a bin, and you've already conceded that it became waste. But many objects become waste not because they lost all value, but because no practical system existed to intercept them at the moment they were discarded.

Take a fourth object alongside the bottle, the carton, and the shirt: a wooden chair set out on the pavement beside a building's bin enclosure. The chair is structurally sound – a loose leg, nothing more. Legally, once placed beside the municipal bins, it falls into the bulky-waste category described earlier (§1), a stream collected periodically and separate from both the green-bin residual stream and packaging. Materially, it's wood, foam, and fabric – three fractions that could, in principle, be separated and recovered. In practice, almost none of that happens.

One of three things usually occurs instead. A neighbour or passer-by takes it before the scheduled collection – an informal reuse transaction leaving no statistical trace in either direction: not counted as waste generated, not counted as waste diverted. Or it waits for the municipal bulky-waste truck, which collects it alongside items that are genuinely broken, and it's recorded only as bulky-waste tonnage – the system has no gate distinguishing repairable

furniture from structurally failed furniture. Or it sits exposed to the weather until no one wants it any longer, and the same statistical outcome – bulky-waste tonnage – records an item that could have been repaired as though it had always been destined for disposal.

Compare this to the bottle, the carton, and the shirt. Each of those, once discarded, at least enters a channel with a name – deposit, orange bin, textile bin – even if what happens after that channel is uneven or opaque. The chair often has no defined channel at all between becoming unwanted and being collected as waste. The interval in between is informal, invisible to any reporting system, and entirely dependent on chance – whether a passer-by happens to want a chair, that particular week.

This is why waste prevention can't be reduced to a moral appeal about sorting correctly. It requires infrastructure that gives objects like the chair a defined channel before the bulky-waste truck becomes the only option: repair points, reuse shelves, food-sharing systems, the kind of textile transparency the previous section found largely missing, composting options, organised bulky-item rescue, local transport support, and civic data collection able to see the difference between "repairable" and "discarded."

The statistical question changes accordingly. Instead of asking only how much waste was recycled after collection, it's worth asking how much material avoided becoming waste in the first place – and how much of that avoidance today depends on informal chance rather than designed infrastructure.

That's the difference between cleanup and system design. Cleanup begins after waste appears. System design asks why the object became waste at all.

Conclusion

This chapter opened with a simple question: what counts as waste? The answer turned out not to be simple. Waste is a legal category, a material category, an operational category, and a statistical category all at once, and the same object can move between several statuses in turn – product, unwanted item, packaging, deposit material, mixed waste, collected stream, sorted fraction, exported commodity, recycled material, landfill residue – as the bottle, the carton, the shirt, and the chair each showed in their own way.

For anyone acting on this information, the practical lesson is to stop asking only whether people sorted correctly, and start asking what the material is, which legal stream it belongs to, where it was placed, who collected it, whether it was accepted, sorted, and turned into a usable fraction, whether it was actually recycled, where the process ended geographically, and – always – what the denominator of any percentage actually is.

Only once those questions are answered does a waste statistic become an argument. The real question this book is asking is not simply how much Israel recycles, but how much material is prevented from becoming waste in the first place, how much is captured before it's lost, and how much ends up as a usable fraction rather than another burden shifted onto a landfill, an incinerator, or another country.

Chapter 2

Who Governs Waste in Israel: Responsibility, Actors, and Conflicts

Ask an average resident who is responsible for Israel's waste, and the honest answer is: no one, entirely. The system is governed by roughly a dozen actors – a ministry, a municipality, two kinds of producer-responsibility corporation, land and enforcement authorities, private contractors – each holding a different, sometimes conflicting, slice of power. That fragmentation is not incidental. It is the direct institutional consequence of a point already established (Ch.1): waste is not one thing but a sequence of statuses, and each status tends to create its own responsible actor.

Understanding this map is not an academic exercise. It tells you where to send a specific question, what data you can reasonably request, and when an answer – "that's not our department" – should be treated as incomplete rather than final.

1. Responsibility follows the stream

A resident asking "who is in charge of my waste" is really asking several different questions at once, because the honest answer depends entirely on which stream the object is in.

If an object enters...	Primary responsible actor / operator
Mixed municipal waste	Local authority + collection contractor
Packaging stream (orange bin)	TAMIR / packaging EPR system
Deposit stream	Deposit operators (ELA, Asufta, Makor Yarok) / retail return points
Construction waste	Contractor / regulated construction-waste chain (from 2026)
Textile donation bin	Municipality / NGO / private textile operator
Illegal dumping	Local authority – municipality, per HCJ 4726/20 (§2 below)
Landfill / sorting facility	Licensed facility operator, under MoEP supervision

Everything that follows expands on one row of that table: who the municipality is, and why its authority is narrower than it looks (§2); who sits above it, and what each actor actually controls (§3–6); who is meant to catch the failures (§7); how the list of actors has grown, reactively, stream by stream (§8); where the incentives built into the structure quietly conflict with one another (§9); how the largest decisions escape this map altogether (§10); and, finally, a practical guide to turning all of this into an actual question addressed to an actual body (§11).

2. Municipal responsibility: the operational core

Legally, most waste questions in Israel come back to one door: the local authority. Article 13 of the Cleanliness Law, 1984, assigns local authorities the duty to organize waste removal within their jurisdiction.^[57] Article 146(8) of the Local Councils Order, 1950, reinforces this by empowering a council to set rules, impose prohibitions, and take measures to protect public health – explicitly including the removal of nuisances and the demolition of dangerous buildings.^[54] Between them, these two provisions are the entire legal foundation for the sentence "the municipality is responsible for your waste."

Under the hood: the 2025 Supreme Court precedent

The clearest test of this framework came in HCJ 4726/20, Ezrahim LeMa'an HaSviva (Citizens for the Environment) v. Minister for Environmental Protection and the Jadeida-Makr Local Council, decided in June 2025.^[34] The petitioner NGO asked the Court to hold the state jointly responsible, alongside the local council, for illegal dumps that had gone unaddressed for years, creating a serious health and environmental hazard.

The state argued that its role is regulatory – enforcement, oversight, supporting funding – not operational; the council did not deny its sanitary powers, but pleaded insufficient budget and inadequate state funding. In a 2-1 majority, Justice Alex Stein, joined by Justice Yosef Elron, held that the duty to clear the dumps rests on the local council alone, grounding the ruling in Article 146(8) and Article 13. Justice Yechiel Kasher dissented: in his view, when residents are left for years without basic sanitation, the state has a constitutional duty to intervene under Basic Law: Human Dignity and Liberty, regardless of where the underlying responsibility sits.

Two things about this precedent matter beyond the case itself. It was not unanimous – the dissent keeps the state's constitutional exposure open in sufficiently extreme cases – and the majority's reasoning rests on exactly the two statutory provisions this section opened with: not abstract doctrine, but their direct, current application to a real dump site.

This division of labour is not only legal theory; the State Comptroller's own language confirms it. The Comptroller's July 2022 audit of municipal waste removal describes organising collection from public space and households as one of the most essential services a local authority must provide^[75] – treating the municipality, not the state, as the default operational address.

The council in HCJ 4726/20 did not dispute this. Its defence was structural rather than legal: it acknowledged its sanitary powers but argued it could not finance the removal of the dumps, and that state funding fell short. This is the hidden-subsidy problem examined at length elsewhere (Ch.14) – the municipality is legally obligated to provide the service, but funding from the state and the EPR system does not reliably cover its actual cost. Both sides of a dispute like this can be right at once: the mayor is formally responsible, and the complaint about insufficient state funding is not merely an excuse.

3. The state: regulator, funder, planner – not collector

Above the municipality sit several state-level actors whose relationship to waste is regulatory and financial rather than operational.

Ministry of Environmental Protection

Sets national strategy, licenses treatment and disposal facilities, administers the Cleanliness Fund, and exercises partial oversight over the producer-responsibility systems discussed below (§4–5). It does not collect waste.

Ministry of Finance and the Cleanliness Fund

The Ministry of Finance is not a sectoral ministry, but it may be the most consequential actor in the system, for one simple reason: money. The Cleanliness Fund, established under the 1984 Cleanliness Law and financed mainly by the landfill levy, exists to fund waste-reduction and cleanliness projects. The State Comptroller's May 2022 report on the Fund found that only around 37 percent of landfill-levy income went toward its own statutory purposes between 2007 and 2019^[73]; that balances of NIS 0.5–3.9 billion sat uninvested and earning nothing across 2011–2020; and that NIS 1.66 billion moved out of the Fund into the general state budget between 2016 and 2020 – which the Comptroller concluded made the Fund function, in practice, as "a budgetary source for financing government operations," contrary to both its statutory purpose and the Cleanliness Law itself.

This is not a closed, historical finding. In February 2023, with the Fund holding over NIS 3.2 billion, the government moved to transfer its management to the Ministry of Interior, with the Finance Minister supporting the move and the sitting Environmental Protection Minister publicly objecting.^[91] Whichever ministry controls the Fund's board controls the closest thing Israel's waste sector has to a discretionary budget – which is exactly why the fight over it keeps recurring.

Ministry of Interior

Sets the general legal and administrative framework within which local authorities operate, including their bylaw-making powers under the Local Councils Order (§2).

Israel Land Authority (RMI)

Controls the land needed to expand landfill capacity – the single most acute constraint identified by the State Comptroller (Ch.6). Its coordination, or lack of it, with MoEP and municipalities is a recurring bottleneck in siting new capacity.

State Comptroller

Manages no part of the system, but its periodic audits – on municipal waste removal, on the Cleanliness Fund, on construction waste – are the main source of publicly available, independently verified data on how the system actually performs. Several figures in this chapter come directly from Comptroller reports rather than from any operating agency's own account of itself.

4. TAMIR is not "the recycling system"

It is easy to collapse a chain of separate facts into one: plastic goes in the orange bin, the orange bin is TAMIR, TAMIR is recycling. Each link is true in isolation and misleading in combination.

TAMIR is not a government body. It is a non-profit public-benefit company, founded by the Manufacturers Association of Israel, whose shareholders are major food and beverage manufacturers – among them Tnuva, Osem, Strauss Group, the Central Bottling Company (Coca-Cola's Israeli bottler), Unilever Israel Food, Hogla-Kimberly, Shufersal, and Tempo.^[87] Under the Packaging Law, every producer or importer of packaged goods must contract with a MoEP-"recognized body" to fulfil its recycling obligations and pay it an annual fee; TAMIR was recognized in December 2011, and that recognition has been renewed twice since.

For over a decade, TAMIR was the only recognized body – an effective monopoly. In 2024, following a petition filed by a competing recycling corporation, Nateli, after four unsuccessful attempts to obtain MoEP recognition, the Jerusalem District Court found that MoEP had "shirked its responsibility" by failing for more than a decade to regulate competition in packaging recycling, and ordered the ministry to open the market. MoEP's own data, cited in the proceedings, showed TAMIR collecting only around 18 percent of the household packaging-waste potential – far below its rate from the commercial sector, which is logistically easier to serve.^[81]

Later chapters examine in detail how TAMIR's design produces this pattern, and why its reported 84-percent recycling rate is measured against a declared base that MoEP's own 2025 scan suggests covers only about half the actual packaging stream (Ch.14–15). What matters here is narrower and purely institutional: TAMIR is a legal-financial operator of one specific stream – packaging under producer responsibility – answerable to its own manufacturer-shareholders, and until 2024 effectively unchallenged in its market position. It does not own Israel's recyclable materials, and the orange bin is only one channel inside the system it runs.

5. Deposit-system operators: a separate stream, a separate cast

Beverage containers under the Deposit Law are handled by an institutionally distinct set of actors, even though both ultimately deal with plastic and metal packaging. Two recognized deposit corporations currently operate the stream: ELA, founded by major beverage manufacturers and holding roughly ninety percent of the market, and Asufta, a competitor established after a 2010 amendment opened deposit collection to competition; a third, Makor Yarok, began operating in 2020.^[84] These corporations run the trucks, urban collection points, and reverse-vending machines that make the deposit system work, and they report separately from TAMIR, with their own collection statistics.

The structural conflict between the deposit stream and TAMIR's – why two systems handling overlapping materials never integrated into one collection channel – is examined at length elsewhere (Ch.15). What matters here is

institutional: a deposit bottle is counted, tracked, and reported by a different actor than an orange-bin bottle, even when the two are made of the same plastic.

6. Private contractors: the invisible operational layer

Much of what a resident experiences as "the municipality's waste service" is, in practice, physically carried out by private companies working under municipal contract – collection and hauling contractors, transfer-station operators, sorting-facility operators, landfill operators, some private and some run as inter-municipal associations. All are licensed by MoEP, but day-to-day operational decisions – routing, staffing, maintenance – belong to the contractor, not the municipality or the ministry.

This has a direct civic consequence, developed further elsewhere (Ch.25): the material may be public waste, generated by residents and regulated by public law, but the operational records that would explain what actually happened to it – which truck, which transfer station, which buyer – sit inside private contracts and private logs, not in any public register. That is one reason public data requests about waste routinely come back partial or empty: the data genuinely is not held by the public body being asked.

7. Enforcement: many bodies, no single picture

Waste-related enforcement in Israel is split across several bodies that share no common database^{[51][86][90]}, which means the country has no unified picture of how much illegal dumping, burning, or littering actually happens, or how effectively it is punished.

Body	Share of fines / scale	Notes
Israel Police	~63% of Cleanliness Law fines	The default enforcer for ordinary littering complaints.
Ne'emanai Nekiyut (volunteer Cleanliness Trustees)	~21% of fines	Civilians trained by MoEP to report littering from vehicles; they do not issue fines themselves – reports become fines only after MoEP processing.
Green Police (executive arm of MoEP)	~30 inspectors nationally, ~NIS 20 million budget	Handles the fuller range of environmental offences, not only waste.
Yanshuf unit (Nature and Parks Authority, MoEP-funded)	14 inspectors nationally	Established 2019 specifically to target illegal waste dumping and burning in open areas.
David unit	Crossing-point enforcement	Targets waste-truck smuggling across the Green Line.
Municipal inspectors	Processed independently per municipality	No shared national database with the other bodies.
Unattributed	~16% of fine records	MoEP's own data shows these records cannot be traced to a specific enforcing body, due to inconsistent paper-form reporting.

The unattributed sixteen percent deserves particular attention, because it is not a footnote – it is a working demonstration of this book's central argument (Ch.1) applied to enforcement data. A fine that cannot be traced to an enforcing body exists in MoEP's ledger but cannot be assigned to any denominator that would let a reader judge which agency is over- or under-performing. The system loses track of its own enforcement statistics for exactly the same structural reason it loses track of material streams: no single gate captures the whole picture, and paper-based reporting between agencies leaves gaps nobody is specifically responsible for closing.

8. A patchwork built reactively

Israel has no single comprehensive waste law – only the foundational Cleanliness Law of 1984, plus a sequence of separate laws adopted reactively, stream by stream, each creating or empowering a new actor rather than extending an existing framework.

Year	Law	What it regulates – and which actor it creates or empowers
1984	Cleanliness Law	Basic municipal responsibility for removal; prohibition on littering; the enforcement powers described in §7.
1993	Collection and Removal of Waste for Recycling Law	Empowers (and can oblige) local authorities to designate recycling infrastructure – an early, narrow version of the actor-creation pattern this chapter traces. ^[83]
1999	Beverage Container Deposit Law	Creates the deposit-operator actor category (§5).
2011	Packaging Law	Creates TAMIR as the recognized packaging EPR operator (§4).
2012	Electronic Equipment and Batteries Law	Creates a parallel EPR structure for e-waste, run by separate recognized bodies from TAMIR.
2026 (in process)	Construction Waste Amendment	Gives a ~7.3-million-tonne/year stream, previously unregulated, its own dedicated actors and enforcement chain for the first time.

The most recent line in that table is not a historical footnote; it is happening now, and it is the clearest illustration of the pattern this chapter has traced. Construction waste – roughly 7.3 million tonnes a year, a larger stream by weight than household packaging – went more than forty years without a dedicated regulatory regime. The State Comptroller documented as early as 2021 that this absence of regulation directly drove widespread illegal dumping of construction debris in open areas and along roads. In June 2026, the Knesset Interior and Environmental Protection Committee approved an amendment to the Cleanliness Law that finally gives this stream its own enforcement chain, folded into the Ministry of Finance's Economic Arrangements Law – the same legislative vehicle now carrying a much larger reform (Ch.16).

The lesson is institutional, not merely historical: a waste stream does not acquire a responsible actor automatically. It acquires one when a law creates that actor – and until then, it defaults to whatever combination of municipal bylaw, general Cleanliness Law provisions, and ad hoc enforcement happens to reach it. Construction waste sat in that default state for four decades. Worth asking, of any stream discussed later in this book: which state is it currently in?

9. Built-in conflicts of interest

The actor map traced so far is not just a list of who does what. Several of its relationships contain structural conflicts of interest – not personal failings, but the predictable consequence of how incentives are distributed across the system.

Actor	Structural incentive
MoEP	Sets recycling targets and also supervises TAMIR, the body that repeatedly falls short of them. This creates an incentive to describe the system's performance in softer terms than the underlying data supports – not concealment, but the ordinary pull of being both referee and institutional owner of the policy architecture being judged.
Ministry of Finance	Both sets how much municipalities receive for waste services and promotes a reform that promises them savings. It also, per the State Comptroller's 2022 finding, drew NIS 1.66 billion out of the Cleanliness Fund into the general budget between 2016 and 2020 –

Actor	Structural incentive
	money earmarked by law for waste-reduction purposes. ^[73] A ministry with this fiscal relationship to the sector is not a neutral scorekeeper of reform outcomes.
Municipality	Legally responsible for removal, but funding may not cover actual costs. The rational response to being simultaneously obligated and underfunded is to supply the minimum defensible level of service, not the optimal one.
Israel Land Authority (RMI)	Controls the land needed to expand landfill capacity, but does not bear direct responsibility for the consequences of the capacity crisis – an asymmetry between the institution that can approve land use and the one that answers publicly for the fallout.
TAMIR	As the sole recognized packaging-EPR body for over a decade, TAMIR had limited structural incentive to prioritize the harder, more expensive household stream over the easier commercial one – and a 2024 court ruling found that MoEP had let this persist by failing to open the market to competition. A recognized monopoly has every reason to report its own performance favourably.
Collection contractors	Paid under municipal contracts that more often reward tonnage collected and service frequency than residual-waste reduction – rewarding volume handled, not waste prevented.

None of this is a reason to trust no one. It is a reason to ask each actor a more precise question – not "why did you fail," but "what does the structure you operate in actually reward you for doing?" A request about collected packaging tonnage is a question for TAMIR (the mechanics of asking it are covered in Ch.25) – but interpreting the answer means remembering that TAMIR, for over a decade the only recognized body in its market, has every institutional reason to present its own performance in the most favourable light the data allows.

10. National infrastructure: exceptional decisions under pressure

Everything described so far is the ordinary, fragmented, local machinery of waste governance. A second, much rarer channel operates by entirely different rules: national infrastructure planning, used to approve the largest facilities – above all, waste-to-energy plants.

The National Committee for the Planning and Building of National Infrastructures, known as VATAL, was established in 2002 under an amendment to the Planning and Building Law. Waste disposal and treatment sites are explicitly listed among the facility types eligible for "national infrastructure" status.^[85] Declaring a specific facility a national infrastructure requires a joint declaration by the Prime Minister, the Finance Minister, and the Interior Minister; once declared, VATAL holds the powers of the National Planning Council directly, bypassing the standard district- and local-committee review that would otherwise apply.

The Neot Hovav waste-to-energy facility – designed to sort roughly 300,000 tonnes of waste a year and generate about 50 megawatts from the non-recyclable residue – was approved through exactly this channel, under a government decision in October 2022. It never went through the ordinary municipal or regional planning process described elsewhere (Ch.6).

This channel matters for two opposite reasons at once. It genuinely accelerates projects that could otherwise stall for years in local approval – part of why waste-to-energy capacity is moving faster than, say, the expansion of conventional landfill space. But the same speed comes at the cost of local civic participation, which is structurally weaker here than in ordinary municipal development; the points of leverage available to residents still exist, because even a fast national process cannot skip public consultation altogether – only compress it.

11. Whom to ask

Understanding this actor map is a practical tool, not an academic one: it tells you where to send a specific question, and helps you recognise when an answer is genuinely incomplete rather than simply evasive.

Question	Who to ask
Why wasn't my green bin collected?	Municipality / sanitation department
Who actually collects my waste, and under what contract?	Municipality, via a data request
Where does mixed waste go after collection?	Municipality + contractor + transfer station
What happens to orange-bin packaging?	TAMIR + municipality
Why is a beverage carton in the orange bin if it's recycled abroad?	TAMIR / MoEP
How much packaging was collected in my district?	TAMIR + municipality
Where is a deposit bottle counted?	Deposit operators (ELA / Asufta / Makor Yarok) / MoEP
Who is responsible for an illegal dump site?	Municipality first (per HCJ 4726/20) – MoEP only if you can show a constitutional-scale failure
Who licenses a landfill or sorting facility?	MoEP
Where is enforcement data published?	Fragmented: police, municipality, MoEP, Nature and Parks Authority – no single source (§7)

Conclusion

The system has no single owner: roughly a dozen actors hold different, sometimes conflicting, pieces of legal and operational power. Legal responsibility for waste removal rests with the local authority, under Article 13 of the Cleanliness Law and Article 146(8) of the Local Councils Order – a position reaffirmed, not merely asserted, by the Supreme Court in June 2025, over a documented dissent. The state's role is regulatory and financial, not operational, and even in that narrower role, the State Comptroller found it diverting dedicated waste funds toward general budgetary use rather than the purposes the law assigns them. Producer-responsibility bodies – TAMIR for packaging, ELA, Asufta and Makor Yarok for deposits – are private or industry-owned corporations operating under a public mandate, not public utilities, and TAMIR's decade-long monopoly was found by a court to be a regulatory failure rather than a neutral market outcome. Enforcement is fragmented across at least five bodies with no shared database, to the point that roughly one fine in six cannot be attributed to a specific agency. New waste streams do not acquire responsible actors automatically – only when a law creates one, and some streams wait decades. And the single largest infrastructure decisions bypass this entire map through a separate, faster, less locally accountable channel.

With the legal framework and the actor map both in place, the sharpest test of how this system makes decisions under pressure comes next: the construction of waste-to-energy plants at the exact moment landfill alternatives are, quite literally, running out.

Chapter 3

Residual Waste, WtE, and Infrastructure Lock-In

A tonne of waste can travel a long way before anyone asks whether it needed to become waste at all. It can be prevented, reused, repaired, composted, sorted, and recycled – or, if none of that happens, it becomes what this chapter calls residual: whatever is left once every upstream option has been tried, skipped, or simply unavailable. Residual is not a natural category. It is a judgment about how well everything upstream worked (Ch.1–2) – and Israel is now building a piece of infrastructure that will define, for a quarter of a century, how much of its waste stream gets that judgment made in its favour.

That infrastructure is waste-to-energy: large facilities that burn non-recyclable waste and generate electricity from the heat. Israel's first is under construction now, in the middle of what officials themselves describe as a landfill crisis. Understanding what that crisis is, why it is producing this particular answer, and what the answer commits the country to for the next twenty-five years, is the subject of this chapter.

1. Residual waste is not a natural category

Waste policy generally recognises an order of preference for what to do with material once it's unwanted: prevent it from arising, reuse what still works, recycle what can be turned back into raw material, recover energy from what's left, and landfill only what remains after all of that. Waste-to-energy sits second from the bottom of that order – one step above the outcome everyone agrees is worst.

"What remains," though, is not a fixed quantity. It depends entirely on how much prevention, reuse, and recycling actually happened first, and how uneven that upstream performance can be (Ch.1). A responsible answer to "is this residual?" requires asking whether the system has genuinely exhausted prevention, reuse, repair, organics diversion, deposit return, and material recycling – or whether it has simply stopped trying and called the result residual by default.

This distinction matters more than it might seem, because the facility this chapter describes is not being built to treat an abstract category called "residual waste." It is being built to treat whatever Israel's system currently produces once packaging, organics, textiles, and everything else has passed through – or failed to pass through – every upstream option first. The quality of that upstream system determines what ends up in the furnace.

2. A landfill crisis, on a specific clock

The immediate reason waste-to-energy is advancing now, rather than as part of some longer-term strategic conversation, is that Israel is running out of places to put waste in the ground. The State Comptroller's July 2025 follow-up audit found that remaining landfill capacity at the start of 2025 stood at 7.67 million cubic metres – not enough for the volume of waste expected by the end of 2026.^[58] That is not a forecast; it is a description of physical space running out on a fixed and near-term schedule.

The pressure from that finding has reached well beyond environmental circles. In November 2025 the Manufacturers Association of Israel – an industrial lobby, not an environmental NGO – petitioned the High Court of Justice, describing the situation as approaching "a point of no return."^[58] Reporting at the time described trucks queuing for hours outside southern landfills, sorting facilities filled to capacity, and drivers dumping waste – sometimes setting it alight – by the roadside rather than waiting. That the warning came from industry rather than activists says something about how broad the concern has become.

This is the context in which any long-term infrastructure decision is now being made – not calm, strategic planning, but a scramble against a specific and approaching deadline. That combination matters, because the facilities discussed in this chapter are built to last twenty-five years, and decisions made under a deadline measured in months tend to look different in hindsight than decisions made with room to think.

3. Why WtE is moving forward now

Israel's first large-scale waste-to-energy plant is being built at Neot Hovav, an industrial zone roughly 15 kilometres south of Beersheba already home to the country's main hazardous-waste facilities. The project moved through its final competitive stage in November 2025, when three bids were submitted to the inter-ministerial tender committee^[31]; on 5 February 2026, the committee – led by the Accountant General, with the Ministry of Environmental Protection and the Ministry of Finance both represented – announced the Shapir–BlueGen–Dekel group as the winning consortium.^[48]

The facility is designed to sort around 300,000 tonnes of waste a year and recover energy from roughly the same volume of non-recyclable residue, generating about 50 megawatts of electricity.^[3] It will operate under a 25-year concession, after which it reverts to state ownership. Its legal basis is the national-infrastructure mechanism described earlier (Ch.2 §10): Government Decision No. 1895 of October 2022 designated it National Infrastructure Plan TATL 107, which is what allowed the project to bypass ordinary municipal and regional planning. As part of the arrangement, Beersheba has committed the whole of its municipal waste stream to the facility, and similar agreements with other regional authorities are expected; officials describe the plant as capable of handling 60 to 70 percent of everything the southern region generates.

Two further facilities are reportedly planned – one serving Jerusalem, one serving southern Israel – each designed for around 650,000 tonnes a year, roughly double Neot Hovav's capacity.^[81] Both are still described as being in a pre-tender stage, with pre-qualification expected sometime in 2025–2026. It is worth treating that timeline with some caution: an earlier attempt at a Jerusalem-area facility, planned between 2019 and 2022 at a similar scale, was terminated after sustained public opposition, including a High Court petition over a proposed site near Kfar Adumim.^[29] Announced capacity and built capacity are not the same thing, and Neot Hovav itself has already been described in earlier planning documents at a smaller scale – 150,000 tonnes and about 20 megawatts – than the figures ultimately awarded.^[92] Numbers attached to projects still years from construction are worth reading as intentions, not commitments.

4. What WtE solves – and what it doesn't

Waste-to-energy earns its place in a landfill crisis because it does a specific, narrow job well: it reduces the volume of waste that needs to be buried, taking pressure off landfill capacity, recovering some energy in the process, and creating treatment capacity for material that has nowhere else to go.

It does not do several other things that are easy to assume it does. It does not reduce how much waste gets generated in the first place – if anything, a facility that needs feedstock has a structural interest in a steady supply of it. It does not put materials back into economic circulation the way recycling does; burning a tonne of plastic destroys it as a material, permanently, in a way that sorting and reprocessing does not. It does not substitute for source separation – a plant built around a mixed residual stream still depends on upstream systems to keep recyclables and organics out of that stream in the first place. And it does not make waste disappear: it transforms mixed waste into three separate downstream products – electricity, and two categories of ash – each of which becomes its own problem to manage (§7). Objects do not vanish in this system; only their status and geography change (Ch.1), and combustion is simply the most physically dramatic version of that change.

5. Lock-in: when residual waste becomes a required feedstock

The central risk in this chapter is not chemical or technical. It is contractual.

A facility built to process 300,000 tonnes a year for 25 years needs, in some form, a guarantee that it will actually receive something close to that volume for those 25 years – otherwise the economics behind a quarter-century of financing don't work. That guarantee typically takes the shape of a take-or-pay arrangement: a minimum tonnage the municipality commits to supplying, or pay for regardless, whether or not that much waste actually exists in a given year.

This creates a structural tension with everything the rest of this book is trying to encourage. If a city becomes serious about waste prevention, reuse, and recycling – precisely the outcomes argued for elsewhere in this book – the residual stream should shrink over time. But a facility financed against a guaranteed minimum tonnage has no built-in mechanism to welcome that shrinkage. Success at the top of the waste hierarchy can become, financially, a problem for a facility sitting near the bottom of it – unless the contract is written to anticipate that outcome rather than penalise it.

This is why the crisis framing matters. Crisis narrows the choice set and increases lock-in risk: when a decision has to be made in months rather than years, there is less room to negotiate contract terms that protect against exactly this tension, and more pressure to sign whatever keeps the immediate landfill emergency from getting worse. None of that makes waste-to-energy the wrong tool. It makes the contract the place where the real decision gets made – and the place civic attention belongs.

6. Capacity versus the national stream

It helps to see the scale of what's being built against the scale of what Israel actually generates.

Facility	Capacity (t/year)	Share of Israel's ~6.26 million tonnes of municipal waste (2023)
Neot Hovav (awarded, Feb. 2026)	~300,000	~5%
Two additional planned facilities (pre-tender stage)	~650,000 each (~1.3 million combined)	~21% combined
Total, if all three are built as currently described	~1.6 million	~25–26%

That combined figure lines up closely with the Ministry of Environmental Protection's own 2030 target, announced in 2020, of treating 23 percent of Israel's waste at energy-recovery facilities – alongside 51 percent recycling and 26 percent landfill, down from roughly 80 percent landfilled at the time.^[80] A more recent government commercial guide, from March 2026, cites an updated 2030 recycling target of 54 percent without restating the energy-recovery and landfill figures alongside it^[81] – which may mean the split has moved since 2020, or may simply mean it wasn't restated. Either way, the WtE share of the target is worth watching rather than treating as fixed: whether Israel ends up needing 23 percent thermal capacity or something different depends entirely on whether the 54 percent recycling target is actually met.

7. Ash, emissions, and new waste streams

Burning waste does not make it disappear. A waste-to-energy facility converts a mixed input into three separate outputs: electricity, bottom ash (the solid residue left in the combustion chamber, typically 20–25 percent of input weight in comparable facilities elsewhere), and fly ash (the finer particulate captured from flue gas, smaller in volume but classified as hazardous waste in most jurisdictions because it concentrates heavy metals and other pollutants). Both ash streams need a destination – usually a specially licensed landfill cell, since ordinary landfilling isn't appropriate for hazardous fly ash.

At the time of writing, no independently sourced figures for expected ash volumes, ash disposal arrangements, or emissions-monitoring protocols specific to Neot Hovav have surfaced in public reporting. Official statements describe the facility as meeting European emissions standards and Israel's Clean Air Act, and note that comparable technology operates at hundreds of sites across Europe – but "comparable technology exists elsewhere" is a different claim from "here is this facility's monitoring plan," and only the second kind of claim should be treated as evidence about Neot Hovav itself.

8. The civic question is not "for or against"

It's tempting to frame waste-to-energy as a binary – support it, because the landfill crisis is real, or oppose it, because burning waste destroys material that could have been recycled. Both positions capture something true, and neither is especially useful for a resident trying to act.

The more useful question is not whether Neot Hovav should exist, since by February 2026 that question was already settled. It's what has to be protected before Israel's residual-waste stream is locked into a 25-year thermal contract – because the contract terms, not the underlying technology, will determine whether this facility ends up treating genuine leftover material or ends up needing genuine leftover material to exist, quietly working against the rest of this book's argument for prevention, reuse, and recycling.

9. Contract safeguards citizens should demand

None of the following is confirmed as current practice at Neot Hovav – the point of this list is what to ask for and watch for, not a report on what has already been secured.

- No feedstock guarantee for materials that belong in recycling, deposit, or organics streams – a take-or-pay volume should exclude, by contract, anything the rest of the system is supposed to be capturing.
- Priority for organics diversion, so that composting and anaerobic digestion capacity is not quietly displaced by a facility that needs tonnage.
- Transparent, regularly published waste-composition monitoring, so residents can see what is actually going into the facility, not just how much.
- Independent emissions monitoring, with data published on a schedule, not released only on request or after an incident.
- A public plan for both bottom ash and fly ash disposal, including where the hazardous fraction goes and who is licensed to take it.
- No financial penalty on the municipality for successful waste reduction – a shortfall against the guaranteed tonnage caused by prevention working should not be treated the same as a shortfall caused by poor service.
- Periodic review of the minimum guaranteed tonnage, rather than a fixed number locked in for the full 25 years regardless of how the upstream system performs.
- Annual public disclosure of the facility's full mass balance – tonnes in, electricity generated, bottom ash, fly ash, and where each output went.

Conclusion

In a landfill crisis, waste-to-energy can look like the only available answer, and in the narrow sense of buying time against a specific 2026 deadline, it may be. But the more of Israel's residual-waste infrastructure gets built and locked into 25-year contracts now, the more carefully the country needs to define what actually counts as residual – because a facility financed against a guaranteed tonnage does not simply treat waste. Left unchecked, it creates a long-term demand for it, at the exact moment when the rest of this book is arguing for a system that produces less.

Chapter 4

The First Mile: Kitchen, Yard, and Bin Site

Waste management does not begin at the truck, and it does not begin with a moral choice about which bin to use. It begins earlier and more quietly than that – in the narrow space between household life and the first public collection point, where food, packaging, organics, textiles, batteries, and bulky items either find a usable channel or collapse into mixed waste. What happens in that space decides a large part of the material's fate (Ch.1), long before any truck, transfer station, or sorting facility comes into view.

This chapter follows that space: the kitchen, the bathroom, the balcony, the stairwell, the yard, and the bin enclosure at the end of it – and asks not "did the resident choose correctly," but "was there a real channel available to choose."

1. More than a packaging decision

Discussions of household waste in Israel tend to collapse quickly into a single question: which colour bin does the packaging go in. That question matters, but it captures only a fraction of what actually happens on the first mile. In an ordinary week, a household generates food scraps and spoiled leftovers, empty packaging, perhaps a deposit bottle, flattened or unflattened cardboard, a broken glass jar, a worn-out shirt, a dead battery, and – periodically – a piece of furniture or an appliance that no longer works. Each of these has its own first-mile options, and each can fail in a different way before it ever reaches a truck.

Reducing all of this to "which bin" produces a narrow picture: a resident either sorts correctly or doesn't. A fuller picture asks what channels actually exist for each material, how close and reliable they are, and what happens when they don't exist at all.

Material	First-mile options	What can go wrong
Food	Eat it, store it, share it, compost it where possible, or place it in the mixed bin.	No organics channel is available in most buildings, and food-sharing infrastructure is rare (Ch.22).
Packaging	Orange bin, blue bin, purple/glass point, or the mixed bin.	Confusion about what belongs where, distance to the right bin, or contamination.
Deposit bottles	Return machine, store drop-off, mixed bin, or litter.	Inconvenient machine locations; simpler to throw it in with everything else.
Paper/cardboard	Blue bin, commercial collection, or the mixed bin.	Wet or greasy paper, bulky unflattened boxes that don't fit.
Glass	Purple/dedicated point, deposit if applicable, or the mixed bin.	Dedicated points are far sparser than orange bins (§2).
Textiles	Donation/reuse, a textile bin, or the mixed bin.	A textile bin is a real channel, but not proof of reuse (Ch.23).
Furniture, appliances	A reuse group, municipal bulky-item collection, or left on the street.	No defined channel between becoming unwanted and being collected – the same gap the chair illustrated (Ch.1 §9).
Batteries, small electronics	A dedicated drop-off point, or the mixed bin.	Drop-off points are few and rarely nearby.

The Ministry of Environmental Protection's 2025 national scan found that only around 6 percent of household packaging reaches a designated packaging container.^[61] That figure should not be read simply as a record of behaviour going wrong. It measures capture into a specific set of channels, and capture depends on how many of those channels are physically available, how far they are, how reliable collection from them is, and how the packaging system itself is designed – not only on what a resident decided at the kitchen counter.

2. Channels are infrastructure, not bins

It helps to describe what each first-mile channel actually captures, rather than treating them as an interchangeable set of coloured containers.

Channel	What it captures	Legal / operational universe	Main risk
Orange	Plastic and metal packaging, beverage cartons	Packaging Law / TAMIR – entry into the packaging system, not a recycling guarantee (Ch.1 §4)	Glass, which breaks and can injure sorters or damage equipment
Blue	Paper and cardboard	Municipal / 1993 recycling law – covers packaging and non-packaging paper alike	Wet or contaminated paper; large boxes left unflattened
Purple / glass points	Glass packaging	TAMIR, but through a physically separate, sparser network	Points are far less common than orange bins – distance itself is a barrier
Deposit machines / points	Beverage containers under the Deposit Law	A separate institutional stream, run by deposit operators, not TAMIR	Convenience and machine density, not awareness
Textile bins	Used clothing and household textiles	Municipal or private operator, largely unregulated (Ch.23)	Collection is not proof of reuse
Bulky-item collection	Furniture, mattresses, appliances	Municipal, on a scheduled basis	No channel exists between the item becoming unwanted and the scheduled collection (Ch.1 §9)
Mixed / residual bin	Whatever no separate channel captured – used or available	Municipal collection, ordinary frequency	Once mixed with organics and residual waste, recovery of anything recyclable becomes harder, costlier, and lower quality

Under the hood: why glass is a separate story

Glass is physically hazardous for a standard sorting line – it breaks during transport and sorting, shards can injure workers, and it can damage machinery built for plastic and metal. That is why it never goes into the orange bin and instead relies on a separate, much sparser network of collection points. The first mile is part of why glass recovery in Israel lags behind other materials (Ch.15 covers the material-specific detail): if the nearest dedicated point is inconvenient, the glass is more likely to end up in the mixed bin regardless of a resident's intentions.

3. When the standard channel doesn't fit: building age and density

The standard model – a yard with a full set of coloured bins – assumes a kind of building that not all housing stock actually is. Israeli building regulations adopted in 2019 set out, for the first time in this level of detail, what a waste room or bin enclosure must provide: minimum dimensions, separate chutes for different streams in larger buildings, ventilation, and safe access. Those requirements apply going forward. They say nothing about buildings constructed before 2019, many of which were never designed with space for more than one or two containers.

This is a structural point about building age and density, not about who lives in a particular building. Two documented cases make the shape of the problem visible without requiring speculation about any specific community. Bnei Brak, one of the country's most densely populated cities, could not fully comply with standard nighttime collection conditions the State Comptroller's office had set for it – conditions written for less densely built environments, including a requirement to avoid compacting waste directly outside any one residence, which is difficult to guarantee on streets that dense. Rahat, examined in the same audit, is served by a different collection arrangement for roughly half its housing units precisely because its built density doesn't fit the standard courtyard-bin model.

The pattern in both cases is the same: it is the physical shape of a neighbourhood, not a judgment about its residents, that decides whether the standard first-mile model works at all. Newer buildings with code-compliant waste rooms give residents meaningfully better access to separate collection than older, denser buildings do – a form of inequality that sits on top of the geographic one (Ch.7) and is rarely discussed on its own terms.

4. Capacity, frequency, and trust

The State Comptroller found that the average container volume available per resident correlates with a municipality's collection costs: more volume generally means more equipment, fuel, and labour per tonne actually collected, particularly where bins run under capacity. Mixed household waste is typically collected two to three times a week across the municipalities examined; collection of recyclable streams is often less frequent, sometimes just once every week or two.

None of this determines behaviour mechanically, but it shapes the convenience structure residents act within. A full bin, a distant bin, a broken or dirty bin, and an infrequent collection schedule are different problems with the same effect: material that was meant for a separate stream ends up in the one that gets emptied more often. A further factor rarely captured in collection statistics is trust – residents who have watched a truck tip separated bins into the same compartment, or who regularly see supposedly separate streams overflowing or uncollected, have a rational reason to stop bothering. That isn't laziness. It's a reasonable response to a channel that has stopped behaving like one.

5. What can be measured – and what can't

A cluster of fifteen local authorities in the Eastern Negev has begun fitting containers with RFID tags, allowing real-time tracking of which bins were actually collected. One member town, Kseife, built a working dashboard of over 1,300 containers in about a week, giving its sanitation department, for the first time, an exact picture of where every bin sits and whether it's in working order. Across the cluster, the same data lets municipalities identify unserved areas before complaints pile up, and – usefully for municipal budgets – verify that a contractor is being paid only for containers the authority actually recognises as its own, rather than ones added without authorisation.

This is a genuine improvement in visibility, and worth taking seriously as a model. But it is visibility into one specific gate: whether a container was collected. It says nothing about what was inside it, whether that content was contaminated, whether the operator accepted it, or what happened to it afterward (Ch.1 §6). RFID can confirm that a bin was emptied. It cannot, by itself, confirm that anything in it was recycled.

6. A civic first-mile audit

Understanding the first mile in your own street doesn't require special access – most of it is directly observable.

- Visit two to five bin sites near you and note what channels are actually present: orange, blue, purple/glass, textile, deposit, bulky-item area – or only a mixed bin.
- Photograph the site and note the date and time.
- Check whether bins are full, overflowing, broken, or blocked.
- Note the collection schedule if posted, and compare it with how full the bins are when you pass.
- Look for contamination – non-packaging waste in the orange bin, food waste mixed into paper.
- Ask the municipality directly where the truck goes after collection, and request the collection contract or schedule if it isn't public (Ch.25 covers how to frame that request).

Conclusion

The first mile is not a moment of individual moral choice. It is the point where household materials meet – or fail to meet – the infrastructure available to them, and a large share of every material's eventual fate is decided there, before any truck arrives.

The Ministry of Environmental Protection's 6-percent figure for packaging capture should be read as a first-mile infrastructure and access problem, not primarily a sorting-behaviour one. Access to a working first mile is uneven – not only between municipalities, but within them, shaped by a building's age and a neighbourhood's density as much as by geography (Ch.7). Container capacity, collection frequency, and resident trust together shape a convenience structure that either supports separation or quietly undermines it. And even where new tools like RFID are making parts of the first mile measurable, what they measure is collection, not outcome – a distinction this book keeps returning to for good reason (Ch.1).

A well-designed first mile does not guarantee that material is recycled. But a poorly designed one guarantees that most of it never gets the chance.

Chapter 5

Collection, Transfer, and Sorting: Where Waste Becomes a Measured Stream

A resident's involvement with a piece of waste usually ends at the bin. The system's involvement is only beginning. Between that bin and the headline recycling statistics quoted in public debate lies a sequence of trucks, weighbridges, transfer stations, and sorting lines that almost no one outside the industry ever sees – and at each point in that sequence, the material either becomes a measured, accountable stream, or it doesn't.

This is the invisible middle. It deserves that name not because nothing happens there, but because everything that happens there is easy for the public to miss: a load can be accepted or refused, a stream can be downgraded by contamination, a "sorted" fraction can turn out to be mostly residual. Chapter 1 described waste statistics as a sequence of gates – generated, collected, accepted, sorted, sent onward, actually recycled, reported. This chapter is about the middle gates in that sequence, and about a fact worth stating plainly: this is where household behaviour becomes contractor data, and where, just as easily, contractor data can quietly stop existing.

1. The truck is the first data device

The truck that empties a bin is not just a vehicle; it's the first instrument that could, in principle, turn a first-mile channel (Ch.4) into a measured stream. What it records – when it does – includes the route driven, the time of collection, which containers were serviced and which were missed, and, where RFID tagging is in use, the identity of each individual bin. None of this data is inherently private. All of it is, by default, held only by the contractor operating the truck.

Whether any of it reaches the municipality, let alone the public, depends entirely on the terms of the collection contract – a theme this chapter returns to directly (§7) and one that later chapters examine in full (Ch.21).

2. The transfer station: the first physical bottleneck

A transfer station is a facility where waste is moved from collection vehicles into larger transport units, or partially sorted into components for recycling or reuse, before continuing on to its next destination. It typically performs several functions at once: consolidating loads from many small trucks into fewer large ones, weighing what comes in, compacting to reduce transport volume, and – in some cases – an initial, coarse sorting pass.

The weighbridge is the detail worth pausing on. It is, in most cases, the first point in the entire chain where a claim about tonnage becomes a number anyone actually measured, rather than an estimate. Everything upstream of it – how much a resident discarded, how full a bin was, how much a truck collected – is inference. The weighbridge ticket is evidence. A transfer station, in other words, is not only a logistics facility. It is the system's first real measurement gate.

3. Primary sorting: an extraction gate, not a recycling gate

Where a transfer station does perform sorting, it is almost always primary sorting: pulling out what is large, visible, and valuable enough to be worth extracting by hand or with simple equipment – large metal objects, big pieces of cardboard, garden waste, visually identifiable hazardous items, sometimes bulky items. Everything else continues on, unseparated, to a landfill, an RDF stream, a waste-to-energy facility, or a specialized sorting complex.

It is tempting to read "the transfer station sorted it" as "the material was recycled." It wasn't, not yet, and often not at all. Primary sorting catches what is large and obviously valuable. It does nothing for contamination, for small or multilayer packaging, or for organics that have already mixed into the residual stream. The rule from Chapter 1 applies here with no exceptions: primary sorting is an early extraction gate, not a recycling outcome.

4. Specialized sorting: where a load becomes a fraction – or doesn't

A specialized sorting complex, sometimes called a materials recovery facility, does what a transfer station cannot: separate a mixed stream into distinct material fractions using optical sorters, magnets, eddy-current separators, and manual quality control. This is where a load of orange-bin packaging actually becomes a PET fraction, an HDPE fraction, a metal fraction – or is rejected outright because contamination made separation uneconomical. Later chapters go deep into how these facilities work and what they can and cannot recover (Ch.9–13); the point that belongs here is simpler: every fraction that leaves such a facility has a rejection rate attached to it, whether or not that rate is ever published, and that rejection rate is one of the most informative – and most rarely disclosed – numbers in the entire system.

5. The remainder: where residual waste is actually made

It's easy to think of "residual waste" as simply whatever a resident failed to sort. Chapter 3 already complicated that picture: residual is not a natural category but the result of a judgment about how much of a stream upstream systems managed to recover. This chapter adds the second half of that argument. Residual isn't only found in a green bin. It is manufactured, gate by gate, by sorting decisions, technology limits, contamination, and the economics of whatever buyer exists for a given fraction that week.

Incoming stream



Accepted fractions → recycler / buyer



Rejected material / tails → RDF, landfill, WtE, or further sorting

Every arrow in this chain is also a document: a route log, an RFID record, a weighbridge ticket, a sorting or rejection-rate report, an invoice, or a sales record for the outgoing material. The chain is auditable in principle. Whether it is audited in practice is a separate question – and usually the more important one.

A tonne that leaves a sorting facility as "remainder" is not a tonne that failed on its own. It is a tonne about which a specific, identifiable set of decisions was made – by a facility operator, under a specific contract, facing specific market conditions for recyclables that week. Those decisions deserve exactly the same scrutiny as a resident's decision at the kitchen counter (Ch.4), and in practice receive far less.

6. Contractors, facilities, and the ownership of data

Chapter 2 already established that much of what residents experience as "the municipality's service" is physically carried out by private contractors, and that the operational records explaining what actually happened to a load sit inside private contracts and private logs, not in any public register. The invisible middle is where that fact becomes concrete.

Data point	Who holds it	Why it matters
Truck route	Contractor, sometimes shared with the municipality	Proves where a stream physically went
Collection time / RFID log	Contractor, or a municipal RFID system where one exists	Proves whether the scheduled service actually happened
Weighbridge ticket	Transfer station or receiving facility	Proves tonnes actually accepted, by stream
Rejection / downgrade rate	Sorting facility	Shows how much of a "separated" stream was in fact usable

Data point	Who holds it	Why it matters
Outgoing fractions	Sorting facility, buyer, or TAMIR	Shows what the material actually became, not what it was collected as
Destination of the remainder	Transfer station or contractor	Shows whether the rest went to landfill, RDF, WtE, or further sorting

The more actors a stream passes through – contractor, transfer station, sorting facility, buyer, TAMIR – the easier it becomes for responsibility to fragment and for any single one of these data points to simply not exist in a form anyone can request. This isn't a claim that facilities are hiding numbers out of bad faith. It's a claim about incentives: nothing in the ordinary operation of this chain requires any actor to publish rejection rates or outgoing-fraction data, and in this round of research no Israeli transfer station or sorting facility was found to publish that data in any public, aggregated form. That absence is itself worth noting – not as proof of concealment, but as a gap nobody is currently required to close.

7. New capacity is not automatically new recycling

Israel is currently investing seriously in the middle of the system. In June 2026, TAMIR and KMM laid the cornerstone for a NIS 70 million packaging-sorting complex at the Hiriya Recycling Park, designed to serve orange-bin collection for roughly half the country's population, sort twelve separate material streams, and run at up to three times the capacity of the facility it replaces.^{[82][14]} Separately, the Cleanliness Fund approved an eight-year program running through 2030 – NIS 900 million for existing and new sorting facilities, plus NIS 144 million for related projects – paying operators NIS 50 per tonne of municipal waste received, rising to NIS 75 per tonne for facilities that also divert organic waste for treatment.^[44] Combined with the Zero Waste Shafdan, Dia, and Neot Hovav facilities and one more planned in the South Sharon region, national sorting capacity is expected to reach roughly 4.5 million tonnes a year, against a national municipal waste stream of about 6 million tonnes.^[44]

All of this is real capacity, and it matters. But capacity is not the same claim as recycling, and this chapter has spent five sections explaining why: a tonne of new sorting capacity still produces accepted fractions, rejected tails, and a remainder, and nothing about building the facility guarantees that any of those three outputs will be measured publicly, let alone published.

Under the hood: who checks the checker

The body managing the money behind this expansion is the Cleanliness Fund, financed by the landfill levy and administered by the Ministry of Environmental Protection. Its own record on transparency is part of the story. The State Comptroller found that only around 37 percent of the Fund's income went toward its own statutory purposes between 2007 and 2019, that the Fund does not evaluate the effectiveness of its own spending, that it has not consistently filed the activity reports the law requires, and that leaving its balances uninvested cost an estimated NIS 964 million in foregone returns.^{[73][12]} The pattern didn't end with that audit: as recently as 2024, funds raised specifically through the landfill levy were approved for purposes with no connection to waste at all, including NIS 32 million toward electric buses, under a legal provision that allows the Fund to spend on "additional purposes."^[43] An institution that has struggled to account for its own money is now the primary funder of the infrastructure this chapter has been describing – worth remembering before treating any investment announcement as, by itself, good news for recycling.

8. Civic mass-balance questions

Understanding the invisible middle in practice means asking for the specific documents that exist at each gate, not general reassurance that the system works.

Question	Ask whom
Which contractor collects this stream, and under what contract?	Municipality
Is every load weighed, and is there a ticket for it?	Transfer station / municipality
Are tonnages reported separately by stream – mixed, orange, blue, garden waste, bulky?	Transfer station
What share of incoming material is rejected or downgraded, and why?	Sorting facility
What fractions leave the facility, by weight, and who receives each one?	Sorting facility / TAMIR
What share of the remainder goes to landfill, RDF, WtE, or further sorting?	Facility / municipality
Under which MoEP permit does the facility operate?	MoEP / facility
Are contractor payments based on tonnes collected, containers serviced, or measured outcomes?	Municipality
Is any of this published in an annual public summary?	Municipality / MoEP

Conclusion

The invisible middle is where a load either becomes evidence or stops being any. A truck route, a weighbridge ticket, a sorting report, a rejection rate, an outgoing-fraction record – each of these could exist, and each, when it does, turns a vague claim about recycling into a specific, checkable one. Chapter 2 explained why so much of this data sits in private hands by default. This chapter has shown what gets lost as a result: not just information, but the ability to tell a genuinely recovered fraction from a tonne quietly reclassified as residual somewhere between the bin and the buyer.

New capacity – a bigger sorting complex, a larger subsidy program – is necessary, but it is not self-certifying. It only becomes evidence of more recycling once tonnes in, accepted fractions out, rejection rates, and remainder destinations are measured and available to be checked, including by the public paying for it.

Chapter 6

Landfill, RDF, and the End-of-Pipe Zone

End-of-pipe is not the place where waste disappears. It is the place where failed prevention and failed recovery become land, fuel, ash, gas, leachate, and long-term liability.

Chapter 5 followed a collected load through acceptance, sorting, and rejection, to the point where whatever remained became a residual stream. This chapter follows that residual stream to where it physically ends up: a landfill, an RDF process, or a waste-to-energy facility. Each of these is a different destination with a different set of consequences, and the end-of-pipe zone is where the system stops asking whether material could still be recovered and starts asking where it can be put, burned, or transformed into fuel.

1. A capacity crisis measured in cubic metres, not metaphor

Israel's landfill shortage is not a forecast or an activist estimate. In its July 2025 follow-up audit, the State Comptroller found that remaining landfill capacity at the start of 2025 stood at 7.67 million cubic metres – insufficient for the volume of waste expected by the end of 2026, even before accounting for emergency expansions still being negotiated at individual sites such as Hagal, Dudaim, and Ganei Hadas.^[72] That is a physical limit with a date attached to it, not a rhetorical flourish.

The warning reached well beyond environmental circles. In November 2025 the Manufacturers Association of Israel – an industrial lobby, not an environmental NGO – petitioned the High Court of Justice, describing the situation as approaching "a point of no return,"^[58] and documenting trucks queuing for hours outside southern landfills, sorting facilities filled beyond capacity, and drivers dumping waste – sometimes setting it alight – by the roadside rather than waiting.

Under the hood: reading the landfill and recycling figures side by side

Public discussion of this crisis draws on more than one figure for both landfill share and recycling rate, and this book has already used more than one of them itself. Rather than picking one silently, it's worth setting them out together, since the differences are mostly about year and methodology, not disagreement about the underlying trend.

Figure	Source / year / denominator	Notes
75.8% / ~76%	State Comptroller, all municipal waste generated, 2022 (Ch.1 §7; also restated in the Comptroller's July 2025 follow-up audit)	The most precisely dated and sourced landfill figure available. Used as this book's reference point. ^{[74][46]}
83%	Calcalist, Nov. 2023, no denominator or methodology stated in the article itself	A later, less precisely sourced figure from a wartime-context news piece. May reflect a genuine 2023 increase, a different waste category, or simply a rounder journalistic estimate – the article does not say which. ^[10]
23.4%	State Comptroller, 2021, OECD-comparable recycling-rate methodology (OECD average ~60%)	A different measure from the 25.6% in Ch.1: an internationally comparable "recycling rate," not the same as "share of municipal waste officially sent to recycling." ^[46]
25.6%	2023 national municipal waste statistics (Ch.1 §7)	Later year, and very likely a different calculation method from the OECD-comparable figure above. The two are not directly comparable without more detail on methodology than either source publishes. ^[74]

Figure	Source / year / denominator	Notes
18% → 26%	State Comptroller, growth in recycling rate over roughly a decade of rising landfill levies	Confirms the same broad trend as the 25.6%/23.4% split: real but slow progress, whichever exact endpoint figure is used. ^[46]

Read together, these figures tell a consistent story even though no two of them are directly comparable: Israel landfills somewhere around three-quarters or more of its municipal waste, depending on the year and definition used, and recycles somewhere between roughly a fifth and a quarter, against an OECD average two to three times higher. The exact percentage matters less than the direction and the gap – and reading multiple sourced figures side by side, rather than picking the most dramatic one, is exactly the discipline this book has been arguing for from its first chapter.

2. Why capacity can't simply expand

The shortage did not appear overnight – the Comptroller flagged it as early as 2021 – and the gap between diagnosis and solution runs through a long chain of institutional obstacles, several of which trace directly back to the fragmented actor map described earlier (Ch.2). Expanding or opening a landfill requires approval from the Israel Land Authority, the Planning Administration, and local planning committees, each stage taking months or years. Individual sites carry their own technical constraints – one southern landfill required special lining because of oil-shale deposits underground, an engineering problem rather than a bureaucratic one – and some sites sit close enough to air bases that the birds a landfill attracts become an aviation-safety issue. Expanding a landfill is, predictably, an unpopular decision for the local authorities asked to host one. And a 2024 government decision calling for an inter-ministerial team to draft a comprehensive framework law produced no such law: the team never convened.

No single actor controls all the conditions needed for new capacity – land, planning approval, technical feasibility, and local consent sit with different bodies, none of which can act alone, and none of which is penalised for the collective delay.

3. Evron: a capacity shock, and a harder question underneath it

The Evron landfill in the Western Galilee handled roughly 360,000 tonnes of mixed municipal waste a year before its remaining volume began running out. A new sorting facility was built alongside it – a NIS 40 million investment, designed to process around 1,300 tonnes a day – and sat ready for more than a year without being switched on, because the region had nowhere to send the organic fraction the sorting process would separate out.^[7] Without a downstream destination for that organic material, sorting everything and then landfilling it anyway made the new facility, for a time, not worth operating at all.

One regional waste-industry figure went further, telling Calcalist in May 2024 that Evron's operators had commercial reasons not to activate the sorting line – landfilling directly remained more profitable for them than sorting first.^[7] Whether or not that specific motive is fully accurate, the structural point survives it: a landfill operator facing a choice between sorting and continuing to bury waste is not a neutral party in that decision, and the same conflict-of-interest logic already applied to TAMIR and the Cleanliness Fund (Ch.2 §9, Ch.5 §7) applies here too. The sorting facility was eventually inaugurated in mid-2025, with the Ministry of Environmental Protection's leadership attending the ceremony.^[49] In the meantime, northern municipalities whose waste could no longer go to Evron had to reroute it – in some cases by rail over roughly 300 kilometres to a Negev landfill – a shift that industry estimates put at more than a 20 percent increase in treatment costs. That cost mechanics deserves its own treatment, which later chapters on pricing and gate fees provide in full (Ch.19–20); what belongs here is simpler: a regional capacity shock does not just move waste, it moves money, and the two are not separable.

4. RDF: energy recovery, not material recovery

Refuse-derived fuel is the least visible of the end-of-pipe routes, in part because it doesn't require a dedicated plant of its own – the residual fraction selected for it is typically burned in existing industrial facilities, cement kilns

among them, rather than a purpose-built waste facility. According to the Knesset Research Institute, organic waste makes up roughly 45 percent of Israel's municipal solid waste, and a further 45 percent or so is residual material left after sorting – material not currently recovered as a recyclable fraction under existing sorting, market, and infrastructure conditions, some share of which becomes RDF feedstock.

That phrasing is deliberate. "Not currently recovered" is a different claim from "not recyclable," and the distinction matters for the same reason Chapter 3 insists residual is not a natural category: what ends up as RDF feedstock is not necessarily unsuitable for material recycling, only unrecovered under the sorting, market, and infrastructure conditions that happened to apply to it. RDF is recovery of energy value, not recovery of material value, and it is worth being precise about which of the two a given tonne represents before calling it "recovery" at all.

Organics deserve a separate note here. Treating them as simply another end-of-pipe stream understates what's really at stake: an organic fraction landfilled or burned for energy is an organic fraction that a composting or anaerobic-digestion system further upstream failed to intercept, and that upstream failure – not the technology used to dispose of the result – is where a lasting fix has to start (Ch.22 covers this in full).

5. Where WtE fits in this picture

Waste-to-energy is a third end-of-pipe destination, competing with landfill and RDF for the same residual stream – but its institutional stakes are large enough that they get their own chapter (Ch.3), which covers its 25-year concession structure, the contractual lock-in risk that comes with guaranteeing a facility a minimum tonnage for a quarter-century, and the specific safeguards citizens should be asking their municipalities to demand. Here it appears only as one row in the comparison this chapter is really about: which destination a given tonne of residual waste ends up at, and what that destination produces in return.

6. Three routes, three sets of outputs

Route	What it does	What it produces	Main risk
Landfill	Buries residual waste in the ground	Methane, leachate, a long-term land liability	Physical capacity, groundwater and air pollution, permanent land use
RDF	Converts a selected residual fraction into fuel, mainly for industrial energy use such as cement kilns	Fuel, plus a rejected fraction that still needs a destination	Can be described as "recovery" while still representing a loss of material value, not material recycling
WtE	Burns residual waste directly in a dedicated facility, generating electricity	Electricity, bottom ash, fly ash, emissions	Contractual lock-in and long-term feedstock demand – covered in full in Ch.3, where the safeguards citizens should demand are set out

None of these three routes make material disappear. Each converts it into something else – buried mass, fuel and rejects, or electricity and ash – and each of those outputs becomes, in turn, something that has to be monitored, stored, or managed for years afterward.

7. The end-of-pipe mass balance

Chapter 5 traced how a collected load becomes accepted, sorted, rejected, or turned into residual material. This chapter's version of that same chain picks up where that one left off.

Residual stream after sorting



Landfill · RDF · WtE



New outputs: buried waste, methane, leachate, RDF fuel and rejects, combustion emissions, bottom ash, fly ash

Objects do not vanish here any more than they did earlier in the chain (Ch.1). What changes is status and geography: a tonne of mixed waste becomes a tonne of buried mass generating methane for decades, or a tonne of fuel and a smaller tonne of rejects, or a fraction of a megawatt-hour and a measured quantity of ash requiring its own licensed destination.

8. Civic questions about destinations

The useful version of this inquiry isn't "for or against" any single technology – it's a specific set of questions about where a specific municipality's waste actually goes, and what evidence exists for what happens to it there.

Where does my municipality's residual waste physically go – which landfill, RDF user, or WtE facility?
Is the load weighed on arrival, and by whom?
How much remaining capacity does that destination have, and until when?
Is the waste sorted before landfilling, or landfilled directly?
What share of the residual stream becomes RDF, and where do the rejects from that process go?
What emissions or leachate monitoring data are publicly available for that site?
Who pays the gate fee, and how is it set?
Under which MoEP permit does the site operate, and when does that permit expire?

9. From physical destination to geographic burden

Landfill capacity is not only a technical question of cubic metres. It is also a geographic one: whose region becomes the endpoint for the country's residual waste, year after year, regardless of where that waste was generated. A meaningful share of Israel's municipal waste is landfilled in the south, often after a long haul from the north or centre – Evron's rerouted waste, travelling some 300 kilometres by rail to a Negev site, is one concrete instance of a much broader pattern. Once we know where residual waste goes, the next question is who lives near those destinations, and why that distribution isn't evenly shared – which is exactly where the next chapter picks up.

Chapter 7

The Geography of Residual Waste and Environmental Justice in Israel

Once material becomes residual, it does not disappear. It goes somewhere. Chapter 5 showed how a collected load becomes measured, accepted, sorted, rejected, or turned into residual material; Chapter 6 showed where that residual stream physically ends up – landfill, RDF, or a waste-to-energy facility, each producing its own long-term outputs of buried mass, fuel and rejects, or electricity and ash. This chapter asks the question that follows naturally from both: where are those destinations located, and who lives beside them?

Residual waste is not only a technical category. It is a geography of burden – and that geography is not natural. It is produced by contracts, permits, land decisions, landfill shortages, sorting failures, and national infrastructure planning, in exactly the same way the earlier chapters showed status and responsibility to be produced rather than given.

This chapter applies the environmental justice framework to that geography. The approach originated in the United States, in the civil rights and environmental movements of the 1980s and 90s, and is now used internationally as an analytical lens for examining who bears an environmental burden and who does not – not as a claim about any particular country's legal obligations, but as a way of asking a question that technical waste-management discussion tends to skip.

1. From destination to burden geography

Chapter 6 established that residual waste has three main destinations, each with its own outputs. Environmental justice asks a version of the same question at each one.

Route	The geographic-justice question it raises
Landfill	Who lives near the burial site itself – the odour, the birds, the truck traffic, the methane and leachate that persist for decades after the site closes?
RDF	Which industrial zones, and which neighbourhoods near them, receive the fuel derived from residual waste – typically burned in facilities such as cement kilns that were sited for industrial, not waste, reasons?
WtE	Where are these facilities built, and who lives close enough to be affected by their emissions, ash logistics, and constant truck traffic, for the 25 years of the concession (Ch.3)?
Ash and rejects	Once bottom ash, fly ash, and RDF rejects are separated out, where do those secondary streams go – and does anyone track them as carefully as the original tonnage?
Long-haul transport	When local capacity runs out, waste travels – sometimes hundreds of kilometres. Which communities live along that route, absorbing the traffic and emissions of a problem that isn't theirs?

End-of-pipe routes are also social routes. A tonne of residual waste is never just processed – it is processed somewhere, by someone's account, near someone's home.

2. The south as national endpoint – read with the same caution as Chapter 6

Public discussion of Israel's waste geography frequently cites a single figure: that some 68 percent of the country's waste is landfilled in the south.^[10] That number comes from the same November 2023 Calcalist article already flagged in Chapter 6 for stating no clear denominator or methodology alongside it. It should be read as an indicator of strong spatial concentration, not as a precise, fully sourced national mass-balance account – the same caution this book has applied to every other headline percentage since Chapter 1.

What is better sourced is the underlying mechanism: the Knesset Research Institute notes that landfilling costs are lower for northern municipalities than for southern ones, a result of tariff and logistics structures tied to where landfill capacity happens to be concentrated. The paradox is real even without a precise percentage attached to it – proximity to disposal sites brings no economic benefit to the region that hosts them, and may even coincide with higher costs for waste generated locally.

One note on what this chapter does not do: an earlier version of this material referenced a specific cost comparison between northern and southern authorities for which the underlying citation was never actually recorded – a gap flagged directly in this book's own source notes rather than quietly filled in. Nothing in this chapter rests on that particular figure.

3. Evron: when a regional shortage exports burden

The clearest illustration of how waste geography shifts under pressure is not hypothetical – it happened in the Western Galilee. The Evron landfill, handling roughly 360,000 tonnes of mixed municipal waste a year, began running out of capacity; a new sorting facility built beside it, a NIS 40 million investment designed to process around 1,300 tonnes a day, sat unused for more than a year because the region had nowhere to send the organic fraction sorting would separate out – and, according to one regional industry figure, because continuing to landfill directly remained more profitable for the site's operators than sorting first (Ch.6 §3).

While that stood unresolved, northern municipalities whose waste could no longer go to Evron had to send it elsewhere – in some cases loaded onto rail cars for a roughly 300-kilometre journey to a landfill in the Negev, a rerouting that industry estimates put at raising treatment costs by more than 20 percent. Every kilometre of that route passes through or beside other communities, absorbing the diesel traffic and transfer-point activity of a capacity shortfall that did not originate with them. Waste geography is not static: when one region's end-of-pipe capacity collapses, the burden does not vanish along with the shortfall. It travels, and it lands on whichever region still has room.

4. The Negev's double burden: national endpoint, local service gap

Within the region that already absorbs a disproportionate share of the country's residual waste, a second and different asymmetry exists, affecting Bedouin communities of the Negev (in Arabic, the Naqab).

Under the hood: the population figures, read transparently

Estimates of the Negev's Bedouin population vary by source and methodology, and this book states them side by side rather than picking one silently. Nagabia, a dedicated Bedouin-population research centre, put the figure at 312,300 as of 2022 – about 3.2 percent of Israel's total population.^[63] Israel's Population and Immigration Authority recorded 301,308 as of June 2023.^[2] The Central Bureau of Statistics' own figure, based on the 2008 census and updated estimates, stood at 279,100 at the end of 2019.^[2] All three sources note that unrecognized villages are inherently difficult to count precisely, which is itself part of the story this section tells.

The same difficulty applies, more sharply, to how many people live without formal recognition. Nagabia's 2024 statistical yearbook put the population of fully unrecognized villages at 86,367 – about 28 percent of the Negev's Bedouin population, against 63 percent in seven recognized urban settlements and 9 percent in eleven recognized villages under two regional councils.^[63] A separate estimate, attributed to geographer Arnon Sofer and circulating in tertiary sources, puts the share living in what it calls "illegal" settlements closer to 45 percent^[37] – a wider definition, not a contradiction of the narrower one. Recognition status, in other words, is itself a spectrum, and the share of the population without it depends on where exactly the line is drawn.

What is not in dispute is the basic mechanism: recognition status governs access to organized municipal infrastructure, and waste collection is one of the services that depends on it. A 2010 multimethod study of the unrecognized village of Umm Batin, published in the Archives of Environmental and Occupational Health, documented that in the absence of organized collection, household and construction waste accumulates near homes and is frequently burned – a practice that falls disproportionately on women, who traditionally manage household waste, and that creates direct exposure to corrosive, toxic, and flammable material. That finding is not an isolated one. Site visits led by the Regional Council for Unrecognized Villages together with the environmental legal organization Adam Teva V'Din, reported in January 2025, documented ongoing accumulation in villages including Alzarnug, Aljurf, and Algrain, with residents explicitly stating they would be willing to pay for regular collection on the same basis as recognized authorities – the obstacle was not unwillingness to participate, but the absence of any authority responsible for organizing it.^[41] As recently as April 2026, a state-backed pilot led by the Ruppin Institute began extending basic municipal services, including sanitation, to Bedouin dispersal areas ahead of full recognition^[56] – evidence that the gap documented for over a decade is not static, even if its resolution remains incomplete.

The State Comptroller's own 2022 audit of the Western Negev cluster of municipalities included, among its five stated audit themes, "gaps in waste collection between Jewish and non-Jewish municipalities and measures to narrow these gaps" – the formulation of an official state body, not an activist assessment, directly acknowledging a measurable gap in municipal service. A useful complication worth noting alongside it: the Israel Land Authority's own land-protection unit has, in separate operations, removed hundreds of tonnes of illegally dumped waste from state land in and around Lakiya, Hura, Rahat, Segev Shalom, and Tel Sheva – meaning the same system that has not organized collection in some of these areas does actively enforce against the dumping that results from its absence.^[47]

Put together, this is not simply a story about one region bearing more burden than others. It is a double asymmetry: the same south that absorbs a disproportionate share of the country's national waste stream also contains communities that, at the local level, receive markedly less organized sanitation service than the national average – over-served as a disposal endpoint, under-served as a place people actually live.

5. Procedural justice: who gets to see what's happening

Chapter 5 established that the data explaining what happens to a load of waste – its route, its acceptance or rejection, its outgoing fractions, the destination of whatever remains – typically sits inside private contracts and facility logs, not in any public register (Ch.5 §6). That fact has a geographic dimension too: the communities living closest to a landfill, an RDF-burning facility, or a WtE plant are the ones with the strongest practical reason to want that data, and often the least formal standing to request it.

Environmental justice, understood this way, is not only about who bears a burden – it's about who can see it, question it, and act on it. A host community that cannot access emissions data, tonnage-by-origin figures, or the terms of the permit governing its nearest facility is being asked to carry a cost it cannot independently verify, still less contest.

6. Civic questions about burden and access

Question	Why it matters
What share of the waste arriving at a nearby facility was generated somewhere else?	Distinguishes a community's own waste from waste imported to it by the system
Were residents near a proposed new site meaningfully consulted before it was approved, or only notified?	Procedural justice – consultation is not the same as participation

Question	Why it matters
Does the host community receive any service improvement, compensation, or infrastructure investment in return for hosting the facility?	Tests whether burden and benefit are ever actually balanced
Is emissions, odour, or leachate monitoring data for the site published anywhere the public can check it?	Exposure without visible monitoring is exposure without accountability
Within the same region, is there also a documented service gap – communities not receiving basic collection at all?	A region can be both a national dumping ground and locally underserved at once

Conclusion

Waste geography is the geography of decisions made upstream – in contracts, permits, land allocations, and the accumulated effect of landfill shortages and sorting failures traced through the previous two chapters. None of it is inevitable, and none of it is politically neutral. If the system sends a disproportionate share of its residual waste to particular regions, those regions should not also be the ones least able to see, question, or benefit from the decisions that put it there – and where, as in parts of the Negev, a region is simultaneously the national system's endpoint and locally underserved by that same system, both halves of that sentence need fixing, not just the more visible one.

Chapter 8

“Collected for Recycling” ≠ “Recycled”: The Evidentiary Scale

The previous chapters have already shown that waste moves through gates, actors, contracts, facilities, and geographies, and that at almost every one of those points, material or data can quietly drop out of the picture (Ch.1, Ch.5). This chapter turns that accumulated insight into a working tool: a method for judging exactly how much of any claim about material fate is actually known, and how much is assumed.

A recycling claim is only as strong as the weakest gate it can document. That is the principle behind everything in this chapter, and it is the principle the next five chapters apply to specific materials – paper, plastics, glass, metal, and beverage cartons.

1. Every claim names a gate – say which one

"Collected" and "recycled" are different events, and treating them as interchangeable is the single most common way a waste statistic misleads without technically lying. The table below lines up against the gates already introduced in this book (Ch.1 §6, Ch.5): each one represents something that has to be independently true, and each has its own kind of evidence.

Gate	What has to be true	Evidence that would confirm it
Placed	The item entered a first-mile channel at all (Ch.4)	Container audit, observation
Collected	A truck actually removed it	Route log, RFID record, contractor record (Ch.5 §2)
Accepted	A facility received the load as that stream, not downgraded on arrival	Weighbridge ticket, intake report (Ch.5 §2, §7)
Sorted	It was separated into a usable fraction, not rejected as tails	Sorting report, rejection-rate report (Ch.5 §4)
Sent onward	The fraction reached a buyer, recycler, RDF user, landfill, or WtE facility	Sales record, transfer record, gate-fee invoice
Actually recycled	A recycler turned it into secondary material or a new product	Recycler output record – the only gate that closes the loop
Geographically placed	The destination, route, and any host community are identifiable	Facility location, permit, monitoring data (Ch.7 §5)

The point of laying the gates out this way is not academic. A source can be excellent evidence for one gate and worthless for another. TAMIR's own reporting is strong evidence for what was collected and declared to it; it is not, by itself, evidence for what a recycler actually did with the material afterward. The Ministry of Environmental Protection's 2025 national scan is strong evidence for the placed and collected gates at the household level; it says nothing about what happened at the gates further down the chain. Confirming one gate never confirms the others – each has to be checked on its own terms.

2. Reading the MoEP 2025 scan precisely

In August 2025, the Ministry of Environmental Protection published the results of a national scan of household waste streams, and the headline figures are worth stating exactly. Around 6 percent of household packaging waste reaches a dedicated separate-collection container.^[61] A large share of the ordinary mixed bin's contents – plastic, cardboard, and other material of this kind – consists of potentially recoverable material that did not enter an appropriate separate first-mile channel.

That second point needs a careful reading, not the shorthand "60 percent should have ended up in the orange bin." Recoverable material belongs in different channels depending on what it is: paper and cardboard mainly in the blue bin, glass through dedicated points or the deposit system, deposit-marked bottles through deposit machines, and only a narrower category of plastic and metal packaging in the orange bin specifically (Ch.1 §1–§2). "Recoverable" and "belongs in the orange bin" are not the same claim, and collapsing them repeats exactly the material-category error Chapter 1 warns against.

The scan's importance isn't that it "proves the system has failed" in a rhetorical sense. It's that it creates an official, directly measured data point – a physical scan of what households actually discard, not a calculation built on what companies declared to TAMIR – against which every collection and recycling claim in this book, and in Israeli public debate generally, can now be tested.

Under the hood: why this measurement is stronger than what came before it

Before this scan, Israeli recycling figures were built almost entirely on TAMIR's own reporting against its "declared base" – the volume of packaging that producers under contract with TAMIR actually declared. That base is estimated to cover only about half the packaging that actually circulates in the country, since MoEP's own data show that more than 40 percent of producers and importers have no contract with TAMIR at all – a gap TAMIR's own CEO, Alon Tal, confirmed publicly in September 2025, stating that only about half of businesses comply with the law. Under that accounting, a widely cited "84 percent recycled" figure for 2021 turns out to describe 84 percent of a declared base covering roughly half the real stream – at best around 42 percent of the full picture, and only for the gates that figure actually measures, not the gates further downstream. The 2025 scan replaces that kind of back-calculation with a direct measurement at the household level: 6 percent is not "recycling from the declared base," it is the share of material physically observed reaching the correct container.

3. The evidentiary scale

To talk honestly about what happens to material after it leaves a household, this book marks every claim about waste flows on a four-level scale. This is not a rating of source quality – it's a rating of how complete the picture is, and sometimes a picture is incomplete not because anyone did sloppy work, but because the relevant data was never made public in the first place.

Level	Meaning	Typical source
✓✓ Confirmed	Documentary evidence at the specific gate in question	MoEP scan, TAMIR report, Comptroller report, licensing registry, weighbridge data
✓ Probable	Independent sources agree, but the full chain isn't documented	Journalism + operator statement + municipal data, read together
~ Reconstruction	No direct data; inferred from adjacent tonnage, capacity, or routes	Indirect estimate – method and uncertainty must be stated
? Unknown	No public data exists for this gate	State it plainly, and note what request could change that (Ch.25)

An activist who treats "probable" as "confirmed" risks being publicly corrected and losing an audience's trust. An official who describes a reconstruction as confirmed creates the impression of a system that documents itself when it

doesn't. Both failures are avoidable with the same habit: naming the gate, and naming the level of evidence available for that specific gate, every time.

4. Where material actually goes: five destinations, not one

Official statistics often compress everything that isn't landfill into an undifferentiated "recycled" or "recovered" category. In fact, material that clears the sorting gate can end up in five genuinely different places, and the difference between them matters.

- Material recycling – the material becomes secondary feedstock or a new product. This is the only path that actually closes the loop.
- Export – shipped abroad, which does not automatically mean recycling; some exported material is reclassified as waste on arrival and landfilled elsewhere (Ch.1 §5 already showed this for textiles).
- RDF – turned into fuel, typically burned in facilities such as cement kilns. Energy recovery, not material recycling (Ch.6 §4).
- WtE – burned in a dedicated facility for electricity, producing bottom ash and fly ash that need their own destination (Ch.3, Ch.6 §5).
- Landfill as tails – even material that passed through sorting can be rejected afterward as contaminated or unsortable residue.

When a report states "60 percent of packaging is recycled" without specifying which of these five paths it means, the number describes an unknown mixture of a genuine loop-closing outcome and four other outcomes that are not the same thing at all – and, per the scan above, it also has to be read against how much material reached this stage to begin with, not just what happened to it once it arrived.

5. Seven questions for any material stream

The scale is only useful if it's pointed at the right questions. This is the minimum checklist this book applies to every stream discussed in the chapters that follow, whether it's plastic from the orange bin, paper from the blue bin, or glass from a dedicated collection point.

- Does the material reach an appropriate first-mile channel, or does it stay in the mixed stream?
- Where does the truck actually go – sorting facility, transfer station, landfill, RDF user, or WtE plant?
- What happens to this specific material at the sorting facility?
- What is the rejection rate – the share the facility downgrades as unsuitable?
- Who is the downstream recycler, and does it have documented capacity and real output?
- Is any of it exported – and if so, to where, and classified as what?
- Is there a public report confirming each of the above, or are we working from a reconstruction?

Where the chapters ahead have answers, they're given with a stated level of evidence. Where they don't, this book says "unknown" and states what kind of request might change that, rather than filling the gap with a plausible-sounding guess.

6. Using the scale in practice

The working method is short enough to repeat at the start of every material chapter that follows.

- State the claim precisely – not "plastic is recycled," but "HDPE bottles from the orange bin in [region] are sorted at [facility] and sold to [recycler]."
- Find the primary source for that specific gate, and mark the level accordingly: an official scan, audited report, or facility record is ✓✓; independent secondary sources pointing the same way are ✓; a calculation from adjacent data is ~; nothing public is ?.
- Put the level in the text itself, not a buried footnote – "according to the MoEP 2025 scan (✓✓), around 6 percent..." or "the downstream fate of this fraction is unavailable in the public domain (?)."
- Where the level is ?, say what request could change that (Ch.25 covers how to frame one).

Conclusion

Recycling evidence is not a single number. It is a chain of gates, and a claim is confirmed only at the gate for which evidence actually exists – not at the gates a reader might reasonably assume follow from it. The MoEP 2025 scan gives this book, for the first time, a directly measured, ✓✓-level answer for one of the earliest and most consequential gates in that chain: how much household packaging reaches a designated channel at all. It does not, by itself, answer what happens afterward, and it should not be asked to.

The chapters that follow apply exactly this discipline, material by material, to find out how far each stream's documented chain actually runs – and where, honestly, it stops.

Chapter 9

Paper, Cardboard, and Deposit PET: Benchmark Streams

If this book has a benchmark for what a relatively well-documented recycling stream looks like, it is paper/cardboard and deposit PET. Not because either system is perfect, and not because the material simply "recycles well" on its own – but because here, unusually, the material has a simple identity, the collection route is clear, a market exists on the other end, and the downstream operators can be named and checked. Applying the evidentiary scale from the previous chapter to these two streams shows what a comparatively strong chain looks like, gate by gate – and, just as importantly, exactly where even a strong chain stops being documented.

1. What makes a stream clean

Four conditions, present together, explain why these two streams document more completely than most of the ones that follow in later chapters.

Condition	Paper/cardboard	Deposit PET
Simple material identity	Visually identifiable at every stage; grades mix without ruining the batch	A single polymer, marked and machine-readable at deposit points
Clear first-mile channel	Blue bin, office bins, commercial collection	Deposit machines and shop drop-off, backed by a cash incentive
Economic incentive / stable market	Steady demand from the cardboard-packaging industry	A refundable deposit at the point of return, plus a certified food-contact end market
Named downstream operator, visible geography	Infinya – Modi'in and Hadera plants, publicly identifiable	Green PET Recycling – Kidmat Galil plant, publicly identifiable

The streams examined in Chapters 10 through 13 will be judged against this same list. Where a stream is missing one or more of these conditions, that absence – not resident behaviour – is usually the first place to look for why the evidence gets thinner.

2. Paper and cardboard: a strong chain, but not one denominator

Israel's leading paper and cardboard recycler is Infinya (formerly Hadera Paper, and before that Amnir), a publicly traded company that converts more than 400,000 tonnes of paper and cardboard waste into new paper rolls for the packaging industry every year – a figure independently confirmed across the company's own reporting, trade press, and Wikipedia's sourced summary of its operations.^{[35][38][6]} It runs two production facilities, in the Modi'in-Maccabim-Re'ut industrial zone and in Hadera, employs roughly 400 people in its recycling division specifically (around 1,500 across the full Infinya group, which also includes paper production and packaging manufacturing), and operates more than 15,000 collection points nationwide.^[35] As of 2025, Infinya operates as a subsidiary of the Veridis Israel group – an ownership structure that has changed since an earlier period under the FIMI investment fund, a reminder that even well-documented companies are moving targets and corporate facts need a date attached to them.^{[33][6]}

That 400,000-tonne figure is one of the most solid single numbers in this book. It is also not, by itself, a measurement of household blue-bin performance, and treating it as one would repeat exactly the error Chapter 8 warns against. Infinya collects from roughly 270 municipalities across the country, but its own account of its operations draws a clear line between two different sources: paper and cardboard collected from local authorities,

and white office paper, cardboard, and plastic collected separately from industry – factories, offices, and businesses. The 400,000-tonne figure is the sum of both. It documents the scale of Israel's domestic paper-recycling industry with high confidence. It does not, on its own, measure what share of the paper and cardboard a household places in its blue bin actually gets there, because the public figures do not break the household portion out from the commercial and industrial ones.

Evidence card: paper and cardboard

Gate	Level	Comment
First-mile channel exists	✓✓	Blue bins, office bins, commercial collection are all documented, visible channels
Domestic operator identifiable	✓✓	Infinya, with two named production sites
Annual processing scale	✓✓	400,000+ tonnes/year, independently confirmed across multiple sources
Household vs. commercial/industrial breakdown	?	Infinya's public figures combine municipal and industrial collection; no published split isolates household blue-bin capture
Rejection / contamination rate	?	Not publicly available at facility level
Final product	✓✓	Paper rolls for cardboard packaging – a named, checkable output
Residues, sludge, and their destination	?	Requires facility-level reporting not currently public

3. Why paper outperforms most plastics

The contrast with the plastics discussed in Chapter 11 is not accidental. Three structural reasons explain why paper recycles more successfully than most plastic as a matter of course, in Israel and internationally: paper of different grades can be mixed and still processed by the same recycler, where HDPE, PP, and PET require separate handling; demand for secondary paper stock is steady, driven by the cardboard-packaging industry, where plastic recyclate competes against the price of virgin material and loses when oil is cheap; and paper is visually identifiable at every stage, where sorting plastic by polymer often requires equipment many facilities don't have. The EU's own paper and cardboard recycling rate has consistently exceeded 80 percent – among the highest of any material category – which is consistent with these being structural advantages, not something specific to how any one country manages the stream.

A physical constraint rarely mentioned

Paper fibre cannot be recycled indefinitely. Each cycle shortens the fibres, and after roughly five to seven cycles they become too short to make new paper. Even a perfectly functioning paper-recycling system therefore needs a constant inflow of virgin fibre from somewhere. This isn't a reason to deprioritise paper recycling – it's a reminder that "recycles well" doesn't mean "solves the problem permanently," and that substituting paper for plastic packaging is not, by itself, the same thing as reducing how much packaging exists in the first place.

4. Deposit PET: a clean stream because the channel is clean

The deposit system for beverage containers, examined in detail elsewhere in this book (Ch.15), belongs among the best-documented streams for a different reason than paper: not because a large operator publishes tonnage figures,

but because the collection mechanism itself removes most of the ordinary sources of contamination before the material ever reaches a recycler.

Green PET Recycling, founded in 2019 by a partnership between recycling operator Zohar Levy and beverage companies Blue Beverages and Yafora-Tabori, operates Israel's main deposit-PET recycling plant in the Kidmat Galil industrial zone, with processing capacity of roughly 3,000 tonnes of PET waste a month – equivalent to around half of Israel's total PET waste stream.^{[66][15]} About 75 percent of its output supplies Israeli industry, with the remainder exported. Its flagship product is food-contact-certified recycled PET (FDA, EFSA, FSSC 22000), produced using solid-state polymerization technology that makes genuine bottle-to-bottle recycling – a used bottle becoming a new beverage bottle – technically and commercially possible.

The deposit system shortens and cleans the middle chain rather than eliminating it. A bottle returned through a reverse-vending machine or shop still moves through collection, weighing, and acceptance before it reaches a recycler (Ch.5) – the financial incentive at the point of return simply means that chain starts with far less contamination and far less risk of the material ending up in a mixed load. It's also not exclusively a closed loop from consumer to recycler: Green PET's own account of its operations lists industrial waste, local authorities, transfer stations, waste sites, and recycling corporations as feedstock sources alongside deposit returns^[66] – a reminder that even a clean stream still passes through the ordinary mid-chain gates, just with a shorter and less contaminated path through them.

Evidence card: deposit PET

Gate	Level	Comment
Legal stream	✓✓	Deposit Law, 1999, as amended
First-mile incentive	✓✓	A refundable deposit at the point of return
Collection channel	✓✓	Shops, reverse-vending machines, deposit operators
Domestic recycler identifiable	✓✓	Green PET Recycling, Kidmat Galil
Processing capacity	✓✓	~3,000 t/month, independently confirmed across company and trade sources
Food-contact rPET / bottle-to-bottle capability	✓✓	FDA/EFSA/FSSC-certified; SSP technology confirmed by multiple sources
Share of output actually achieving bottle-to-bottle (vs. other end uses)	?	Not broken out in public reporting
Rejection / contamination at intake	?	Not publicly available at facility level

One caution specific to this stream, flagged for later chapters rather than resolved here: food-contact rPET and genuine bottle-to-bottle recycling describe deposit PET specifically. They do not automatically describe deposit glass, deposit metal cans, or the deposit system as a category – each of those has its own material chemistry and its own downstream fate, taken up separately when this book reaches glass and metal (Ch.13).

5. Even benchmark streams produce tails

Neither of these streams is contamination-free, and a complete evidence chain should say so rather than stopping at the headline tonnage. Paper and cardboard processing generates its own residual outputs: wet or food-contaminated paper, plastic sleeves, tape, staples, and – further downstream, at the pulping stage – sludge and short fibres too degraded to reuse. Deposit PET processing generates labels, caps, off-colour or non-bottle PET, and residues from the washing and flaking stages. None of these rejects are currently reported publicly at facility level for either stream, which is itself worth noting: even the best-documented material chains in this book stop being documented at exactly the point where the material becomes waste again.

6. Geography is part of why these streams are legible

Part of what makes paper/cardboard and deposit PET easier to evaluate than most other streams in this book is not only that the material is cleaner – it's that the geography of processing is unusually visible. Infinya's two plants and Green PET's single plant are named, located, and checkable in a way that, as later chapters show, is not true for where sorted material from more complicated streams actually ends up (Ch.7). Naming a facility doesn't guarantee complete evidence about what happens inside it, as the previous section just showed, but it is the necessary first step that several of the streams ahead lack entirely.

7. The upstream constraint, with the right denominator each time

It would be dishonest to end this chapter without returning to the limit examined in Chapter 8: however well the downstream chain works for these two streams, that only matters for the share of material that reaches the right channel to begin with. But applying a single number to both streams – as if one figure described the same thing for each – repeats the exact error Chapter 8 was written to prevent.

The Ministry of Environmental Protection's 2025 scan shows low capture of household packaging into designated separate-collection channels generally, but paper/cardboard, deposit PET, and orange-bin packaging are three different measurements, not one. For household packaging broadly, the relevant question is capture into a designated container. For paper and cardboard specifically, part of the stream is packaging and part isn't, and Infinya's own figures mix household with commercial and industrial sources, so the household capture rate specifically isn't isolated in what's publicly available. For deposit PET, the relevant measure is the deposit return rate through deposit channels – an entirely different mechanism from container capture, since the incentive operates at the point of return rather than the point of disposal. Treating these as one number would erase exactly the distinction this chapter has been built around.

What can be said with more confidence is why these two streams likely still outperform most others even given that constraint: a deposit is direct cash, and a large flattened box is inconvenient to store at home, both of which push a resident toward disposing of the material properly even without perfect infrastructure. That's a plausible partial explanation, not a documented one – and it doesn't reduce the need to keep improving first-mile access for both streams (Ch.4).

Conclusion

The cleanest streams in this book are not clean because residents behave better where paper and deposit bottles are concerned. They are cleaner because the system gives the material a simple identity, a clear route, a buyer, and – more than any other streams examined so far – a name and an address at the other end. Even then, the documented chain runs out at a specific, identifiable point: the split between household and commercial sources for paper, the share of PET that actually becomes a new bottle rather than something else, and the rejects and residues neither company publishes. The chapters that follow apply the same evidentiary discipline to streams where that chain runs out much sooner.

Chapter 10

The Orange Bin: Anatomy of the TAMIR Stream

Paper/cardboard and deposit PET were relatively clean benchmark streams because the material had a simple identity, the channel was clear, and the downstream operator could be named (Ch.9). The orange bin is a different kind of object entirely. It is not a material stream – it is a legal packaging-collection category, defined by the Packaging Law rather than by chemistry, and it contains several materials with genuinely different fates once they arrive together in the same truck.

Every time something is placed in the orange bin, a journey begins. Where exactly it leads depends on what went in, how clean it was, and what happens at the far end of the chain – the end the person who set it in motion has no access to. This chapter follows that chain as far as publicly available evidence allows, and says plainly where it stops.

The system this chapter describes faces a proposed overhaul under the 2026 Economic Arrangements Law, which would abolish the Packaging Law's current structure and end TAMIR's role as monopoly operator (Ch.16). That doesn't make this chapter's account obsolete – understanding how the current system works, including where it fails to document itself, is the baseline against which any new architecture has to be judged.

1. What belongs in the orange bin – and what "belongs" means

The orange bin accepts packaging waste, not all waste – a distinction that matters because it is specifically packaging that falls under the Packaging Law and is financed through TAMIR.

YES	Plastic packaging – bottles, flasks, containers, crates
YES	Metal packaging – tin and aluminium cans, lids
YES	Beverage cartons – Tetra Pak, Combibloc and equivalents
YES	Packaging film and bags (technically acceptable, though sorted less reliably)
NO	Glass – purple bin
NO	Paper and cardboard – blue bin (Ch.9)
NO	Food waste – contaminates the entire batch
NO	Deposit-marked bottles (חייב בפיקדון) – return to shop or RVM (Ch.9, Ch.15)
NO	Clothing, electronics, hazardous waste

A material marking – the number 1 to 7 inside a small triangle – identifies the polymer, but it is not a recycling instruction. It doesn't say whether the item is packaging, whether it belongs to a different regulated stream such as deposit or glass, whether it's small enough to survive sorting, or whether black or multilayer construction will make it invisible to the equipment that's supposed to catch it. The more useful two-part question is: is this packaging, and if so, does it belong to a different stream first (deposit, glass, paper/cardboard)? Only once both are answered does the orange bin apply.

That distinction points to the principle this chapter keeps returning to: orange-bin acceptable is not the same claim as orange-bin recyclable. Acceptance means the item has entered the packaging-collection system. It says nothing yet about whether it will actually be recovered – that depends on everything that happens after the truck arrives, which is where the rest of this chapter goes.

2. From bin to facility

According to TAMIR's own guidance, orange-bin packaging is collected separately by a dedicated truck and transferred to a sorting facility – a declared system design, not necessarily a confirmed fact for any specific street. Whether a specific neighbourhood's orange bin is actually collected on that dedicated route, rather than folded into another truck's run for logistical reasons, is the kind of local detail this chapter can describe in general but not confirm case by case; municipal or contractor route data would be needed to check it for any one address (Ch.5 §2).

The general design matters because it's the reason sorting instructions emphasise cleanliness: a dedicated truck keeps orange-bin material away from food waste and construction debris at the collection stage, so one contaminated load degrades only that load, not a mixed stream.

How many sorting facilities actually process the orange bin?

Sources genuinely disagree, and the disagreement is itself worth stating rather than smoothing over. TAMIR's own public FAQ says packaging goes to "2 sorting stations located in Rishon LeZion and Afula." Infospot, an industry tracker cited throughout this book, and a December 2024 Mako report both describe three facilities – Rishon LeZion, Afula, and a smaller site in Eilat. A fourth, substantially larger facility is now under construction at the Hiriya Recycling Park, within the Dan Region Association's territory near Rishon LeZion: first announced in May 2024 as a partnership between TAMIR and KMM, its cornerstone was laid in June 2026, with a confirmed NIS 70 million investment, sorting capacity for twelve separate material streams, service to roughly half of Israel's population, and up to three times the throughput of the facility it's replacing (Ch.5 §7, Ch.9). Given the industry-press consensus and the more detailed, dated confirmation of the new facility, this book follows the three-facility count for the current network, soon to be four, and treats TAMIR's own "2 stations" page as an unupdated outlier rather than a more authoritative source. The disagreement is a useful reminder on its own: even a basic infrastructure fact like a facility count can be source-dependent and needs a date attached to it.

3. What happens at the sorting facility

Per TAMIR's own description, the sorting process runs through eight stages: unloading, where the stream's actual composition becomes visible for the first time; manual pre-sorting to remove large contaminants and hazardous items; conveyor separation with air jets pulling out bags and film; NIR optical sorting, where infrared sensors identify polymer type and direct PET, HDPE, and PP into separate hoppers – a method that cannot detect black or dark plastic, which absorbs infrared instead of reflecting it; magnetic separation for steel and eddy-current separation for aluminium; manual quality control as a final check before baling; baling into standard 700–1,000 kg units; and transfer to recyclers, most of which TAMIR states operate within Israel. Four of the eight fractions are sorted technologically, without human contact; the rest require manual intervention, which is exactly why contamination slows the whole line down rather than affecting only the contaminated item.

4. Fraction map: what goes where

Mapping each fraction against what's actually known about its destination shows the pattern this chapter is built around: acceptance into the system is well documented, and downstream fate mostly is not.

Fraction	What's accepted	Sorting method	Where it goes – evidence level
Rigid plastic (HDPE, PP)	Cleaning-product bottles, shampoo bottles, non-deposit bottles	NIR optics, manual inspection	Local recyclers, partial export – ~ Reconstruction
PET (non-deposit)	Trays, non-standard bottles	NIR optics	Partial export; lower value than deposit PET (Ch.9) – ? Unknown

Fraction	What's accepted	Sorting method	Where it goes – evidence level
Flexible plastic	Bags, film	Air separators	Predominantly RDF or landfill (Ch.6) – ✓ Probable
Metal – aluminium	Aluminium cans and packaging (non-deposit)	Magnets, eddy currents	Recycled locally – ✓✓ Confirmed (TAMIR data)
Metal – steel	Tin cans, lids	Magnets	Recycled locally – ✓✓ Confirmed
Beverage cartons	Juice and milk cartons	Manual sorting	Specialist recycler, partial export – ~ Reconstruction
Dirty / mixed residual	"Tails" after sorting	—	RDF or landfill (Ch.6) – ✓ Probable

Two fractions – aluminium and steel – carry a confirmed local destination. Everything else in this table is either a reconstruction, a probable pattern, or genuinely unknown. That imbalance is not an accident of incomplete research; it is the systemic opacity Chapter 8 described, made concrete for one specific stream.

5. Tails: what never becomes secondary raw material

Even after a load clears sorting, part of it does not become usable feedstock. These are the tails – and for the orange bin, understanding them matters as much as understanding what gets sorted correctly, because the orange bin's real output is not one thing but two: accepted material fractions, and tails.

Food contamination degrades batch quality and bale value, even when a single dirty item doesn't ruin the whole load on its own. Black and dark plastic is invisible to NIR sensors and defaults to the "unidentified" stream, headed for RDF or landfill – a technical limitation of the sorting method, not a failure of whoever put the item in the bin. Multilayer and laminated packaging – sachets, stand-up pouches, metallised wrappers – gives NIR a mixed signal and isn't reliably sorted for the same underlying reason. Items smaller than roughly 3–5 centimetres fall through conveyor gaps or miss the air stream entirely. And some technically sorted material, polystyrene among the clearest examples, has little or no market demand even once correctly separated, and ends up as RDF regardless of how well it was sorted.

What share of the total stream ends up as tails is not published anywhere in the public domain – evidence level: unknown. The missing number here isn't only "how much was recycled," but "how much became tails, and where did they go." Both halves of that question matter equally, and only the first one usually gets asked.

What the tariffs reveal about stream quality

TAMIR's 2023 tariff data shows a striking gap between commercial and household processing fees – for PP plastic, roughly NIS 135 per tonne commercial against NIS 642 per tonne household, nearly a fivefold difference. That gap is a direct signal about stream quality: household material costs significantly more to process, because of higher contamination, smaller volumes per collection point, and more complex logistics. It also confirms a structural point examined at length elsewhere in this book (Ch.2 §4, Ch.14): TAMIR has a rational economic incentive to prioritise the commercial stream over the household one, within the system as currently designed.

6. What's known downstream – and what isn't

TAMIR states publicly that most recyclers are in Israel. That is a geography claim, and geography claims need geography evidence: named receiving facilities, tonnes by fraction, and identifiable final products (Ch.7). Without those, it remains a general operator statement rather than a documented chain.

TAMIR's own materials describe plausible destinations – plastic bags becoming film for agriculture and insulation, cleaning-product bottles becoming plant pots and technical parts, metal food cans being smelted into construction steel – descriptions broadly consistent with how these material chains are known to work elsewhere (Ch.11). But these are illustrative examples, not a mass balance. No specific recycler is named alongside confirmed capacity or output figures for the orange-bin stream as a whole.

One of the arguments made for the 2026 reform is that a national waste regulator would require annual public reporting on downstream fate, stream by stream (Ch.16). If that element survives into the final legislation, some of what this chapter has to mark "unknown" today would become a matter of public record – a real test of whether the reform changes substance or only structure.

7. Practical sorting notes – as quality gates, not moral failures

A handful of habits genuinely affect stream quality, and it's worth stating them as what they are – design constraints the system hasn't solved, not tests a resident can fail.

- Rinse, but don't scrub. Removing visible food residue is enough; thorough washing wastes water for no sorting benefit.
- Don't seal items inside closed bags – the sorting line can't see what's inside one. Loose in the bin, or the bag itself empty and open.
- A cap left on a bottle doesn't impede sorting, and keeps a small item from being lost on the conveyor.
- Items under roughly 3–5 cm will most likely fall through the process uncaught; collecting several in a slightly larger container before discarding helps.
- Black plastic and heavily multilayered packaging are functionally unsortable by the equipment currently in use. Officially, they're accepted as packaging; technically, NIR is unlikely to catch them, and they become tails. That's a design failure in the sorting technology, not a sorting failure by the person who used the bin.

Conclusion

The orange bin is a necessary first gate, and its early stages are the best-documented part of the chain this chapter has followed: what belongs in it, how it's collected, and how sorting technically works are all matters of public, checkable record. What happens after sorting is a different story. Rejection rates, the identities of specific downstream recyclers, actual output volumes, and the geography of tails are, for most fractions, reconstructed at best and unknown at worst.

The orange bin is not where packaging becomes recycled. It is where packaging becomes sortable – and it is the point where this book's evidence chain either continues, for a shrinking number of well-documented fractions, or breaks, for most of the rest. The next two chapters follow the plastic fractions that emerge from this system in more material-specific detail (Ch.11–12), to see how far each one's chain actually runs.

Chapter 11

Rigid Plastic, HDPE, PP and PET: Different Fates for Similar Bottles

A shampoo bottle and a water bottle look the same. Both plastic, both in the orange bin. But their subsequent fate is fundamentally different, and the gap between them begins not at the sorting facility, but at the filling point.

Chapter 9 showed what a relatively strong evidence chain looks like: paper/cardboard and deposit PET, where the material is simple, the channel is clear, and the downstream operator can be named. Chapter 10 showed why the orange bin itself resists that kind of simplicity: it is not a material stream but a packaging-collection category, and acceptance into it says nothing about eventual recovery. This chapter follows the rigid plastic fractions that emerge from that category once sorting has already happened – HDPE, PP, and non-deposit PET – and asks how far each one's evidence chain actually runs. Deposit PET appears here only as the comparison point already established in Chapter 9, not as a subject in its own right.

The moment plastic leaves the orange-bin sorting line, it stops being "plastic" and becomes HDPE, PP, PET, or tails – several material-specific evidence chains hidden under one everyday word.

1. Why "plastic" was never one stream

HDPE (high-density polyethylene) is hard and slightly matte – shampoo, conditioner, dish soap, and bleach bottles, and most 5-litre cleaning jerrycans. PP (polypropylene) is more flexible, usually marked "5" in a triangle – bottle caps, fruit crates, some food containers. PET (polyethylene terephthalate), marked "1," is the clearest plastic to consumers and the most familiar to recyclers, but exists in Israel as two parallel streams that are chemically identical and systemically unrelated: deposit and non-deposit.

All three are thermoplastics, meaning they melt and re-solidify without breaking down chemically – mechanically recyclable in principle. What determines whether that principle becomes practice is not chemistry but everything this chapter and the last one have described: sortability, market demand, contamination, and whether anyone is actually documenting what happens next.

2. Deposit PET as benchmark, non-deposit PET as contrast

The deposit system's advantage was already examined in detail (Ch.9): a refundable deposit at the point of return produces a cleaner, more uniform stream before the material ever reaches a recycler, and Green PET Recycling's Kidmat Galil plant turns a meaningful share of it into food-contact-certified rPET. Non-deposit PET – trays, blisters, non-standard bottles that enter the orange bin alongside HDPE and PP – is chemically the same polymer, but almost nothing else about it is the same.

Feature	Deposit PET	Non-deposit PET
First-mile channel	Deposit return (machine or shop)	Orange bin
Contamination	Low	Higher – mixed with other packaging
Uniformity	High – mostly beverage bottles	Mixed – trays, blisters, non-standard bottles
Downstream operator	Named: Green PET Recycling (Ch.9)	Not clearly documented
Food-contact rPET	Documented	Not automatic – needs separate verification
Evidence overall	Strong through processing; output-share gap remains	Weak from sorting onward

The practical consequence: extending the deposit system to larger bottles in 2021 didn't solve the orange bin's plastic problem so much as change its composition. The most valuable PET moved into the deposit circuit; the less valuable, less clean PET stayed behind in the TAMIR stream. The two should never be mentally merged just because a label says "PET" on both.

3. HDPE and PP: comparable acceptance, different evidence

HDPE is, on paper, the strongest orange-bin plastic case: rigid, large, usually light-coloured, and reliably identified by NIR sorting. PP is a step down in consistency – good for larger mono-material items like crates and tubs, weaker for small caps and thin-walled containers, and more exposed to price competition from virgin resin, since demand for PP recyclate is less steady than for HDPE's. Both pass through the same basic chain: NIR sorting, magnetic separation for any residual metal, manual quality control, baling into 700–1,000 kg units, then shredding, washing, and extrusion into granulate for sale.

Gate	HDPE	PP	Non-deposit PET
Accepted in orange bin	✓✓ – rigid packaging	✓✓ – if packaging	✓✓ – non-deposit packaging
Sortable by NIR	✓✓ – large, rigid, good detectability	✓ / ~ – large items better; small caps and tubs worse	✓ – if clean, detectable, not too small or dark
Sorted fraction exists	✓ – TAMIR process describes it	✓ – but public data thin	✓ – but chemically identical to deposit PET only in the polymer, not the system
Named downstream recycler	? / ~ – examples exist, no public mass balance	? – not public by fraction	? – not clearly documented
Market demand	~ – steadier than PP	~ – more variable, competes with virgin resin	~ – generally lower value than deposit PET
Rejection / tails rate	?	?	?
Geography of output	? / ~ – "mostly Israel" unconfirmed without named facilities	?	?

The pattern across both columns is consistent: acceptance and sortability are well documented; everything past the sorting line thins out fast.

4. Who recycles HDPE and PP in Israel: what's actually known

Named operators exist, and naming them is worth doing even where the picture stays incomplete. K.B. Recycling Industries, a family-owned company based in Beit Shean operating for over two decades, runs a truck fleet, two sorting facilities, and a plastics recycling plant producing granulate from LDPE, HDPE, and PP – though its capacity and downstream products aren't published in open sources. KMM, one of the longest-established players in Israeli recycling and also, as Chapter 5 and Chapter 10 both noted, a partner in the new Hiriya sorting complex, states its own plastics capacity at 100,000 tonnes a year, exporting whatever finds no local buyer, though again without published downstream detail. Keter Group, the world's largest manufacturer of home and garden plastic products, is a genuine secondary-market anchor: it already uses recycled content for roughly 40 percent of its raw material, with a stated target of 55 percent by 2025, and ran a 2021 pilot in Kiryat Ono collecting used plastic furniture and household items directly from residents' sidewalks for compaction, sorting, and reprocessing^[17] – since extended

into a recurring partnership with the Scout movement across several cities. Keter's post-consumer collection is a separate stream from the orange bin, but it is an important, real buyer on the demand side of the market.

One name that surfaces repeatedly in industry references – "SuperPipe Israel," reportedly a manufacturer of pipes from secondary HDPE – could not be independently confirmed during the research for this chapter. It is left here, marked unverified rather than dropped, because a name industry sources repeat but this book cannot confirm is itself useful information for anyone able to check it directly.

Beyond named companies, Israel's OECD review notes the country imports roughly 80 percent of its virgin plastic resin^[20] and exports most of what it collects for recycling – meaning a meaningful share of sorted HDPE and PP likely leaves the country for Turkey, EU states, or Asia, though systematic data on export destinations and downstream fate isn't published. This export orientation is structural, not incidental: Israel has historically built trading relationships for collected plastic rather than domestic reprocessing capacity, which is also part of why downstream disclosure requirements never developed the way collection requirements did. Nothing in Israeli law currently obliges anyone to publish where a specific sorted fraction ends up or what it becomes.

5. The upstream problem is larger than the downstream one

Everything in this chapter so far concerns what happens to plastic after it reaches the orange bin. The Ministry of Environmental Protection's 2025 scan reframes how much that question actually matters: only around 6 percent of household packaging reaches a designated container at all (Ch.1 §7, Ch.8 §2). For the clear majority of shampoo bottles, cleaning-product flasks, and food containers Israeli households discard, the relevant question isn't what happens after sorting – it's why the item never reached a first-mile channel in the first place (Ch.4).

That framing matters for where effort is best spent. Even a perfect downstream chain – no rejection, no export losses, full recycling of everything sorted – would only resolve the fate of the roughly 6 percent that gets that far. The MoEP scan is also, per the evidentiary scale, the most solidly confirmed fact in this entire chapter: a direct official measurement, where nearly everything downstream sits at "probable," "reconstructed," or "unknown." That doesn't make downstream questions unimportant – a specific recycler's rejection rate is still worth requesting (Ch.25) – but it does mean upstream access improvements are likely to move more total tonnage than downstream sorting improvements alone.

How much plastic, and how much of it is recycled – read carefully

Two external figures are worth stating precisely rather than casually. UNIDO's SwitchMed programme puts Israel's total plastic waste generation, across all sectors – packaging, agriculture, industry, not households alone – at roughly 1 million tonnes a year, of which about 6 to 9 percent is recycled depending on the reporting year; less than 100,000 tonnes of recycled resin is produced domestically against an estimated local industry demand of over 120,000 tonnes.^{[77][78]} Separately, an industry analysis (CDS-Luthi) states that plastic makes up about 18 percent of household waste by weight, and that Israel recycles roughly 7 percent of its plastic waste against an OECD average of 39 percent.^[17] These two figures are broadly consistent with each other and with the MoEP scan above, but they measure different things – one an economy-wide total, the other a household-waste-composition share – and neither should be read as a household-specific tonnage figure. No source found in this research specifies household plastic waste generation as a standalone number with a clear methodology; general economy-wide and composition-share figures are the best publicly available evidence.

6. What turns rigid plastic into tails

Even correctly sorted HDPE and PP can still fail to reach a recycler. Food contamination lowers a batch's grade even when it doesn't disqualify it outright. Black and dark plastic is invisible to NIR and defaults to the mixed stream or RDF – a sorting-technology limitation, not a resident's mistake. Small items under roughly 3–5 cm fall through gaps or miss the air separators. Multilayer construction – plastic bonded to metal or another polymer – gives NIR a mixed signal it can't resolve. And even clean, correctly sorted material can simply have no buyer that week, particularly when virgin resin prices fall and recyclate loses its price advantage. TAMIR's declared-base figures put overall

plastic-packaging recycling at roughly 30 percent, but which specific polymer that covers, how much genuinely closed the loop versus went to export or RDF, isn't described anywhere in the public domain.

7. Design decides fate before the bin

A UNIDO pilot under the SwitchMed programme illustrates this cleanly. A standard Israeli yoghurt or salad cup is typically a PP body with a PET sleeve – a combination that gives NIR sorting a mixed signal it reads as unsortable. Switching the sleeve to a polyolefin material makes the same cup fully detectable, at marginal cost to the producer; UNIDO estimated that broad adoption of this single change across Israeli brands could make roughly 2,100 additional tonnes of plastic recyclable each year.^[20] The same principle applies more broadly: mono-material construction, light or transparent colouring, removable rather than heat-fused labels, and a minimum size are all design choices made long before a resident ever sees the product, and each one either keeps the material legible to the sorting system or guarantees it won't be. The EU's PPWR regulation has begun making "recyclable by design" a market-entry requirement; Israel has no equivalent yet. This connects directly to packaging design as a producer-responsibility question, taken up in full later in this book (Ch.17).

Conclusion

Plastic does not fail as one material – it fragments into several evidence chains, some longer than others and most publicly incomplete. HDPE and some PP packaging are among the more recoverable fractions leaving the orange bin, but their actual recycling still depends on size, colour, contamination, market demand, and a documented buyer – not on the polymer type alone. Non-deposit PET carries the clearest risk of mental shortcut: chemically identical to the deposit stream examined in Chapter 9, but institutionally a different world once it loses the deposit channel. The question worth asking of TAMIR or a municipality is rarely "is this plastic recyclable" – it's which specific polymer from a specific facility reaches a named recycler, in what volume, and as what output. For most of what this chapter has covered, that answer still doesn't exist in public form. The next chapter turns to the fractions the system handles worse still: flexible, multilayer, and small packaging, most of which has, by any available evidence, already become RDF or landfill tails before the question is even asked.

Chapter 12

Flexible, Multilayer and Small Packaging: Designed Tails

A crisp packet is not one recyclable plastic. It's typically five or six layers of different materials – plastic, aluminium foil, another layer of plastic – bonded together precisely so they can never be pulled apart again. The same design that keeps a product fresh guarantees that the packaging protecting it cannot survive mechanical recycling.

Chapter 9 showed what a benchmark stream looks like: simple material identity, a clear channel, a named downstream operator. Chapter 10 showed that the orange bin doesn't automatically deliver any of that – it's a packaging-collection category, not a material stream. Chapter 11 followed the rigid plastic fractions that at least have a chance of clearing every gate in that category. This chapter follows the packaging for which one or more of those gates fails not because of anything a resident did, but because of how the package was built.

Flexible and multilayer packaging does not fail because people sort it badly. It fails because the package was designed to protect a product in ways that destroy its identity as a recyclable material. For this category, the first recycling gate isn't the orange bin. It's the design desk – before the product was ever filled.

1. Not one stream, but several failure modes

"Problem packaging" isn't a single material. It covers flexible film (LDPE) – bags, wrapping film, frozen-food bags; multilayer packaging – crisp packets, coffee bags, spice and cosmetics sachets, bonded to resist moisture, light, and oxygen; stand-up pouches for dried fruit, nuts, and pet food; and small packaging – caps, small bottles, single-serving formats – too small for standard sorting equipment regardless of what they're made of. These formats don't share a chemistry. They share a set of different reasons for ending up in the same place.

2. Why standard sorting fails this category

UNIDO's work on plastic circularity in Israel identifies flexible packaging specifically as the hardest fraction for standard sorting lines to process, for three compounding reasons: low density means lightweight material is poorly separated by weight and shape on equipment built for rigid items; a tendency to wrap around moving conveyor and sorting-line parts causes equipment downtime and requires specialised air separators; and multilayer construction – several different polymers bonded into one object – gives NIR sensors a signal they can't resolve into a single material.^[78]

Per TAMIR's own data, only clean PE-type plastic bags are separately extracted during orange-bin sorting. Multilayer packaging and PP-based film don't form a distinct marketable stream and default, by the structure of the system, into residual.

3. Where it actually goes

Format	Why it fails	Evidence for Israel	Likely destination
Clean PE film	Low density, wraps around equipment – but a single, identifiable polymer	✓ – TAMIR data confirm PE-type bags are separately extracted	Limited mechanical recycling into technical granulate (bin bags, technical film) – evidence on volumes: ~ / ?
Multilayer (sachets, crisp packets, pouches)	Several bonded polymers give NIR a mixed signal it can't resolve	? – no public evidence of a marketable TAMIR fraction for multilayer material	RDF or landfill – ✓ Probable, since the sorting structure makes any other outcome

Format	Why it fails	Evidence for Israel	Likely destination
			unlikely, though direct volumes aren't published
PP-based and mixed films	Not the PE-type TAMIR data confirm are extracted separately	? – not documented as a distinct marketable stream	Defaults to residual – ✓ Probable
Small loose packaging (<3–5 cm)	Falls through sorting-line gaps or misses air separation, regardless of polymer	✓✓ – a physical sorting-equipment limit, well established (Ch.10 §7)	Tails, then RDF or landfill – ✓ Probable
Contaminated flexible packaging	Food or product residue degrades the batch even where the polymer itself is recoverable	✓ – consistent with the contamination pattern already established for rigid plastic (Ch.11 §6)	Residual – ✓ Probable

The pattern across this table is not "everything fails equally." Clean PE film has a real, if narrow and poorly quantified, chance at mechanical recycling into technical granulate for non-food applications. Everything structurally multilayer effectively doesn't, regardless of how carefully it was sorted – the failure is upstream of the sorting line, not downstream of it. Small items fail for a reason that has nothing to do with material at all: they're too small for the machinery, full stop, unless attached to something larger. A cap left on a bottle travels with the bottle. A loose cap in the bin is, in practice, gone before sorting even starts.

4. What a resident can and cannot fix

This category is where the moral framing around household sorting breaks down most clearly, and it's worth stating the boundary directly rather than leaving it implied.

A resident can help with	A resident cannot solve
Removing heavy food residue before disposal	The bonded, multilayer structure of a laminate
Keeping caps attached to bottles where possible	The physical limits of NIR sorting equipment
Not sealing mixed contents inside closed bags	The absence of a marketable buyer for a given fraction
Choosing bulk or refill formats where available	A producer's original packaging-design decision

A resident can protect a stream from unnecessary contamination. A resident cannot redesign a laminate.

5. The solution is upstream: reduction, reuse, and recyclability-by-design

Unlike rigid plastic, where better sorting and stronger downstream markets can genuinely move the needle (Ch.11), the technological fix available on the recycling side of flexible and multilayer packaging is narrow bordering on nonexistent. That doesn't leave only one lever, though it's tempting to say so – it leaves three, in the order this book's waste hierarchy already established (Ch.3): reduce or avoid the format entirely where a simpler one would do; shift toward reuse, refill, or bulk options, especially viable for dry foods, cleaning products, and cosmetics; and, for whatever packaging remains necessary, redesign it to survive sorting – mono-material construction instead of bonded layers, detectable colours, removable rather than heat-fused labels.

A UNIDO-documented mono-material redesign – replacing a multilayer structure with a single polymer, PE instead of PE-plus-aluminium-plus-PP – makes packaging technically sortable by standard NIR equipment^[20] (Ch.11 §7

showed a concrete Israeli example of this same principle). The EU's PPWR regulation has set recyclability-by-design requirements targeting exactly this packaging category for 2030, an acknowledgment that downstream investment alone won't solve it. The Ellen MacArthur Foundation identifies flexible and multilayer packaging as a globally "non-recyclable by design" category – a structural problem general to the format, not specific to how Israel happens to sort its waste. Israel has no equivalent design requirement yet, which is precisely why this chapter's category stays systemically unsolvable at the sorting-facility level until upstream regulatory pressure arrives – the subject Chapter 17 takes up as a producer-responsibility question in its own right.

Conclusion

Flexible, multilayer, and small packaging systematically fails on density, identifiability, and market demand – but not all in the same way or to the same degree. Clean PE film has a narrow, real path to recycling. Multilayer structures effectively don't, no matter how a resident sorts them. Small items fail on physical size alone. TAMIR's sorting system extracts only clean PE-type film as a distinct stream; everything else in this category defaults to RDF or landfill by the structure of the system, not by anyone's individual failure.

Chapter 13 turns to a different kind of contrast: glass, metal, and beverage cartons, where the material itself is often highly recyclable in principle, and the gap this book keeps finding between acceptance and actual recovery comes instead from collection geography, economics, and institutional routing.

Chapter 13

Glass, Metal, and Beverage Cartons: Three Lessons in Material Fate

Glass can be recycled indefinitely without any loss of quality. That is exactly why Israel's recycling rate for non-deposit glass – somewhere between 4 and 8 percent – is one of the most telling numbers in this book: it shows, more starkly than any other material examined so far, that technical recyclability and actual recycling are different things.

This chapter closes the material-by-material review that began with the benchmark streams in Chapter 9. Glass, metal, and beverage cartons are grouped together not because they resemble each other, but because each tests a different condition for recycling success – and together they make the case this whole section of the book has been building, one material at a time: a material's fate is never explained by its chemistry alone.

1. Glass: the best technical case, the worst practical result

Glass never goes in the orange bin, and for a specific physical reason: it breaks during transport and sorting, the shards injure sorting staff, and it damages equipment calibrated for plastic and metal. It requires its own dedicated collection points instead – which means glass runs into the same first-mile problem examined earlier in this book (Ch.4) in its starkest form. Glass doesn't fail at the recycling plant. It fails before it ever reaches one, because a purple bin or dedicated point is reliably less accessible than an orange bin on the same street.

Deposit glass and non-deposit glass tell almost opposite stories. Deposit-marked bottles – roughly 120,000 tonnes a year – see a comparatively high recycling rate, for the same reason deposit PET does (Ch.9): a financial incentive at the point of return produces a clean, motivated stream regardless of what the material is made of. Non-deposit glass – jars, sauce and jam containers, headed for the purple bin or a specialist point – tells a different story: only around 30,000 tonnes a year is declared to TAMIR, well below the actual volume of glass packaging circulating in the country, and the estimated recycling rate sits at 4 to 8 percent – the lowest of any material this book has examined, for a material that is, chemically, the easiest of all of them to recycle.

Why technical recyclability doesn't help here

Glass can be recycled without limit, unlike paper (which loses fibre length) or most plastics (which degrade with each heat cycle) – provided it's correctly sorted by colour, a genuine technical advantage confirmed by European industry data. The barrier isn't technical at all. Glass is heavy, which makes transport expensive per tonne; fragile, which demands careful handling; and loses value the moment colours mix. Without a deposit-style incentive and without collection infrastructure as convenient as the orange or blue bin, even a technically ideal material underperforms materials that are, chemically, far harder to recycle. Infrastructure accessibility predicts outcomes more reliably than a material's technical properties – the same conclusion Chapter 4 reached from the opposite direction.

2. Metal: the most economically aligned case

Metal is one of the few streams in this book where ordinary market economics pushes in the same direction as recycling policy, largely without needing EPR subsidies or deposit incentives to make it happen. Aluminium and steel hold real secondary market value regardless of how well or badly the formal EPR system functions, which keeps metal collection profitable even through informal channels. Separation is mechanically simple – magnets pull ferrous metal, eddy currents pull aluminium – without the complex optical sorting rigid plastic requires. Like glass, metal doesn't degrade with recycling. And Israel has a genuine informal scrap market running alongside the formal one; TAMIR publishes separate tariffs for metal from commercial and household streams, and authorises specialised operators, including firms such as Recycom that work with metal exclusively.

That combination makes metal the most economically aligned case in this book, not a simple one. Good mechanical separation and real market value are not the same as a documented mass balance: how much household packaging metal actually moves through TAMIR versus informal scrap channels, what exact tonnages each accounts for, and

who the downstream buyers are for the household-specific stream are questions this chapter cannot answer with public data, even where the broad outcome – metal gets recycled – is more reliable than for almost anything else examined so far.

3. Beverage cartons: accepted, sorted, and downstream-opaque

Beverage cartons – Tetra Pak-type packaging for milk and juice – are one of TAMIR's sorted orange-bin fractions, which creates a misleading impression of simplicity. Technically, a beverage carton is a composite: layers of paperboard, polyethylene, and aluminium foil bonded together to protect the contents (Ch.1 §4 already introduced this same material in a different context). Recycling it requires hydropulping, a specialised process that separates the paperboard fibre – which can then join the ordinary paper stream – from the polyethylene-aluminium layer, which needs its own separate processing or energy recovery.

Beverage cartons are, in effect, the polite version of the orange-bin problem this book identified in Chapter 10: the instruction is clear, and the downstream evidence is not. Despite cartons being listed among TAMIR's own sorted fractions, no systematic public data on which specific Israeli operator processes this material, or by what method, turned up anywhere in the research for this chapter. European data from the Alliance for Beverage Cartons and the Environment confirms that hydropulping technology exists and operates at scale elsewhere – but that confirms the technology is real, not that equivalent infrastructure exists in Israel. Evidence level: unknown. This is a clean, specific example of the kind of question worth putting directly to TAMIR (Ch.25): who recycles beverage cartons collected from the orange bin, and what happens to the polyethylene-aluminium layer once the paperboard is separated out?

4. Three materials, three lessons

Gate	Glass	Metal	Beverage cartons
Technical recyclability	✓✓ – infinite, if colour-sorted	✓✓ – infinite	✓✓ – technology exists internationally
Accepted collection channel	✓✓ – deposit / dedicated points, not orange bin	✓✓ – orange bin, plus informal scrap market	✓✓ – one of TAMIR's sorted orange-bin fractions
First-mile accessibility	~ – dedicated points far sparser than orange bins (Ch.4)	✓✓ – same channel as other packaging	✓✓ – same channel as other packaging
Volume: declared/documentd	✓ – deposit ~120,000 t/yr; non-deposit declared ~30,000 t/yr (Infospot)	✓ – TAMIR publishes commercial/household tariffs	✓ – listed as a sorted fraction
Named downstream operator	~ / ? – colour-sorting and buyer not confirmed for non-deposit stream	✓ – authorised metal contractors (e.g. Recycom)	? – no Israel-specific processor found in open sources
Actual recycling rate	✓ – non-deposit estimated 4–8% (Infospot)	~ / ? – no household-specific mass balance	? – unknown

Glass proves that recyclability without access still fails. Metal proves that market value can accomplish, almost incidentally, what deliberate policy elsewhere in this book struggles to achieve. Beverage cartons prove that a clear sorting instruction and a genuinely documented downstream chain are two different accomplishments, and a system can deliver the first without ever managing the second.

Conclusion

This chapter closes the material-by-material review that ran from Chapter 9 through Chapter 13. Across paper, deposit PET, the orange-bin system, rigid and flexible plastics, and now glass, metal, and cartons, one pattern has held without exception: recycling success is never explained by a material's chemistry alone. It depends on the combination of collection-channel access (Ch.4), economic incentive, sorting technology, and – running through every chapter in this section – whether anyone actually documents what happens after the bin.

With that material-by-material picture complete, the book turns next to the people who make it happen: the labour behind the trucks, the sorting lines, and the informal collection networks this section has repeatedly mentioned but not yet examined directly.

Chapter 14

The Sorting Line: Labour Behind Recovery

Some societies make discarded material unusually visible. Japan requires clear or semi-transparent bags for household waste in many municipalities, precisely so that what a household throws away is not entirely private. Police investigators, in an unrelated but revealing practice, search a suspect's garbage as a matter of routine, because what a person discards, believing it gone for good, often tells the truth their words don't. Waste is one of the few materials people hand over specifically because they no longer want to look at it, touch it, or answer for it. Someone else, further down the chain this book has followed since Chapter 4, still has to.

Chapters 9 through 13 followed what happens to paper, plastic, glass, metal, and beverage cartons after they leave a household. At almost every gate in that chain – sorting, quality control, rejection – this book has already mentioned, without dwelling on it, that a person is doing the work. This chapter dwells on it.

1. Why labour belongs in a waste handbook

Chapter 5 described the invisible middle as the place where household behaviour becomes contractor data, or stops being data at all. It is also the place where household behaviour, imperfect packaging design, and a weak first mile all become someone else's task. Chapter 10 already described that task in technical terms: manual pre-sorting to remove large contaminants, manual quality control before baling, hand intervention wherever automated equipment can't resolve what it's looking at. This chapter asks the question that description left open – not what happens on the sorting line, but who is standing there when it does.

2. Where human labour enters the chain

Manual work isn't scattered randomly through the sorting process – it clusters at specific, identifiable points: unloading and the first visual read of what actually arrived; removing large or obviously hazardous contaminants before the line can run; hand-sorting fractions that optical and magnetic separation can't resolve on their own; a final quality check before baling, catching what slipped through everything upstream; and – a category this book keeps returning to – dealing with tails, the rejected material that machinery has already given up on. Chapter 10 named these gates individually. Set side by side, they describe a specific kind of work: judgment applied to material that has already defeated a machine's judgment once.

3. Automation still needs hands

It's tempting to assume this is a temporary problem – that better robots will eventually finish the job. The evidence suggests something more structural. Waste is not a standard industrial input. A factory robot excels with standardised parts: consistent shape, known orientation, clean surfaces, predictable arrival on a conveyor. Municipal waste is close to the opposite of that on every count – crushed, wet, folded, torn, mixed, partly hidden under other objects, and rarely presented twice in quite the same way.

International research on robotic sorting is candid about where the limits sit. Even in facilities built specifically to automate sorting, a well-documented 2024 case at a major recycling centre in Washington State – a \$40 million overhaul incorporating volumetric scanning and optical sorting – found that plastic bags and film routinely defeat the equipment, wrapping around belts and drums and forcing shutdowns; pulling them free is itself a hazardous manual task.^[16] The same reporting notes that material-recovery-facility workers face materially higher rates of nonfatal injury and illness than the general workforce, from accidental needle sticks and glass cuts to the machinery itself.^[16] A recent academic review of automated sorting technology puts a number on the underlying problem: even well-automated facilities still discard up to a fifth of genuinely recyclable material, because the "cluttered, overlapping, and deformable nature" of a real waste stream keeps defeating detection systems trained on cleaner^[67]

data. Industry sources describing the newest robotic picking arms are explicit that these systems are best understood as a complement to human sorters, not a replacement for them – engineered to take over repetitive, high-volume picks so that people are left for the harder judgment calls, not removed from the line altogether.^[23]

That pattern has a shape worth stating plainly: automation removes people from the easiest parts of the job first. The exceptions – contamination, ambiguous objects, hazards, anything that doesn't match a training set – stay human. Robots work best once a waste stream has already been made machine-readable. Chapters 10 through 12 spent considerable space on exactly what makes a stream machine-readable and what doesn't: mono-material construction, light colour, minimum size, a clean surface. Packaging that lacks those properties doesn't just sort badly. It creates a task for a person, at the exact point where a machine's confidence runs out.

4. Who works the sorting line in Israel: what's known and what isn't

For Israel specifically, public information about who performs this work is markedly thinner than information about facility capacity, tonnage, or sorting technology – a gap this book has learned, over the preceding chapters, not to read as an accident.

Claim	Evidence level	
Sorting facilities rely on manual labour alongside machinery	✓✓ / ✓	TAMIR's own process description (Ch.10)
Which specific tasks are manual vs. automated	✓ / ~	Inferred from TAMIR's eight-stage process
Number of workers per shift, per facility	?	Not published
Wage levels for sorting-line work	?	Not published
Direct employment vs. subcontracted labour	?	Not published
Injury, illness, or exposure data	?	Not published
Turnover rates	?	Not published
Worker representation or union coverage	?	Not published
Employment of people with disabilities / disabled veterans in the sector	✓✓	Named, self-reporting social enterprises (§5)

One genuinely distinctive Israeli pattern did surface clearly in this research, and it deserves to be stated with the complexity it actually has, not flattened into either a success story or an accusation. Several named social enterprises operate specifically in waste and e-waste collection, sorting, and dismantling, and specifically employ people with disabilities recognised by Israel's National Insurance Institute, alongside disabled IDF veterans. Yerok Hevrat ("Green Social"), part of the Shekulo Tov group, provides recyclable-waste collection services and employs, by its own description, only people with recognised disabilities.^[89] Ecologia LeKehila Muganah ("Ecology for a Protected Community") dismantles and sorts e-waste using workers with special needs, refurbishing computers for sale at reduced prices and for donation to families who can't otherwise afford one – a programme publicly praised by Israel's Minister of Welfare, and one that in 2024 provided a refurbished computer to a wounded IDF reservist during his rehabilitation.^[25] Michzur Yarik Israel (G.R.I.) directly and indirectly employs people with disabilities and disabled veterans, and partners with a special-education school to prepare students for future employment through e-waste dismantling workshops.^[59]

These organisations present this work, consistently and publicly, as social inclusion – meaningful, dignified employment for people who face real barriers to entering the labour market elsewhere, explicitly framed by government officials as a national asset rather than a burden. That framing is genuine, and the employment it

describes is real. It sits alongside a second, equally genuine observation this book is not in a position to resolve: this is also some of the least visible, least prestigious physical work in the entire economy – handling what has already been rejected once, by people society already tends not to see in most other workplaces. Both things can be true about the same arrangement at once. Dignified inclusion and a pattern of directing undesirable, low-status labour toward populations with the least market power to decline it are not mutually exclusive descriptions – they can be, and here plausibly are, the same fact seen from two directions. This book states both directions rather than choosing between them, because the public record supports the observation and does not support a verdict.

What the public record does not currently support at all is any picture of labour conditions at TAMIR's own main sorting facilities – Rishon LeZion, Afula, Eilat, and the new Hiriya complex (Ch.10). Whether that workforce includes migrant or foreign workers, as it does in several other physically demanding, low-visibility Israeli sectors such as caregiving and construction, is a reasonable question this research could not confirm or rule out. It is left here explicitly as open, not implied one way or the other.

5. Working conditions as part of recycling performance

Without Israel-specific data, this book cannot describe local conditions and won't invent them. What can be stated is the general, internationally documented profile of this kind of work: exposure to odour, dust, and noise; a line speed set by throughput targets rather than by the pace careful sorting requires; direct contact with contaminated packaging; the risk of sharp or hazardous items arriving mixed into an ordinary load; and a genuine dependence on protective equipment and training being both provided and used. These are known occupational concerns in sorting and recycling work internationally, documented well enough in the research already cited in this chapter. The Israeli question this book can pose but not yet answer is narrower and more specific: whether, and where, these risks are measured, published, and built into the contracts that govern this work – the same test Chapter 5 already applied to tonnage and rejection rates, now applied to the people handling both.

6. The chain of responsibility this labour sits inside

Chapter 12 showed that a producer's packaging-design choices determine, before a resident ever sees the product, whether that packaging can survive mechanical sorting. This chapter adds the link that argument left implicit: when a design choice makes packaging unsortable, that failure doesn't vanish into the technology. It becomes a task for whoever is standing at the point where the machine's confidence runs out.

A producer designs packaging that a sorting line can't reliably read. A resident is told, correctly, that it belongs in the orange bin. The sorting facility's equipment does what it can and hands the rest to a person. That worker absorbs the contamination, the ambiguity, and the exceptions the design created – and whatever they can't recover becomes tails, then RDF or landfill, then somebody's local environmental burden (Ch.6–7). The sorting worker stands at almost the exact midpoint of that chain: downstream of a design decision they had no part in, upstream of an environmental and geographic burden they don't bear directly. Bad design doesn't disappear into technology. It becomes someone's sorting task, and this book has now traced where that someone actually stands.

7. Civic questions about labour

The same discipline this book has applied to material claims – name the gate, ask for the evidence, say plainly what's unknown – applies here.

Question	Why it matters
How many workers are on the sorting line per shift?	Indicates how much of the process actually depends on manual labour
Which specific tasks are manual, and which are automated?	Shows exactly where human judgment substitutes for machine failure

Question	Why it matters
Are workers directly employed by the facility operator, or subcontracted?	Determines who is accountable for conditions and safety
What protective equipment is provided, and is training documented?	Basic safety evidence
Are exposure or injury statistics collected and published?	Without measurement, conditions cannot be evaluated or improved
What is staff turnover on the sorting line?	A possible indirect indicator of working conditions
Are labour standards written into municipal or TAMIR collection and sorting tenders?	Tests whether procurement rewards conditions, not only tonnage
Is worker safety reported alongside tonnage and recycling-rate figures?	Tests whether labour is treated as part of system performance

Conclusion

A recycling system is usually judged by tonnes recovered and percentages reported. This chapter has argued it should also be judged by what it asks of the people who make that recovery physically possible – and that, for Israel specifically, this book can currently describe where that labour sits in the chain far more confidently than it can describe who performs it, under what terms, or with what protection. Some of what is known is genuinely admirable: real organisations providing real, dignified work to people the broader labour market too often excludes. Some of what is unknown is a gap this book is unwilling to fill with a comfortable guess in either direction. Recycling is not only a material chain. It is a labour chain, and for large parts of it in Israel, that chain currently has no public account attached to it at all.

With the material streams and the labour behind them both examined, the book turns next to the policy architecture that was supposed to govern all of it at once: how the EPR mechanism through TAMIR was designed, and why, on the evidence assembled so far, it has struggled to deliver what it promised.

Chapter 15

EPR in Practice: How TAMIR Was Designed and Why It Failed

A system designed so that its success is measured by what it reports about itself is not a recycling system. It is a reporting-about-recycling system. That distinction is the subject of this chapter.

Part III of this book followed packaging materials through the gates where they succeed or fail – capture, sorting, downstream recycling, geography, tails (Ch.9–13) – and the chapter just before this one showed that many of those failures don't simply vanish into technology; they become someone's manual sorting task (Ch.14). This chapter asks a different question about the same evidence: why did the institutional architecture built specifically to prevent these failures not prevent them?

A functioning EPR system carries five distinct obligations: cover the full packaging stream, not a self-selected fraction of it; fund the real cost of municipal collection; give producers a genuine incentive to design recoverable packaging; make the downstream chain transparent enough to verify; and avoid quietly shifting cost onto residents, municipalities, sorting workers, and the communities that host the end-of-pipe burden. This chapter examines Israel's EPR system, run through TAMIR since 2011, against each of those five obligations – not to relitigate whether EPR is the right principle, which this book does not dispute, but to show precisely where its Israeli implementation stopped measuring what it was supposed to measure.

This chapter was written while a 2026 Economic Arrangements Law reform was under active discussion, proposing to end the Packaging Law's current structure and TAMIR's role as sole recognised operator (Ch.16 covers that reform in detail). Reform proposals of this kind evolve during negotiation, and this chapter does not treat any specific version of it as settled. What it asks instead is why a system that operated for fifteen years produced results that made its overhaul politically possible at all.

1. How the system was designed

The 2011 Packaging Law required every producer and importer of packaged goods in Israel to contract with a "recognised body" (guf mukar) licensed by the Ministry of Environmental Protection to fulfil recycling obligations on their behalf. TAMIR – founded by the Manufacturers Association of Israel as a non-profit company – became that recognised body, and remained the only one for over a decade (Ch.2 §4).

The mechanism runs in a closed loop, on paper: a producer contracts with TAMIR, declares the weight of packaging it places on the market each quarter, and pays a per-kilogram treatment fee by material type; TAMIR uses those fees to pay municipalities for organising source separation and to pay for sorting, then reports its recycling performance back to the Ministry. The loop is real. It is simply closed only around whatever packaging entered it in the first place – and that qualification is where every failure in this chapter originates.

2. What EPR was supposed to make visible

The point of an EPR system is not only to finance collection. Done properly, it makes the entire packaging chain accountable in a way the market alone doesn't: who placed packaging on the market, who paid for its treatment, where it was actually collected, what fraction was genuinely recovered, what became tails, and whether a producer's design choices raised or lowered that cost. Each of the five failures below is, in essence, a place where that accountability chain went dark.

3. Failure one: the declared base

TAMIR reports its performance against its "declared base" – the volume of packaging declared by companies actually under contract with it. The Ministry's own estimate puts that base at roughly half of all packaging entering

the Israeli market annually. A widely cited "84 percent recycled" figure for 2021 is therefore 84 percent of about half the real stream – at best around 42 percent of the full picture, before any downstream losses at sorting are even factored in (Ch.9 §2 already flagged this same calculation).

Why a declared base creates structural optimism

The denominator here isn't "all packaging sold in Israel" – it's "packaging declared by whoever already joined the system." That framing lets reported recycling figures rise simply because new companies sign contracts, even if the actual tonnage recovered doesn't change at all. Eurostat has documented a related pattern across several European EPR systems, where collection gets counted as recycling and system coverage stays incomplete; the State Comptroller flagged this exact methodological trap in a 2021 report, without a systematic recalculation following it.

4. Failure two: free riders

Per Ministry data, more than 40 percent of Israel's packaging producers and importers still have no TAMIR contract at all – meaning no fee paid, no packaging declared.^[40] TAMIR filed its first legal action against uncontracted companies only in 2024, thirteen years after the law took effect. The reason this persisted so long is straightforward: the statutory fine, NIS 150,000, was trivial against the savings from simply not paying treatment fees on hundreds of tonnes of annual imports, and a free rider's packaging still ends up in the same orange bins everyone else is funding – a textbook collective-action problem, where the system's benefits are shared but its costs fall only on whoever chose to pay them.

5. Failure three: commercial optimisation over household capture

TAMIR has collected roughly 75 percent of its packaging from the commercial sector and only about 25 percent from households – despite households generating an estimated 70 percent of all packaging placed on the market. The operational logic is simple: collecting from shopping centres and warehouses is cheaper and easier than collecting from tens of thousands of residential entrances, and commercial volume alone is enough to meet the law's statutory targets. The result is a system that satisfies the letter of the law while leaving most of what the law was meant to address largely untouched – a pattern this book already traced from the household side (Ch.4) and the orange-bin side (Ch.10).

The Ministry's 2025 national scan gives this failure a specific, direct number: around 6 percent of household packaging reaches a designated collection channel at all. That figure should be read precisely, not as a verdict on household behaviour – Chapter 8 already established that low capture reflects infrastructure and access at least as much as individual choice – but as confirmation, from the regulator itself, that fifteen years of this EPR system have not produced high household-level capture.

6. Failure four: downstream opacity

A less-discussed failure sits past collection entirely. Chapter 10's fraction-by-fraction review found that for most orange-bin material, rejection rates, the identity of specific downstream recyclers, actual output volumes, and the eventual geography of tails are simply not published. TAMIR can report what was collected and whether it met a declared-base target. The public cannot, from anything TAMIR or the Ministry currently publishes, reconstruct a full fate chain for a given fraction – from bin to bale to buyer to product. An EPR system that can report compliance without being able to document material fate has made itself accountable for the wrong thing.

7. Failure five: a weak signal to packaging design

Chapters 11 and 12 showed that some packaging is structurally difficult or effectively impossible to recycle under current sorting technology – multilayer sachets and pouches most clearly, but also small items, dark plastics, and mixed-polymer formats. A functioning EPR system should make that kind of packaging measurably more expensive

for the producer who chose it, creating a real incentive to redesign. Where treatment fees are set by broad material category rather than by how readily the sorting system can actually recover a given format, the polluter-pays principle weakens into something closer to a flat reporting fee: a producer pays for the category, not for the specific design decision that turned a package into tails before it ever reached a recycler.

8. Failure six: cost and burden shifting

TAMIR compensates municipalities for organising source separation, but that compensation covers only part of the real cost; the difference is absorbed by the municipality's general budget – arnona, the property tax every resident pays, not only those whose packaging is properly declared to TAMIR (Ch.2 §3 already established this same hidden-subsidy pattern for the Cleanliness Fund more broadly). The exact size of that gap has not been published systematically anywhere in the public domain. The burden doesn't stop at the municipal budget line, either: contamination, difficult formats, and unsorted tails become manual sorting labour, line stoppages, and rejected material that ends up as someone's local environmental burden (Ch.6–7, Ch.14). A complete accounting of what this EPR system costs would have to price not only tonnes, but the labour and risk required to recover or reject them – and no such accounting currently exists.

9. Monopoly status, and its limits as an explanation

TAMIR was the sole recognised operator for more than a decade. A 2024 Jerusalem District Court ruling, following a petition from a competing recycler after four failed attempts to obtain recognition, found the Ministry had "shirked its responsibility"^[8] by failing for over ten years to regulate competition in packaging recycling, and ordered it to open the market (Ch.2 §4) – a finding that this lack of competition was a regulatory failure, not simply an accident of TAMIR's institutional position. A separate class action, brought by a cement importer, alleged TAMIR used its monopoly position to collect retroactive treatment fees – reportedly NIS 645,000 – as a condition of contracting; this claim comes from a single secondary source and should be read as a documented allegation, not an adjudicated fact.

It's tempting to treat competition as the fix for everything in this chapter. It probably isn't. Opening the market may reduce the complacency that comes with having no rival, but competition by itself doesn't correct declared-base accounting, doesn't catch free riders, doesn't strengthen the design signal, doesn't make downstream data public, and doesn't close the municipal funding gap. Those are architectural problems, and a new operator built on the same architecture would likely reproduce most of them.

10. What the figures actually show

Claim	What it proves	What it does not prove
84% recycling, 2021	TAMIR's result measured against its own declared base	Actual recycling of all packaging placed on the Israeli market
480,000 tonnes declared	The scale of the contracted, reporting system	The total packaging entering the market – the declared base covers roughly half of it
>40% of producers/importers uncontracted	A real, quantified free-rider problem	The exact tonnage those producers place on the market
~75% commercial / ~25% household collection	Systematic optimisation toward the easier, cheaper stream	Success at reaching the household stream, which generates most packaging by volume
~6% household packaging capture (MoEP 2025)	Low first-mile capture into designated collection channels	The downstream recycling rate – capture and recycling are different gates (Ch.1 §6–§7, Ch.8)

Claim	What it proves	What it does not prove
Rejection and tails rates	A documented reporting gap	That tails don't exist – only that their scale isn't published

Read this way, the table's real message isn't that any single figure is wrong. It's that each one answers a narrower question than the headline version implies – exactly the discipline this book has applied to every material chapter since Chapter 8, now applied to the institution that was supposed to be reporting on all of them.

Civic questions for the municipality and TAMIR

How many tonnes of packaging were collected from orange bins in your city last year, and what share of the estimated local packaging stream does that represent?
What is the gap between TAMIR's compensation to your municipality and its actual cost of organising source separation?
Which specific companies in your region have no TAMIR contract, and what enforcement action, if any, has followed?
What is the rejection rate at the sorting facility serving your city, and is it published anywhere?
Does your municipality's contract with TAMIR or its sorting contractor include any labour-standard or safety-reporting requirement (Ch.14)?

Conclusion

Israel did not fail because it adopted extended producer responsibility. The principle – producers pay for the packaging they place on the market, not taxpayers – is sound, and works elsewhere. It failed because the architecture built to implement that principle made producers responsible for a self-reported fraction of the stream, while leaving the real stream, household capture, downstream fate, design incentives, labour burden, and municipal costs only partly visible – and then measured its own success using exactly the numbers it had chosen to make visible.

The next chapter turns to the reform now on the table in response to these failures, and asks the same question this book has asked of everything else: not whether the reform sounds right, but what, specifically, it would make newly visible – and what it might, deliberately or not, leave just as dark as before.

Chapter 16

The Deposit System and TAMIR: Two Streams That Failed to Integrate

Two systems operating in parallel do not add up to one plus one. They add up to one plus one minus friction, minus coverage gaps, minus whatever material ended up in the wrong stream entirely.

Israel's packaging system rests on two mechanisms that emerged independently and were never institutionally integrated: the deposit system, running since 1999 on a consumer cash incentive, and the EPR system through TAMIR, running since 2011 on producer-paid treatment fees (Ch.15). This chapter isn't about either system failing on its own terms – the deposit system works well within its scope, and TAMIR's problems were already examined at length. It's about what happens at the seam between them: the deposit system proves that money and a clean channel work. Its unresolved conflict with TAMIR proves that a successful stream can still weaken the larger system around it if no one is responsible for how the two fit together.

1. Why deposit works, briefly

The mechanism is simple: a consumer pays a deposit on a beverage container at purchase and gets it back on return, via a shop or a reverse-vending machine. That incentive, examined in detail earlier in this book as a benchmark case (Ch.9), produces a cleaner, more uniform stream than voluntary sorting ever manages on its own – not because deposit consumers care more about the environment, but because money is at stake. This chapter doesn't re-argue that case. It asks what the deposit system's success does to the system built alongside it.

2. Same materials, different legal worlds

The deposit system and TAMIR don't simply "compete for the same materials," as that framing implies a symmetry that doesn't exist. They overlap by material and diverge by everything that actually determines an object's fate: which law governs it, which actor collects it, what incentive applies, how it's reported, and against what denominator its performance gets measured.

Material	Under the deposit system	Under TAMIR (orange/purple bin)	What this means
PET	Transparent beverage bottles, 1.5–5 L since Dec. 2021 – clean, high-value, bottle-to-bottle capable (Ch.9)	Trays, non-standard bottles, non-beverage packaging – mixed, contaminated, lower value (Ch.11)	Chemically the same polymer; institutionally two different materials with two different fates
Aluminium / metal	Deposit-marked cans – high secondary value, clean stream	Mixed metal, foil, non-deposit cans – still recoverable, but a different reporting stream (Ch.13)	The most valuable metal fraction is captured before it ever reaches TAMIR's system
Glass	Deposit-marked bottles – ~120,000 t/yr, high recycling rate	Jars, sauce and jam containers – only ~30,000 t/yr declared, 4–8% recycling rate (Ch.13)	The same material, chemically infinitely recyclable either way, performs completely differently depending only on which system captures it first

The pattern across all three materials is the same: the deposit system captures the cleanest, most valuable fraction of each one before it ever reaches TAMIR's collection channel, leaving TAMIR with what's left – more mixed, more contaminated, and structurally harder to recycle at a profit. When the deposit was extended to larger bottles in December 2021, this asymmetry sharpened further: the most valuable large-format PET moved into the deposit

circuit, and the orange bin was left with a lower-value residual mix, without any compensating adjustment to TAMIR's funding model.

3. Glass: where the gap is most visible

Glass makes the structural mismatch impossible to miss. Deposit-marked glass bottles – roughly 120,000 tonnes a year – see a comparatively high recycling rate. Non-deposit glass – jars, sauce and jam containers, headed for TAMIR's purple bin – tells the opposite story: only around 30,000 tonnes a year gets declared, well short of the real volume in circulation, and the resulting recycling rate sits at 4 to 8 percent, among the lowest of any material this book has examined for a substance that is chemically infinite in its recyclability (Ch.13 covers the material case in full). Under the current incentive structure, TAMIR has no particular reason to invest in better purple-bin glass logistics: the material is heavy, fragile, and expensive to move, and nothing in the system rewards fixing that.

4. The denominator problem, made explicit

This book has repeated one instruction since its first chapter: name the denominator before trusting a percentage (Ch.1 §7). Nowhere does that instruction matter more than here, where two entirely separate reporting systems produce figures about materials that sound comparable and aren't.

Claim	Its denominator
Deposit return rate	Containers covered by the Deposit Law specifically
TAMIR recycling rate (e.g. 84% in 2021, 97% in 2022)	TAMIR's own declared base – packaging reported by contracted producers, roughly half or less of the full market
Household orange-bin capture (~6%, MoEP 2025)	Household packaging reaching a designated separate-collection channel
Glass recycling rate	Must specify deposit glass or non-deposit glass – the two are not comparable (Ch.13)
Overall packaging recycling estimate (28–55%, 2024 committee range)	Total estimated packaging placed on the market, now revised upward to 800,000–1,600,000 t/yr – a wide range reflecting genuine measurement uncertainty

A 2024 inter-ministerial committee tasked with examining competition in the packaging sector – convened after a Jerusalem District Court ruling ordering the Ministry to act (Ch.15 §9) – put the honest range at somewhere between 28 and 55 percent of Israel's packaging actually recycled, after revising the Ministry's own estimate of total packaging generated sharply upward, to somewhere between 800,000 and 1.6 million tonnes a year.^[40] That range is wide because the underlying uncertainty is real, not because anyone is hedging. TAMIR's own more recent reporting shows 439,000 tonnes recycled against 451,000 tonnes declared for 2022 – 97 percent of the declared base, an improvement on the 84 percent figure for 2021^[40] – but the declared base itself is the same fraction of the real market it always was, so a rising percentage against a shrinking-in-relative-terms denominator says less than it appears to. Deposit success cannot be used to prove TAMIR is succeeding, and TAMIR's documented weaknesses cannot be used to describe deposit PET, deposit glass, or deposit metal, which are institutionally separate materials with separate evidence chains.

5. Money: opaque by design, not by accident

Each system's finances are legible on their own terms and nearly impossible to compare against each other. In the deposit system, the operator (ELA or Asofta) receives the deposit from the beverage producer, refunds it to whoever returns the container, and keeps the difference between deposits collected and deposits redeemed to cover its

operating costs. In the TAMIR system, producers pay treatment fees, TAMIR compensates municipalities for organising collection, and – as this book has already shown – that compensation covers only part of the real cost, with the gap absorbed by *arnona*, the general municipal tax (Ch.15 §8).

Neither system's risk exposure is symmetric, and neither absorbs shocks from the other automatically. When oil prices fall, virgin plastic undercuts recycled plastic and TAMIR's sorted material loses value. When the volume of unsortable tails grows, landfill costs rise. When the deposit system expands its scope, as it did in 2021, it removes exactly the material TAMIR's financial model depended on most – high-value PET – while TAMIR's operational overhead stays the same. No regulatory mechanism governs that transition. The two systems simply absorb whatever the other one does to them, with no shared ledger and no one accountable for the interaction.

6. What real integration would require

Integrating the deposit system and TAMIR wouldn't mean merging them into one – the deposit system's strength comes specifically from being a separate, simpler, cash-incentivised channel. It would mean a coordination mechanism doing three things neither system currently does on its own: a clear, unambiguous rule for which packaging belongs to which stream, so a deposit-marked container found in the orange bin has a defined status rather than becoming an accounting grey zone; a financial rebalancing mechanism, so that when the deposit system's scope changes and pulls value out of TAMIR's stream, TAMIR's funding adjusts rather than silently absorbing the loss; and unified downstream reporting, so a single mass-balance figure for a material like PET or glass could actually be constructed by adding the two systems' data together – which, as things stand, cannot be done, because each measures a different population against a different denominator.

Other countries have built exactly this kind of coordination. Germany's deposit system and its EPR mechanism (*Grüner Punkt*) operate in parallel but with a clear legislative boundary between streams and unified reporting to the regulator, so a producer knows in advance which system handles a given package. The Netherlands coordinates its EPR and deposit-refund systems through a single body responsible for a shared mass balance across all streams, publishing one annual report covering every fraction. These international comparisons are carried forward from the original research for this book without independent re-verification in this specific pass, and are offered as illustrations of what coordination can look like structurally, not as a claim that either system should be copied wholesale.

7. When fragmentation becomes someone's labour

This gap isn't only a reporting inconvenience. When a deposit-marked bottle ends up in the orange bin instead of a deposit machine – through confusion, inconvenience, or simple habit – it doesn't just get "mis-sorted" in some abstract sense. It adds a real task to a sorting line already handling TAMIR's harder, more contaminated residual mix, and it lowers the quality of the batch it lands in (Ch.14). Institutional fragmentation between two well-designed systems becomes, at the point where they actually meet, somebody's extra work.

Conclusion

The problem this chapter has described isn't that Israel operates both a deposit system and an EPR system. Plenty of countries run both successfully. The problem is that Israel built two separate reporting worlds for materials that physically overlap, with no public bridge connecting them – no shared denominator, no financial rebalancing when one system's scope changes, and no single mass-balance report a citizen could actually use to answer the simple question of how much packaging, in total, gets recycled in this country.

The next chapter turns to the reform now under discussion in response to failures across both systems (Ch.15–16), and asks the same question this book keeps returning to: not whether the reform sounds right on paper, but what it would actually make visible that isn't visible today.

Chapter 17

The 2026 Reform: Israel's Waste Sector at a Crossroads

A reform proposed, opposed, and withdrawn within the same two-month window is not an administrative formality. It is an acknowledgement, from every side of the argument, that the existing system has exhausted itself.

This chapter describes a process still in motion at the time of writing. Israel's waste-sector reform has been under active discussion since 2024, has been drafted, publicly attacked, and pulled from legislation at least once, and its central parameters – who regulates the sector, which laws survive, how the money moves – remain unresolved. This chapter says so plainly rather than smoothing over the uncertainty, and explains why civic participation matters most precisely in a moment like this one.

The previous chapters showed that Israel's waste problem was never one failure but a stack of them: material fate that stops being documented past a certain gate (Ch.8–13), sorting labour that absorbs whatever design and infrastructure fail to resolve (Ch.14), an EPR system that measured success against a fraction of the real stream (Ch.15), a deposit system and TAMIR that were never institutionally integrated (Ch.16), and packaging design decisions made with no financial pressure to consider recoverability at all (Ch.18). This chapter asks what a reform would actually need to do to address that whole stack – not just the piece currently making headlines.

1. Three drivers, arriving at once

Three things are true in Israel simultaneously, and together they make reform difficult to avoid and difficult to get right. The State Comptroller's July 2025 follow-up audit confirms remaining landfill capacity is insufficient for the volume of waste expected by the end of 2026 – a physical constraint, not a projection (Ch.3, Ch.6). The regulator's own 2025 national scan confirmed low household capture into designated collection channels after fifteen years of the existing EPR system (Ch.1 §7, Ch.15). And after years of stalled discussion, the Ministry of Finance moved the reform into the Economic Arrangements Law – the legislative vehicle Israel typically uses to force through structural change alongside the annual budget, whether or not the underlying policy work is finished.

2. Two reform logics, not one

Israel currently has two distinct, parallel proposals in play, and treating them as a single “reform” obscures a real difference in approach. The Ministry of Finance proposal tries to rebuild the sector from above: a single framework law replacing today's patchwork of separate statutes, a new National Waste Authority with power to set tariffs and license every operator in the chain, and a “closed market” model in which every service – collection, sorting, processing, landfilling – carries a regulated tariff covering its real cost, explicitly modelled on how Israel already regulates water. The MoEP memorandum, by contrast, tries to repair packaging EPR from within: it keeps the Packaging Law, raises its targets toward European levels, opens the market to competition beyond TAMIR, and adds a household-collection minimum and a producer registry (Ch.18 §5) – reform by tightening an existing structure, not replacing it.

These aren't presented as alternatives voters will choose between. They are two live legislative tracks, drafted by two different ministries, that could both partly succeed, both stall, or collide with each other over jurisdiction – over, for instance, whether TAMIR's role should be reformed under the Packaging Law or dissolved entirely into a new Authority's licensing regime.

3. What the Finance Ministry proposal actually contains

Its published elements: one framework law covering every waste category, replacing today's separate statutes for packaging, deposit containers, and electronic waste once the first implementing regulations under the new law are issued; a National Waste Authority housed within MoEP, empowered to set tariffs, license every collector, sorter,

and landfill operator, plan long-term capacity, set service standards, and publish an annual public report; abolition of the landfill levy and the Cleanliness Fund, replaced by direct, cost-based service payments – justified in part by the State Comptroller's own finding that the Fund's accumulated surplus, now on the order of NIS 4 billion, has functioned largely as a source of interest-free loans for the Ministry of Finance's own budgetary needs rather than as the environmental instrument it was designed to be^[39] (Ch.15 §8); and a public registry of every licensed operator in the sector, aimed partly at the organised-crime element the Ministry says has entered waste collection and disposal.

The Ministry's own projection: savings of over NIS 450 million a year for local authorities in the reform's first year, plus further savings from a reformed landfill levy^[53] – a figure worth treating the way this book treats every other headline number (Ch.1 §7): a projection under one specific reading of the reform, not a guarantee, and not yet independently verified against an alternative estimate.

4. The clearest risk: a capacity-first reform

The sharpest criticism of the Finance Ministry's draft, and the one this book takes most seriously, is not about tariffs. It's about sequencing. Under the current draft, MoEP's own commitment – made only a few years earlier – to landfill no more than 20 percent of Israel's waste by 2030 is pushed back to 2050, a two-decade postponement embedded in the same document that proposes the new Authority.^[55] Greenpeace Israel's director has raised a second, more structural concern: the reform's stated solutions to the capacity crisis lean heavily toward new waste-to-energy plants, built with little visible reference to the internationally recognised waste hierarchy – prevention at source, reduction, reuse, repair, and only then recycling or energy recovery (Ch.3 already established this same hierarchy for this book). His specific warning is worth stating precisely: co-locating a sorting facility with an incineration facility under the same concessionaire creates a direct financial incentive to route high-value organic material toward power generation rather than composting or anaerobic digestion, because the concessionaire profits either way and burning is operationally simpler.^[55]

This is the single most important test of the entire reform, restated as plainly as this book can manage: a regulator built primarily to guarantee landfill and WtE capacity is not the same thing as a regulator built to reduce how much residual waste the country produces in the first place. Both can be called a National Waste Authority. Only one of them makes the rest of this book's argument obsolete in the way it should be – by making prevention work well enough that end-of-pipe capacity stops being the most urgent question.

5. Three positions, three internal logics

The Ministry of Finance's central argument is that the current system's opacity and fragmentation have created space for inefficiency and, in its own account, organised crime – and that a single regulator with public, cost-based tariffs corrects this structurally rather than case by case. Its weakness: environmental outcomes are named as goals but not written into the law as binding, measurable targets.

MoEP's central argument is that the Finance proposal removes environmental policy from environmental hands and hands it to an authority optimising for cost, not ecological outcome – while MoEP itself has overseen the existing EPR system for fifteen years, by its own data producing exactly the low household-capture result documented in Chapter 15. Both things can be true about the same ministry's objection at once.

Environmental organisations, Greenpeace Israel among them, argue that neither ministry's proposal puts the waste hierarchy at the centre of the law, and that a tariff regulator without binding prevention and reuse targets will optimise service cost, not material recovery – a position that is analytically sound but politically harder to legislate quickly under crisis conditions than either ministry's simpler pitch.

6. The reform scorecard

Rather than asking whether the reform “sounds right,” this book applies the same test it has applied to every material stream, every institution, and every claim since Chapter 8: what, specifically, would this reform make newly visible that isn't visible today – and where might it simply relocate the same opacity under a new name?

Test	The question that matters
Evidence	Will rejection rates, downstream recyclers, output volumes, and the destination of tails become publicly reportable (Ch.8–13), or will the new regulator simply publish a new headline percentage with the same undisclosed denominator?
Finance	Does the reform close the hidden municipal subsidy – the gap between what producers pay and what collection actually costs (Ch.15 §8) – or does it just relabel who writes the cheque?
Design	Does anything in the new architecture create financial pressure on producers to make packaging more sortable before it's manufactured (Ch.18), or does it price membership in the system rather than the design decisions that determine a package's fate?
Labour	Does the reform say anything at all about who performs manual sorting, under what conditions, and with what safety reporting (Ch.14) – or does it standardise tariffs while leaving the labour chain exactly as invisible as it is today?
Geography	Will residents be able to find out where their residual waste physically ends up – which landfill, RDF user, or WtE facility – and who lives beside it (Ch.6–7)?
Governance	Is the new regulator's mandate environmental as well as tariff-setting, with real authority to enforce the waste hierarchy – or is it, in substance, a landfill-and-tariff authority wearing an environmental title?
Prevention	Does the law give prevention and reuse legal priority over landfilling and energy recovery, with consequences attached – or does it fund capacity for the back end of the system while leaving the front end exactly as under-resourced as Chapter 3 and Chapter 14 already found it?

A regulator only resolves the fragmentation this book has traced if it is required to publish data, enforce the hierarchy with real consequences, set tariffs transparently, coordinate the actors already mapped in Chapter 2, and prevent cost-shifting onto municipalities, workers, and host communities. Absent those specific requirements, a National Waste Authority can just as easily become a more efficient coordinator of the same end-of-pipe logic this book has spent sixteen chapters describing – better at moving waste around, no better at making less of it.

7. A corrected and honest timeline

Understanding the current moment requires understanding how many times this reform has already started and stalled.

Date	Event
2020	MoEP publishes its National Strategic Plan 2030 – targets for reducing landfilling, raising recycling, and building WtE capacity.
2021	State Comptroller documents the landfill-capacity crisis for the first time and issues initial recommendations.
Sept. 2024	Ministry of Finance publishes its first version of the reform. The sector's response is sharp rejection. ^[39]
Oct. 2024	Reform tone “softened” following industry pressure; the government instead approves a slimmer decision creating an inter-ministerial team to draft guiding principles for a framework law. That team never convened. ^[39]
Nov. 18, 2025	MoEP publicly calls the Finance Ministry's proposal a machtaf – a grab. ^[39]

Date	Event
Nov. 25, 2025	Ministry of Finance withdraws the reform from that year's Economic Arrangements Law bill. ^[39]
Dec. 2025	MoEP publishes its own memorandum reforming the Packaging Law from within (Ch.18 §5) – a second, narrower track moving in parallel with whatever the Finance Ministry proposes next.

At the time of this writing, the most recently confirmed development is the Finance Ministry's November 2025 withdrawal of its proposal from that year's Economic Arrangements Law, alongside MoEP's own December 2025 memorandum moving forward on a separate track. Events beyond this point were not independently confirmable during this research and are not presented as settled – a reader encountering this chapter later than mid-2026 should treat everything past this timeline as needing a fresh check, not an assumption that nothing further happened.

8. What civil society can do right now

Reforms drafted under crisis conditions move faster, and worse, when no one outside government is asking specific questions of the draft text. Genuine leverage points exist at this stage: both ministries have run public consultation rounds, and any further legislative version will likely open another; the Knesset Interior and Environmental Protection Committee holds open hearings on waste legislation and can be asked to hear a specific organisation's position; the Union of Local Authorities has already taken a public position on the reform and can be lobbied by local activists working through their own municipality; and the transparency obligations any version of this reform would create are themselves a data-request opportunity worth using now, before the law exists, using the civic tools this book sets out later (Ch.27).

The most effective form this takes is rarely a general statement of support or opposition. It's a specific, narrow demand attached to specific legislative language – “this law must require rejection-rate publication by facility” is harder to quietly drop than “we support environmental reform.”

Conclusion

A reform born from landfill panic must not become a reform that simply manages landfill panic more efficiently. The test of the 2026 reform is not whether it creates a new regulator – both proposals on the table do that, in different forms. The test is whether the new architecture makes the parts of the system this book has spent sixteen chapters uncovering – costs, tails, labour, geography, design failure, and prevention – genuinely governable, or whether it leaves them exactly as invisible as they are today, under a more confident-sounding name.

The chapters that follow examine what a design-first, prevention-first alternative could actually look like in practice: what the EU's PPWR framework offers as a more developed model (Ch.19), and, later in this book, the financial and infrastructural changes that would need to accompany any of it for prevention to become more than a stated goal.

Chapter 18

Packaging Design: Why Some Materials Are Born Recyclable

Nobody asks a packaging designer whether their product is recyclable. They're asked whether it sells, whether it protects the contents, and what it costs to produce. The first recycling decision is made before the package ever reaches a bin – at the design desk, by someone optimising for entirely different questions.

Chapters 9 through 13 traced specific material streams, showing where each succeeds and where it fails. This chapter moves up one level: what determines, well before a container is ever chosen, whether a given piece of packaging lands in the successful category or the problem one? The answer is design – and design decisions determine a package's fate more powerfully than any amount of subsequent investment in sorting or recycling infrastructure.

1. Recyclability is a property of design, not of material

It's tempting to think in binary terms – plastic bad, paper good – but the material chapters already showed this doesn't hold: HDPE recycles far better than multilayer packaging despite both being "plastic," while technically ideal glass recycles worse than metal for reasons of logistics and economics, not chemistry (Ch.9–13). The more precise statement is that recyclability is determined by how a material is combined, coloured, labelled, and sized into a single object – not by the material as such.

Designer's decision	Recoverable option	Harder-to-recover option	Why
Material composition	Single polymer (mono-material)	Several bonded layers of different materials	NIR sorting can't reliably distinguish a composite material (Ch.11–12)
Colour	Transparent or light	Black or dark	Black pigment absorbs infrared; the sensor doesn't register a signal at all
Label	Removable paper label	Heat-shrink sleeve from a different polymer	A non-removable, foreign-material label contaminates the batch during recycling
Size	Large, above ~5 cm	Small, under ~3 cm	Physically falls through sorting equipment regardless of what it's made of (Ch.4, Ch.10)

Case study: the yoghurt cup that couldn't be seen

A UNIDO pilot under the SwitchMed programme, already introduced in this book's plastics chapters (Ch.11), makes the principle concrete. A standard Israeli yoghurt or salad cup is typically a polypropylene body wrapped in a PET sleeve – two different polymers combined for branding and shelf appeal, with no thought given to sorting. That combination gives an NIR scanner a mixed signal it reads as unsortable, and the cup defaults to tails. Replacing the PET sleeve with a polyolefin sleeve – chemically compatible with the cup body – makes the same product fully detectable, at marginal cost to the producer. UNIDO's estimate: broad adoption of this single change across Israeli brands could make roughly 2,100 additional tonnes of plastic recyclable every year, from a change no consumer would likely notice.

2. What "non-recoverable" actually means, and doesn't

It's tempting to describe some packaging as "born unrecyclable" – but that phrase claims more certainty than the evidence supports, and it treats very different failures as though they were the same failure. The more accurate

description is that a given format is non-recoverable under current Israeli sorting and market conditions, and that qualification matters because the underlying cause differs by category.

Multilayer packaging – sachets, laminated pouches, the yoghurt-cup sleeve above – comes closest to a genuine design failure: no realistic downstream fix exists once those layers are bonded, and the only path to improvement runs through the design stage itself (Ch.12). Black plastic and packaging below the sorting-line's size threshold are closer to equipment limitations that happen to be permanent under current technology, not properties baked irreversibly into the material (Ch.4, Ch.10). Glass outside the deposit system is different again – a case this book has already shown to be substantially about access and economics rather than design or detectability at all (Ch.13, Ch.16): glass is heavy, fragile, and expensive to move, and no design change to a jam jar would fix that. Collapsing these three causes into one phrase – "born unrecyclable" – would blur a distinction the rest of this chapter depends on.

3. Who decides, and why recoverability isn't their priority

The decisions examined above aren't made by recycling operators or municipalities. They're made by producers and their packaging suppliers, at a stage institutionally and physically separate from everything else this book has described. A producer choosing a PET sleeve for a yoghurt cup is optimising for vivid branding, cost, and shelf protection – not for whether an infrared sensor two years later will register a clean signal.

The costs of that choice don't fall on the producer. They fall on the collection system – the sorting facility, the municipality, and ultimately the taxpayer through arnona (Ch.15 §8) – and, this book has now shown, on a person. Bad packaging design doesn't disappear into technology. Chapter 14 traced exactly where unsortable, ambiguous, or contaminated material actually goes once a machine's confidence runs out: into a worker's hands, as an extra sorting task, a line stoppage, or a rejected batch. A design decision made at a product-development meeting becomes, months later, someone else's physical work – a link in the chain this book has been tracing since long before this chapter, now closed.

This is a textbook case of externalised cost, and the "polluter pays" principle behind extended producer responsibility should have corrected it. Chapter 15 already showed why Israel's implementation of EPR through TAMIR didn't: a declared base covering roughly half the real market and weak enforcement against free riders don't add up to financial pressure precise enough to change any single product's design. An EPR system that charges by broad material category, rather than by how readily a specific format can actually be sorted, is not really pricing recyclability at all – it's pricing membership in the system. Put plainly: EPR without a design-modulation mechanism is a weak EPR, whatever else it gets right.

4. Reading the evidence on design

Claim	Level	Comment
Mono-material construction improves sortability	✓✓	Well-established sorting-technology principle (Ch.11)
Black/dark plastic defeats NIR detection	✓✓	Confirmed repeatedly across this book's material chapters (Ch.10–12)
A shrink sleeve of a different polymer degrades batch quality	✓ / ✓✓	Consistent with contamination patterns already documented (Ch.9, Ch.11)
Packaging under ~3–5 cm is physically unsortable	✓	A mechanical limit of sorting-line equipment, not a material property (Ch.4, Ch.10)
Actual adoption of recyclable-by-design changes by Israeli producers	?	Mostly unknown – no systematic public tracking of design-level compliance found

The pattern is familiar from every material chapter in this book: the technical mechanism by which design defeats sorting is well documented and repeatedly confirmed. Whether Israeli producers are actually changing their designs in response – as opposed to simply absorbing the treatment fee and continuing as before – is close to entirely unknown, because nothing currently tracks design-level compliance the way TAMIR tracks tonnage.

5. What's changing: the November 2025 memorandum

In November 2025, the Ministry of Environmental Protection published a memorandum of law – a draft bill for public comment – amending the Packaging Law, with comments due by December 10, 2025.^[30] This is a document distinct from, but running in parallel with, the broader Ministry of Finance reform (Ch.17): where the Finance reform aims to replace the sector's architecture from above, the MoEP memorandum aims to repair packaging EPR from within – tightening the existing law rather than abolishing it.

Its core elements: a gradual rise in recycling targets aligned with European standards, with the general target climbing from 60 percent today to 70 percent by 2031, alongside material-specific sub-targets; new treatment-fee obligations^[41] for reusable packaging, currently exempt entirely; a minimum requirement that at least 50 percent by weight of what a producer recycles come from the household stream specifically – a direct response to the commercial-sector optimisation this book has already documented (Ch.15 §5); an opening of the market to multiple recognised operators alongside TAMIR, under a new coordination mechanism; and a public, customs-linked registry of every producer and importer of packaged goods, closing the free-rider gap that Chapter 15 described at length.

Israel today	MoEP memorandum proposal	EU (PPWR)
60% general recycling target	70% by 2031, with material-specific sub-targets	70% by 2030

That target alignment is not incidental – it is a deliberate move toward the European figure this book examines in detail next (Ch.19). Raising the overall recycling target creates indirect pressure toward more recoverable design, since a higher target is harder to reach without improving how sortable packaging is at the point of manufacture. What the current memorandum does not do is set direct, binding design requirements the way the EU's PPWR does – that remains the sharper instrument, examined in the next chapter.

Conclusion

Sorting technology, resident behaviour, and recycling facilities all operate only within the boundaries packaging design has already set for them. No amount of investment downstream can sort a sleeve NIR can't read, recover a fraction too small for the equipment, or separate a laminate that was bonded specifically so it couldn't be pulled apart. The first recycling decision is made before the package reaches the bin – and everything this book has described since Chapter 4 has been, in some sense, a system working within whatever room that first decision left it.

The next chapter turns to the most developed existing model for regulating that first decision directly: the European Union's Packaging and Packaging Waste Regulation, and what of it Israel's current reform proposals do and don't actually borrow.

Chapter 19

What Israel Can Learn from PPWR

PPWR is not a perfect law. It passed through what EU policy commentators describe as one of the most heavily lobbied legislative processes in the bloc's history, and the final text is a diluted version of the European Commission's original 2022 proposal. Even diluted, though, it contains mechanisms that don't exist in Israel in any form at all.

The previous chapters identified a specific, recurring set of missing mechanisms in the Israeli system: no binding design rules (Ch.18), no full producer registry closing the free-rider gap (Ch.15), no formal coordination between the deposit system and TAMIR (Ch.16), and no permanent, funded evidence base beyond a single 2025 scan (Ch.1, Ch.8). This chapter isn't a tour of European legislation. It's an examination of which specific PPWR mechanisms turn those exact gaps into binding legal obligations – and which parts of Israel's problem PPWR doesn't touch at all.

1. Why form matters before content does

The EU's Packaging and Packaging Waste Regulation – formally Regulation (EU) 2025/40 – was adopted by the Council in December 2024, published in the Official Journal on January 22, 2025, entered into force on February 11, 2025, and applies from August 12, 2026 after an eighteen-month transition. It replaces the previous Packaging Directive (94/62/EC), and that replacement is itself the first lesson worth taking seriously.^[27]

A directive sets a shared goal and leaves each member state to write its own national law to reach it – which is exactly the pattern that produced significant divergence in how EU countries handled EPR rules and labelling under the old directive. A regulation applies directly, the same text, in every country at once. Israel's own legislative history is a directive-style pattern taken to its logical extreme: the Cleanliness Law of 1984, then separate, reactive laws for collection (1993), packaging (2011), electronic waste (2012), and deposits (1999, amended twice), each adopted in response to a specific problem rather than as part of one coherent structure (Ch.1 §1, Ch.2 §8). The choice PPWR represents – one directly applicable framework instead of an accumulating patchwork – is exactly the fork in the road the 2026 reform now faces (Ch.17).

2. Design becomes a legal gate, not a suggestion

Chapter 18 showed that the first recycling decision is made at the design desk, long before a package reaches a bin – and that nothing in Israel's current system creates financial pressure on that decision. PPWR is significant here not because it's European, but because it's one of the first major regulatory systems anywhere to treat design as a binding legal obligation rather than voluntary producer goodwill.

Under Article 6, packaging must be designed so its components can be separated for recycling, using methods that actually exist in collection and sorting infrastructure – not a theoretical recycling process. From January 2030, packaging that falls below a defined recyclability grade will not be permitted on the EU market at all, and EPR fees are set to be modulated by that same grade, so a harder-to-recycle design costs its producer more, directly and predictably, rather than being absorbed into a flat, category-wide fee the way TAMIR^[50]'s tariffs currently work (Ch.15 §3). Article 10 adds a minimisation requirement – packaging must use the minimum necessary material, with specific limits proposed for empty space in transport and e-commerce packaging, though the exact threshold varies across sources reviewed for this chapter and shouldn't be treated as settled to the percentage point.^[32] Article 7 phases in minimum recycled-content quotas for plastic packaging, rising through the early 2030s and verified through a mass-balance accounting methodology, and Article 5 restricts PFAS and other substances of concern in food-contact packaging.^[32]

This is precisely the layer of regulation that Chapter 18 found absent from Israel's own November 2025 memorandum: that document raises recycling targets and opens the market to competition, but sets no mandatory design criteria comparable to Article 6.

3. Coordinating deposit and EPR, not choosing between them

PPWR requires EU member states to establish deposit-return systems for single-use plastic and metal beverage containers by 2029, unless a country already achieves high separate collection without one. Israel already has exactly the kind of system PPWR is pushing the rest of Europe toward – it has run one since 1999, and this book has already examined both its strength as a clean, incentivised stream and its unresolved structural conflict with TAMIR (Ch.9, Ch.16).

The lesson here isn't that Israel needs a deposit system. It already has a good one. The lesson is coordination: PPWR expects formal mechanisms linking deposit and EPR reporting, precisely the mechanism Chapter 16 found missing in Israel – a shared mass balance, a clear rule for where a deposit-marked container found outside its own stream gets counted, financial rebalancing when one system's scope changes. If the 2026 reform creates a national regulator (Ch.17), that coordination function is a natural, low-cost part of its mandate – Israel doesn't need to build a new deposit system to benefit from this lesson, only a bridge between the one it already has and the EPR system running alongside it.

4. Registry and evidence: turning declarations into obligations

EPR element	Israel today (Ch.15)	PPWR (EU)
Single or multiple operators	TAMIR alone since 2011; competition proposed but not yet law (Ch.17–18)	Multiple scheme operators by default, under national oversight
Base for targets	Declared base – roughly half the real market	Direct producer reporting via a single registry
Enforcement against free riders	Weak until 2024–2025; first lawsuits only in 2024	Registration is a condition of market access, not an optional declaration
Financing of monitoring	Not formalised – MoEP's 2025 scan was a one-off	Producers obligated to fund recurring composition surveys and container labelling

The bottom row deserves particular attention. PPWR obliges producers to finance ongoing surveys of mixed-waste composition – the exact category of measurement MoEP conducted as a single national scan in August 2025 (Ch.1 §7, Ch.8 §2). Under the European model, that kind of survey stops being a one-off regulatory initiative and becomes a permanent, producer-funded obligation. Israel now has one data point precise enough to build a book's methodology around. PPWR's lesson is that one data point should become a recurring practice, not a landmark exception.

5. What PPWR checks – and the diagnostic table this chapter is built around

The Israeli gap	The PPWR mechanism that answers it
Packaging is designed with no recycling gate in mind (Ch.18)	Recyclability-by-design (Art. 6): components must be separable, and design must account for real collection and sorting methods
No penalty exists for packaging that's structurally hard to recycle (Ch.11–12, Ch.18)	Recyclability performance grades A/B/C; packaging below Grade C (under a 70% threshold) barred from the EU market from Jan. 2030, with EPR fees modulated by grade

The Israeli gap	The PPWR mechanism that answers it
Deposit and TAMIR were never institutionally integrated (Ch.16)	Deposit-return systems required EU-wide by 2029, with formal coordination expected between DRS and EPR schemes, not two unconnected reporting systems
Declared base and free riders (Ch.15)	A single, direct producer registry as a condition of market access – not a voluntary declaration a company can simply skip
MoEP's 2025 scan was a one-off exercise (Ch.1, Ch.8)	Producers financially obligated to fund regular, ongoing surveys of mixed-waste composition – turning a single national scan into a permanent, funded practice

6. What PPWR doesn't solve

It would be dishonest to present PPWR as a complete answer to everything this book has documented. Two limits matter most.

First, honesty about the regulation itself: many of its most consequential details – the precise recyclability-grading methodology, specific design-for-recycling criteria – are delegated to secondary legislation the European Commission isn't required to adopt until January 2028, meaning full operational clarity arrives years after the headline law.^[50] A multi-tier grading system this complex also assumes real institutional capacity to verify and audit compliance – a capacity question Israel would need to answer explicitly, not assume solved by copying the text.

Second, and more specific to this book's argument: PPWR makes packaging more machine-readable. It does not, by itself, make sorting labour visible. Nothing in the regulation as reviewed for this chapter requires member states to report on who performs manual sorting, under what conditions, or with what safety data – the exact gap Chapter 14 found almost entirely undocumented in Israel. A country could adopt every mechanism in this chapter's checklist and still leave sorting-line labour exactly as invisible as it is today. Good design regulation is not a substitute for labour reporting, safety standards, or subcontracting transparency; it simply reduces how much of the hardest, most ambiguous sorting work still needs to be done once the packaging in question has been designed properly in the first place.

Conclusion

The lesson of PPWR is not that Israel should become Europe. Copying a diluted, still partly unfinished European regulation word for word would import its compromises along with its mechanisms. The lesson is narrower and more useful: recycling targets without binding design rules, a real producer registry, formal deposit/EPR coordination, and a permanently funded evidence base remain structurally weak, regardless of how ambitious the target percentage sounds. Israel should copy the function these mechanisms perform, not necessarily their European form or their exact grading thresholds.

PPWR is worth Israel's attention not because it's European, but because it regulates exactly the points where this book has found Israel's own system losing responsibility: design, registration, coordination, and who pays for knowing what's actually happening. It leaves at least one of those points – labour – as unresolved as Israel's own system does. Any new Israeli law should be read against both facts at once.

Chapter 20

How Much Waste Costs: Mass Balance, Gate Fees and Hidden Costs

The previous chapters followed how material physically moves through the system — from kitchen to landfill or recycler, and, more recently, through the hands of whoever sorts it. This chapter and the next ask a different question: how much does all of that cost, and who actually pays for it.

The answer is not simply "the taxpayer pays for collection." Money moves through several parallel channels in Israel's waste system, some opaque even to the municipalities writing the cheques — and that opacity is not incidental. Chapter 8 established that every claim about material fate names a gate, with its own evidence and its own denominator. The same discipline applies to cost: every shekel spent on waste also belongs to a specific gate, and a "price per tonne" that doesn't say which gate it's pricing is exactly the kind of number this book has learned to distrust.

Gate	Possible cost	Where this book examined it
First mile	Bin type, collection frequency, access, contamination	Ch.4
Transfer	Hauling, weighing, compaction	Ch.5 §2–§3
Sorting	Gate fee, labour, rejection handling	Ch.5 §4–§5, Ch.14
Tails	RDF, landfill, or WtE for whatever sorting couldn't recover	Ch.6, Ch.10–12
Downstream recycling	Transport, buyer uncertainty, price exposure to virgin material	Ch.9–13
End-of-pipe	Landfill levy, gate fee, ILA royalties	Ch.6–7, this chapter
Hidden costs	Labour risk, undocumented rejection rates, geographic burden	Ch.7, Ch.14

1. Two municipal cost segments

Municipal waste cost divides into two structurally different pieces. The intra-municipal segment covers collection from the bin site to the transfer station — dependent on distances, bin type, collection frequency, and volume (Ch.4). The extra-municipal segment covers everything from the transfer station onward: transport, gate fees at sorting facilities or landfills, and the landfill levy. This chapter focuses mainly on the extra-municipal segment, because that's where costs are rising fastest and most visibly, directly tied to the capacity crisis examined earlier in this book (Ch.3, Ch.6).

2. The landfill levy: incomplete incentive, not a failed one

Article 15 of the Cleanliness Law establishes the landfill levy — a per-tonne charge on waste placed in a landfill. In practice, a landfill operator pays it, but the cost passes through to the municipality, either directly or folded into a collection contractor's tariff. Introduced roughly a decade ago specifically to create financial pressure away from landfilling and toward recycling, its revenue flows to the Cleanliness Fund, whose own troubled record — only around 37 percent of its income used for its statutory purpose, and roughly NIS 1.66 billion diverted to the general state budget between 2016 and 2020^[73] — this book has already examined in detail (Ch.2 §3, Ch.15 §8).

The more useful question isn't whether the levy is a good idea. It's whether it functions as a complete incentive or an isolated, partial one. A levy that taxes landfilling in isolation, with no coordinated relationship to gate fees, sorting economics, or land-use royalties, can penalise the wrong step in the chain as easily as the right one — which is exactly what the next section shows happening in practice.

3. The pay-twice paradox, understood through tails

This is arguably the most counterintuitive financial fact in this book, and Chapter 8 through 12 already supplied the vocabulary needed to explain it properly: sorting doesn't fail because it's irrational. It fails financially because sorting produces two outputs — accepted fractions and tails (Ch.5 §5, Ch.10 §5) — and the tariff system charges a municipality for both while rewarding neither prevention nor high-quality capture strongly enough to offset the cost of the second.

A tonne of mixed waste sent straight to a landfill is billed once: the gate fee plus the landfill levy. The same tonne sent first to a sorting facility is billed at the sorting facility for the full tonne — and then, for whatever share of that tonne becomes residual tails rather than a saleable fraction, billed again at the landfill for that portion. Given how much of any given stream ends up as tails rather than a marketable fraction (Ch.10–12 traced this fraction by fraction, and it is substantial for anything beyond the cleanest rigid plastics and metals), the municipality can end up paying for sorting the entire stream and then paying landfill charges on most of it anyway.

Why this looks like municipal disinterest and isn't

When a municipality seems reluctant to expand sorting or source separation, that reluctance is often not a lack of environmental will. It's a rational response to a tariff structure that, as currently built, can penalise the intermediate step toward recycling rather than reward it. This is the same hidden-subsidy logic already examined in this book (Ch.2 §3, Ch.15 §8): financial incentives at the municipal level and environmental goals declared at the national level point in different directions, and a mayor balancing a budget will generally follow the incentive that's actually in front of them.

4. What design failure costs, concretely

Chapter 18 showed that a package's fate is largely decided at the design stage, before it ever reaches a bin. That decision has a price tag this book can now attach to it directly. Multilayer sachets, black plastic, undersized packaging, and mismatched-polymer sleeves (Ch.12, Ch.18) don't just "recycle poorly" as an abstract description — each one raises the sorting facility's rejection rate, adds to the manual labour required to catch and remove it (Ch.14), and increases the tonnage that ends up as tails, which then carries the landfill levy, the gate fee, and, where relevant, RDF processing charges.

A package that is born difficult does not become cheap simply because it enters the orange bin. It becomes expensive somewhere else — at the sorting line, in a worker's added task, and in the landfill invoice for whatever that worker couldn't save. None of that cost appears on the product's price tag or the producer's treatment fee under the current tariff structure (Ch.15 §7), which is precisely why Chapter 18 argued that EPR without a design-modulation mechanism prices membership in the system rather than the design decisions that actually drive cost.

5. A concrete cost shock: Evron

Chapter 6 already followed what happened when the Evron landfill's capacity ran out and a new sorting facility sat unused for over a year for want of downstream capacity for the organic fraction it would separate out. The cost consequence of that episode is worth restating in financial terms specifically: industry estimates put the cost of rerouting Western Galilee waste roughly 300 kilometres by rail to a Negev landfill at more than a 20 percent increase in treatment costs for the municipalities affected^[7] — a real, documented figure, arrived at independently of any single official's account of it. The sorting facility that might have reduced this pressure was eventually inaugurated in mid-2025 (Ch.6 §3). The lesson generalises beyond this one case: a regional capacity shock doesn't

just move waste. It moves money, on a timeline set by infrastructure decisions the affected municipalities didn't control.

6. ILA royalties: a third, uncoordinated price signal

A further cost layer sits outside both the landfill levy and ordinary gate fees. In early 2025, the Union of Local Authorities formally asked the Israel Land Authority to cancel a royalty mechanism for land use at landfill sites, arguing that landfill operators had begun passing this cost — set some years earlier — directly on to municipalities. By the Union's own estimate, the cumulative additional burden runs to roughly NIS 150 million a year nationally, with an explicit warning about financial strain on local authorities. MoEP has itself acknowledged it has no direct regulatory control over landfill gate-fee pricing, and has referred the matter to the national Inter-Ministerial Price Committee for recommendations.

Put together, the price of landfilling a tonne of waste in Israel is assembled from three independent sources — the landfill levy (a national instrument under the Cleanliness Law), gate fees (set by whichever operator runs the site), and ILA royalties (a separate, newly contentious channel) — and no single body regulates or even coordinates them as a whole. Israel does not have one waste price. It has several uncoordinated price signals, arriving on the same invoice as if they were one number. This is the financial expression of exactly the coordination gap Chapter 16 already found between the deposit system and TAMIR — the same structural absence, showing up again in a different part of the system.

7. What the 2026 reform proposes to change

The Ministry of Finance's reform proposal, examined in full in the previous chapter (Ch.17), explicitly proposes abolishing the landfill levy and the Cleanliness Fund, replacing them with direct, regulated service payments across the extra-municipal segment, set on a full-cost basis by the proposed National Waste Authority. In principle, that logic addresses the pay-twice paradox directly: if every service in the chain — sorting, transport, landfilling — carries a tariff that reflects its real cost, the structural penalty for taking an intermediate step toward recycling should, in theory, disappear.

MoEP's counter-argument, already noted in the previous chapter, is that abolishing the landfill levy removes the only environmental price signal currently in the system, however imperfect, without any guarantee the new regulator's tariff-setting will carry an equivalent environmental priority. Whatever the reform ultimately decides, this chapter's argument stands independently of that outcome: the pay-twice paradox, the ILA royalty dispute, and the uncoded burden of design failure and sorting labour are concrete, current problems that any new pricing architecture has to address explicitly — not assume a new regulatory body will resolve automatically, simply by existing.

Conclusion

The problem this chapter has traced isn't only that waste costs a great deal. It's that the system prices disposal far more clearly than it prices prevention, prices tails far more clearly than the design failures that created them, and prices tonnes far more clearly than it prices responsibility. A tariff structure built this way will keep rewarding the wrong step in the chain no matter how large the headline reform turns out to be, unless it is rebuilt around the same gates this book has used to trace material and evidence throughout.

The next chapter follows the same money one step further — not how much the system costs in total, but who specifically writes each cheque: arnona, contractors, TAMIR, and residents — and how those flows relate, or fail to relate, to one another.

Chapter 21

Who Pays: Arnona, Contractors, TAMIR and Residents

"Ask a resident who pays for waste collection and the answer will often be: I do, through arnona. That is true - but only partly. A waste system becomes politically opaque when the actor who pays, the actor who caused the cost, and the actor who can reduce it are three different people."

Chapter 20 showed that waste cost is not one number. It is a chain of costs attached to gates: collection, transfer, sorting, rejection, transport, landfill, RDF, WtE, data gaps, labour, and design failure. This chapter asks the next question: who is asked to carry those costs?

The short answer is not "the resident" and not "the municipality". Israel's waste-finance system is built from several money circuits that overlap only partially: arnona, municipal contracts, TAMIR compensation, producer treatment fees, contractor invoices, and consumer deposits. Each circuit pays for a different part of the system, and none of them gives the resident, municipality, or regulator a complete picture of the real bill.

The central rule of this chapter is simple: the person who pays is not always the actor who caused the cost, and not always the actor who can reduce it.

1. Payment is not the same as responsibility

Waste finance is often discussed as if it were only a question of budget size. In practice, the more important question is alignment. A fair system should connect three things: who creates a cost, who receives the invoice, and who has the power to reduce the cost next time.

In the current system those three points are often separated. A producer may choose a package that becomes expensive to sort; the municipality may pay for collection and disposal; the resident may cover the gap through arnona; the sorting worker may handle the difficult material; and a landfill region may receive the residual burden. None of these actors has the same capacity to change the original design decision.

That separation is why the question "who pays?" is not a narrow financial question. It is a question about responsibility.

2. Four visible money circuits

Israel's municipal waste system is financed through four main visible circuits. They are often discussed separately, but residents experience them together.

Money circuit	Who pays first	What it covers	What it does not cover
Arnona	Resident / property holder	Municipal collection and budget gaps	Not earmarked; no material or downstream detail
TAMIR payments to municipalities	TAMIR via contracted producers/importers	Partial compensation for packaging separation	Not full household cost; not all real-stream packaging
Damei tipul	Contracted producers/importers	Treatment fees by material and declared weight	Free riders; design externalities; undeclared costs
Consumer deposits	Consumer, refunded on return	Separate return circuit for deposit containers	Municipal/TAMIR gaps outside deposit stream

The practical result is that a resident may pay for waste in several ways at once: through arnona, through product prices that include producer compliance costs, and through deposits on beverage containers. These payments do not flow into a single transparent bill, and they do not finance the same parts of the system.

3. Arnona: the municipality as payer of last resort

The local authority is legally responsible for organising waste removal within its jurisdiction. In everyday budget terms, this means that the municipality is the payer of last resort. When collection costs rise, when disposal becomes more expensive, when TAMIR compensation does not cover the real household-stream cost, or when a regional capacity shock forces longer transport, the gap usually reaches the municipal budget first.

Arnona is a general property tax, not a waste bill. That matters. Waste expenditure competes with education, parks, social services, street maintenance, cultural services, and every other municipal responsibility. When the waste bill rises, the effect is not confined to the sanitation department.

This structure also creates inequality between municipalities. A wealthy municipality can absorb rising waste costs with less visible damage to other services. A municipality with a weaker tax base faces a sharper trade-off: better sanitation service, or money for other needs. The same waste-price shock therefore has different civic consequences depending on the local tax base.

Under the hood: arnona hides the waste bill

Because arnona is not earmarked, most residents do not see how much of their payment goes to collection, transfer, sorting, landfilling, street cleaning, bulky waste, or covering gaps left by producer-responsibility systems. The cost is real, but the bill is politically invisible.

4. TAMIR and damei tipul: partial responsibility inside a declared system

TAMIR operates as a financial intermediary under the Packaging Law. Producers and importers that contract with TAMIR declare the weight of packaging they place on the market and pay damei tipul, or treatment fees. TAMIR then transfers part of these funds to municipalities and pays for parts of the collection, sorting, and recycling chain for packaging.

This is producer responsibility in principle. In practice, it is producer responsibility inside a partial declared system. TAMIR pays for packaging that enters its accounting frame; it does not automatically pay for every packaging-related cost that appears in the real municipal waste stream.

The tariff gap between commercial and household streams shows the problem clearly. For PP plastic, published treatment-fee data show roughly NIS 135 per tonne for the commercial stream and NIS 642 per tonne for the household stream. Part of this reflects real differences: household collection is more dispersed, less dense, and more contaminated. But it also creates a structural incentive to prioritise the easier commercial stream over the harder household stream.

Cost or burden	Where it appears	Why TAMIR compensation may not fully cover it
Household packaging collection	Municipal budget / contractor invoice	Many small points, lower density, higher contamination, more complex logistics
Rejected material / tails	Sorting facility, then RDF / landfill / WtE route	Rejection rates and tails destinations are not systematically public and may not be priced back to product design
Free-rider packaging	Same bins and sorting facilities as compliant packaging	Producer did not pay damei tipul, but the system still handles the package
Difficult design	Sorting line, labour, downtime, tails	Fees by broad material category do not always capture actual recoverability or labour burden

5. Producer payments: when the polluter pays only partly

The polluter-pays principle is supposed to shift costs upstream: the actor placing packaging on the market should carry the cost of managing it after use. The principle weakens when three things happen at once: the declared base covers only part of the real stream, free riders avoid payment, and treatment fees do not strongly distinguish between packages that are easy and difficult to recover.

The result is not simply underfunding. It is cost shifting. A producer that complies but uses hard-to-sort packaging may still externalise part of the sorting, labour, rejection, and residual cost. A producer that does not comply at all shifts even more. The packaging still enters bins, trucks, sorting facilities, and tails, but the financing chain does not follow it back to the actor that placed it on the market.

A stronger EPR system would not only ask whether a producer paid by material weight. It would ask whether the package was designed to be captured, sorted, recycled, and documented without pushing hidden costs downstream.

6. Consumer deposits: a cleaner payment loop, but a separate one

The deposit system is financially different from arnona and TAMIR compensation. The consumer pays a deposit at purchase and can recover it by returning the container. The municipality is not the main payer in that circuit, which is one reason deposit PET performs more cleanly than much of the ordinary packaging stream.

That advantage should not be confused with integration. A successful deposit stream does not automatically solve the finance of the remaining orange-bin or mixed-bin stream. It removes some high-value, high-visibility material from municipal handling, but the material left behind may be less valuable, more contaminated, or more expensive to manage.

Deposit leakage also has a cost. When deposit-marked bottles enter the orange bin or mixed bin, they do not simply become a sorting error. They add work to collection and sorting systems that were not designed to finance that material through the deposit-return circuit.

7. Contractors and facilities: paid to perform services, not necessarily to reduce residual waste

Private contractors and facility operators are central to the money chain. Municipalities pay contractors to collect, transfer, transport, sort, or dispose of waste; TAMIR and other producer-responsibility bodies may pay for parts of specific streams; facility operators charge gate fees for what they accept.

The key question is what contracts reward. If a contract pays primarily for tonnes collected, kilometres driven, containers emptied, or tonnes accepted at a facility, the financial signal is operational throughput. It is not prevention. It is not residual reduction. It is not better design. It is not necessarily higher-quality capture. This does not mean contractors behave badly. It means they respond to the service they are paid to deliver. A system that wants less residual waste has to make residual reduction, data reporting, rejection-rate transparency, and contamination reduction part of the contract itself.

8. Shifted costs: where the invoice does not show the burden

Some waste costs are paid as invoices. Others are carried as work, risk, lost capacity, local burden, or missing data. The table below maps several common shifts.

Cost created by	Often paid by	Hidden carrier
Difficult packaging design	Municipality, TAMIR-funded system, resident via prices/arnona	Sorting workers, facility downtime, tails routes
Weak household capture	Municipality and residents through arnona	End-of-pipe regions receiving residual waste
Contamination and tails	Sorting facility, contractor tariff, municipality	Workers handling rejects; landfill/RDF/WtE destinations
Free riders	Compliant producers, municipalities, residents	The public system that still handles unpaid packaging
Deposit containers entering wrong stream	Orange-bin or mixed-waste system	Sorting line and lost deposit-circuit efficiency
Lack of downstream data	No one receives a clear invoice	Residents and municipalities unable to evaluate performance

The most important point is that hidden costs are not imaginary costs. They are costs whose bearer is not named in the formal tariff. A labour burden, a rejected fraction, a longer transport route, or a host-region impact may not appear as a separate line in the resident's bill, but each is part of what the system asks someone to carry.

9. Why residents experience the system as unfair

Residents are often addressed as if their main role is behavioural: sort better, rinse correctly, use the right bin, return deposit containers. Those actions matter. But the payment structure asks residents to do more than behave correctly. It asks them to finance gaps they did not create and often cannot see.

A resident pays through arnona for municipal collection and the general waste budget. The same resident pays indirectly through the price of goods, where producer compliance costs may be embedded. The same resident pays a deposit on certain beverage containers, even though that circuit is separate from municipal finance. Meanwhile, the resident has no single document showing how these payments interact, what part of the bill reflects preventable design failure, or whether money paid into one circuit reduces pressure on another.

This is why transparency is not a luxury. A system that asks residents to pay at several points has to show what each payment actually finances.

10. What the reform must make visible

The 2026 reform debate matters for this chapter because any future regulator will have to decide not only how much waste services cost, but who should pay for which part of the chain. A full-cost tariff system may solve some problems if it makes sorting, transport, disposal, and alternative treatment costs explicit. It may reproduce other problems if it prices only service delivery while leaving design failure, labour burden, rejection rates, and geographic externalities invisible.

A serious finance reform should therefore make at least six things public: municipal waste expenditure by function, TAMIR or successor-body compensation by stream, rejection and tails rates by facility, producer compliance and free-rider enforcement, deposit/EPR leakage, and labour-related reporting for sorting facilities.

The goal is not only to know who paid last year. The goal is to align future payment with the actor best able to prevent the next cost.

11. Financial questions for a civic data request

A resident group, NGO, councillor or municipal official can turn the abstract question "who pays?" into concrete requests.

How much did the municipality spend on waste management last year, and how is that expenditure divided between collection, transfer, sorting, disposal, bulky waste, street cleaning, and administration?

How much did the municipality receive from TAMIR or any successor packaging-responsibility body, by stream and by year?

What is the estimated gap between TAMIR compensation and the municipality's real cost for organising household packaging collection?

Which contracts pay by tonne, by route, by container, by service availability, or by performance outcome?

What costs are linked to tails or rejected material from sorting facilities serving the municipality?

How much of the municipality's waste cost is connected to landfill levies, gate fees, long-distance transport, or capacity shocks?

Are labour costs, safety requirements, PPE, subcontracting, and incident reporting included in sorting or collection contracts?

What data exists on deposit containers found in orange-bin or mixed-waste streams, and who pays for handling them when they leak out of the deposit circuit?

Conclusion

The answer to "who pays for waste?" is not one actor. Residents pay through arnona, product prices, and deposits. Municipalities pay contractors and facilities, and often cover the gap between formal producer responsibility and real service cost. Producers pay damei tipul only inside the declared system, while free riders and design externalities shift costs outward. Contractors and facilities are paid according to contracts that may reward throughput more clearly than prevention. Workers, host regions, and future infrastructure carry costs that may not appear as invoices at all.

A fair waste-finance system should make the actor who can prevent a cost responsible for paying it, rather than pushing that cost downstream to residents, workers, municipalities, or landfill regions.

The next chapter turns from financing flows to contract design: how tenders, KPIs and PAYT principles can shift incentives from tonnes collected to residual waste reduced.

Chapter 22. Contracts, KPIs and PAYT: From Tonnes to Residual Waste Reduced

"The tender must define the expected outputs and not the processes." This phrase from the Neaman Institute report sounds obvious. But applied to Israeli waste collection contracts, it describes a gap that explains a great deal.

Chapters 19 and 20 examined the financial structure of the waste system – how much it costs and who pays. This chapter concludes Part V by asking the question that follows logically from the two previous: if money exists in the system (albeit chronically insufficient), how exactly does it "speak" to those making operational decisions? What do existing contracts actually incentivise – and what do they not?

The answer lies in how KPIs (key performance indicators) are structured in contracts between municipalities and contractors – and why in their current form they measure primarily the quantity of waste collected, not the quality of how it is handled.

1. The Problem: Contracts Pay for Tonnes, Not for Outcomes

Most Israeli municipal waste collection contracts are structured as volume-based service contracts: the contractor receives payment for the number of trips, tonnes collected, or a combination of both.

This sounds reasonable – but creates a structural incentive towards quantity rather than quality.

◆What "Pay-per-Tonne" Contracts Incentivise – and What They Do Not

Incentivise: as many trips as possible, trucks as full as possible. The contractor's interest is to collect a large volume.

Do not incentivise: reducing stream contamination – the contractor is indifferent to how clean the contents of the orange bin are. A clean bottle and a dirty plastic bag count equally in the tonnage.

Do not incentivise: reducing total waste volume – the contractor earns less if there is less waste.

Direct consequence: the contractor is neutral or hostile towards programmes that reduce waste generation or improve source-separation quality. Its financial model is built on volume.

The State Comptroller documented in 2022: among the local authorities examined, performance measurement against waste-reduction targets was limited or absent; contracts focused on collection frequency and vehicle availability rather than environmental outcomes.

2. The Alternative: Outcome-Based Contracts

The Neaman Institute, analysing the feasibility of PPP contracts for large waste treatment facilities in Israel, formulated a key principle: "The tender must define the expected outputs and not the processes."^[70]

Applied to collection and removal contracts, this means shifting from "how much was collected" to "how much residual waste was generated," "what share of sorted material comes from the total stream," or "what is the quality of the stream entering the sorting facility."

KPI "per tonne"	KPI "per outcome"
Number of trips per collection point	Percentage of correctly sorted material from the collection point's stream
Tonnage of mixed stream collected	Year-on-year reduction in mixed stream (residual waste) tonnage
Equipment availability (% working days)	Rejection rate of orange bin material at the sorting facility
Response time to 106 complaint	Stream contamination (% non-sortable items in dedicated bins)
Technical compliance with routes	Citizens actually using source-separation bins (measured by sampling)

◆Under the Hood: International Context – How Other Countries Do This

Germany and the Netherlands require contracts with explicitly stated targets for improving recycling performance – the contractor faces penalties if performance falls below the threshold.

The UK is transitioning to outcome-based performance contracts in a number of municipalities: a minimum recycling percentage is enshrined in the contract, not just in municipal intentions.

Taipei implemented PAYT (see below) precisely through changing incentives at the consumer level – and achieved a 35% reduction in waste volume and a 2.6-fold increase in recycling.

OECD directly recommends: municipal waste contracts should include targets for reducing residual waste and increasing recycling rates, not only volume and service parameters.

3. PAYT – "Pay As You Throw": The Principle and Its Limitations

What PAYT Is

PAYT (Pay As You Throw) is a system under which residents and organisations pay for waste management in proportion to how much they actually discard: by weight, volume, or number of special bags or containers.

The logic is simple: if you pay for each unit discarded, you receive a direct financial incentive to generate less waste and sort more (since recycling is usually cheaper or free).

Three main PAYT models:

Model 1: Special bags. Residents purchase official numbered bags for mixed waste; sortable materials are deposited in standard containers for free. Popular in South Korea and Taiwan.

Model 2: Chipped containers with scales. Each household has an RFID-tagged container; every time it is emptied, weight or number of lifts is automatically measured. The bill reflects actual use. Used in the Netherlands and parts of Belgium.

Model 3: Partial pricing. A base volume of collection is included in a fixed charge; anything above that is charged additionally. A compromise that reduces the risk of social resistance.

Why PAYT Works – and What It Requires

Taiwan (Taipei) – the best-documented example: the introduction of special bags led to a 35% reduction in waste volume and a 2.6-fold increase in recycling. Key enabling conditions: clear and simple rules, parallel coverage of all source-separation access points, and neutralisation of increased costs for low-income households (who statistically generate less waste).

◆The Main Risk of PAYT: Illegal Dumping

PAYT creates a direct incentive against paying – and therefore potentially against legal disposal. Illegal dumping ("fly-tipping") and waste burning have been documented to increase in a number of municipalities following PAYT introduction without a sufficient monitoring and enforcement system.

The Knesset Research and Information Center's document on the economic costs of waste burning explicitly warns: the rising cost of landfilling (Chapter 19) already creates an incentive for illegal burning. PAYT may intensify this effect if not accompanied by a proportionate enforcement system.

In Israel's context, where waste management enforcement is already fragmented (Chapter 2), introducing PAYT without first strengthening the monitoring and penalty system is potentially counterproductive.

4. PAYT in Israel: An Honest Assessment of Prospects

At the time of writing, no systematic PAYT exists in Israeli municipalities. Arnona is a non-earmarked tax based on floor area, not on volume discarded. This means there is no direct financial incentive for the resident towards waste reduction under the current system.

Can PAYT be introduced in Israel? Technically – yes. Organisationally and politically – it is a more complex question.

Factor	In favour of PAYT	Against / requires resolution
Legal basis	Municipalities have powers to set charges for municipal services	Requires changing an arrangement traditionally embedded in arnona – politically sensitive
Monitoring infrastructure	RFID systems for containers are already being tested (Chapters 4, 5)	No national household-level waste measurement system
Social justice	PAYT is documented to be fairer for lower-income residents (less waste = less paid)	Requires protective mechanisms for households facing genuine hardship; perception as a new tax
Illegal dumping	Manageable with sufficient monitoring	Current enforcement of illegal burning and dumping is insufficient for PAYT (Chapter 7)
2026 reform	Creating a regulator potentially simplifies PAYT standardisation	Until the reform is adopted, no national platform for a pilot

Conclusion: PAYT is not an imminent but a potentially achievable instrument for Israel given certain preconditions. The priority for the next 3–5 years is not full-scale PAYT but a more realistic step: transitioning from volume-based KPIs to outcome-based KPIs in existing contracts.

5. What Can Change Right Now: A Realistic Plan

There is no need to wait for the 2026 reform or the introduction of PAYT to improve incentives in the system. A number of changes are possible at the level of an individual municipal contract right now.

Four elements of an improved waste collection contract:

1. A target indicator for residual waste reduction.

Example: "The contractor undertakes to contribute to a 5% year-on-year reduction in residual mixed waste tonnage in the serviced area." This is not 100% within the contractor's control – but its presence in the contract changes the incentive.

2. An incoming stream quality indicator for the orange bin.

Example: "The rejection rate at the sorting facility for this district's stream shall not exceed X% – if exceeded, a payment adjustment is triggered or additional resident education obligations apply."

3. Data as part of the service.

The contractor must provide quarterly data on tonnage by stream, observed violations (what ended up in the wrong bin), and statistics on 106 complaints. Without data, meaningful oversight is impossible.

4. Diversified payment: fixed + variable.

The base portion of payment covers service provision (fixed). The variable portion covers achieving quality targets and residual waste reduction (bonus or penalty). This does not abolish volume pricing but adds a results-based component to it.

6. How Citizens and NGOs Can Influence Contracts

A contract between a municipality and a waste collection contractor is a public tender. In Israel, tender specifications are publicly published, and adopted contracts are requestable under the Freedom of Information Law (Chapter 25).

This opens specific leverage points for civic organisations.

Chapter 23. Organics and Food Waste: From a Methane Problem to a Local Resource

"Organics is the largest fraction in the waste bin. It is also the most harmful when landfilled. And the most valuable when handled correctly. The gap between what exists and what is possible is most visible here."

Food and plant waste accounts for approximately 40–45% of all solid household waste in Israel by weight. It is the single largest fraction – more than plastic, cardboard and glass combined. And it is precisely this fraction that the waste management system handles worst of all.

This is not accidental. Organics is a chemically active material. Under anaerobic landfill conditions it decomposes releasing methane – a greenhouse gas approximately 28 times more potent than CO₂. It is estimated that proper treatment of Israel's organic waste could reduce national greenhouse gas emissions by approximately 5.5%.

This chapter describes the problem, the available technologies, and where workable solutions are beginning to emerge in Israel.

1. The Problem: Organics in the Mixed Bin

The vast majority of Israeli households discard organics in the mixed (green) bin. This organic material makes up a significant portion of the 60% of mixed bin content that, per MoEP's 2025 scan (Chapter 8), could technically be handled differently.

◆Why Organics in the Mixed Bin Is a Double Loss

First loss: organics contaminate all other fractions in the mixed bin – plastic, cardboard and metal in the same container become wet and dirty, reducing their recyclability.

Second loss: organics landfilled decompose anaerobically and release methane. Most Israeli landfills are not equipped with full landfill gas capture systems, meaning a significant proportion of methane enters the atmosphere. The alternative – anaerobic digestion or composting – turns organics into a resource (biogas, fertiliser) rather than an environmental threat. This is not a distant prospect but a technology operating in dozens of countries and beginning to appear in Israel.

2. Technologies: What Science Can Do

Anaerobic Digestion (עיכול אנאירובי)

Anaerobic digestion is the decomposition of organic material without oxygen, under the action of microorganisms. It is a natural process used in industrial installations called bioreactors or digesters.

What goes in and what comes out:

INPUT: household organic waste, food waste, garden waste (*gezem*), sewage sludge, manure.

PROCESS: in a sealed bioreactor at controlled temperature and humidity, microorganisms decompose organics over several weeks.

OUTPUT – two valuable products:

→ **Biogas (ביוגז):** a mixture of methane (~60%) and CO₂. Used directly as fuel or for electricity and heat production.

→ **Digestate (חומר מעוכל):** nutrient-rich organic matter. When input stream quality is high (source-separated at origin) – a valuable agricultural fertiliser. From a mixed stream – quality is lower and market demand is limited.

◆Under the Hood: Why Input Stream Quality Determines Everything

MoEP explicitly prefers anaerobic digestion over simple composting – as a more climate-valuable process (captures methane rather than releasing it) and as a source of renewable energy.

But there is a fundamental constraint: if organics have been mixed with plastic, glass and metal, the digestate will contain their fragments. Such a product cannot be used as agricultural fertiliser and has no market. In Europe it is designated CLO (Compost-Like Output) – material that has technically undergone processing but is ecologically unusable.

Conclusion: anaerobic digestion works well only with source-separated organics (brown bin). From a mixed stream, even after mechanical sorting, output quality is critically reduced.

This is why Israeli MBT (mechanical-biological treatment) installations processing the mixed stream produce low-quality digestate – not due to a technology failure, but due to input material quality.

Composting (קומפוסטציה)

Composting is an aerobic (oxygen-present) decomposition process for organics. The result is compost – a nutrient-rich organic fertiliser. Technologically simpler and cheaper than anaerobic digestion, but produces no energy.

Characteristic	Anaerobic digestion vs Composting
Product 1	Biogas (energy) None
Product 2	Digestate (fertiliser) Compost (fertiliser)
Input stream quality requirement	High (source separation) Medium
Installation cost	High Medium
Land area	Compact (closed reactor) Large (open rows)
Odour emissions	Low (closed process) Significant if unmanaged
MoEP Israel position	Priority Acceptable as supplement

3. What Exists in Israel Now: An Honest Picture

The situation in Israel in 2025–2026 is a combination of considerable potential, limited existing infrastructure, and two new major facilities under construction.

Current state of organic infrastructure in Israel:

OPERATIONAL CAPACITY (~530,000 tonnes/year):

→ **Tovlan** (Beka'at Hayarden): the largest facility, receiving ~90% of all organics treated in the country. Operates on primitive open composting rows. Product: CLO, not standard compost. Facility under MoEP investigation.

→ **Dudaim**: second operational facility; composting.

UNDER CONSTRUCTION (opening ~2026–2027):

→ **Zero Waste Shafdan, Rishon LeZion**: anaerobic digestion, 400,000 tonnes/year. Largest facility of this type in the country.

→ **Dia (Tzealim, Eshkol Regional Council)**: GES/YSB, 250 million NIS, 200,000 tonnes/year. 25-year concession. Combination: anaerobic digestion + composting of digestate in sealed concrete chambers.

THE GAP:

After the new facilities open, total capacity will reach ~1.2 million tonnes/year.

Israel's actual annual organic waste generation: ~3 million tonnes/year.

Capacity deficit for meeting 2030 targets: 7–12 additional facilities still required.

◆A Disturbing Precedent: What Happened to Organics During the Veridis Crisis

In 2022 MoEP imposed restrictions on Veridis – the largest organic processing operator. Veridis halted intake of organics.

Immediate result: capacity for organic waste treatment in the country virtually disappeared. Sorting facilities were forced to send the organic fraction directly to landfill, paying the full landfill levy rate.

This exposed a systemic problem: dependence on one or two facilities with no reserve capacity creates fragility across the entire system. A single major player and a single major facility is not a system – it is a single point of failure.

4. Small-Scale Formats: What Works Now

Not all solutions require large infrastructure. A number of Israeli municipalities and clusters have implemented small-scale organic formats operating close to the point of waste generation.

Kibbutz Yagur (יגור): a small anaerobic digester converts kitchen food waste into biogas for water heating and fertiliser for kitchen gardens. This is the first documented example of "on-site" community-scale urban organic waste digestion with biogas production in Israel.

Zichron Ya'akov (זכרון יעקב): food waste is transported to an anaerobic facility in the Hefer Valley; garden waste goes to a separate composting facility. A local facility for composting garden waste and high-quality organic waste at source is planned.^[52]

These examples show: small scale, orientation towards source separation, and proximity to the point of waste generation are the conditions under which organics becomes a resource rather than continuing to be landfilled. This is the principle described in Chapter 24 on circularity hubs: local infrastructure close to the resident.

5. What Is Needed for This to Work Systemically

Academic literature and MoEP documents agree on one point: the key condition is source separation of organics (separation in the brown bin at the resident level), not mechanical extraction from the mixed stream. This requires the greatest investment – not in technologies but in first-mile infrastructure (Chapter 4): convenient brown bins in every courtyard, frequent collection (organics decompose quickly), and clear labelling.

Note: connection to other chapters

MoEP's 2025 data (6% of packaging in dedicated bins) shows that the first-mile problem (Chapters 4 and 8) applies to organics with the same force as to plastic. Expanding brown bin coverage is an infrastructure project, not just an educational campaign. The 2026 reform (Chapter 16) potentially opens funding through a new regulator for accelerating the construction of organic treatment capacity – currently financed primarily through the Cleanliness Fund, whose future is uncertain under the reform.

Chapter 24

Clothing, Toys, Furniture: Ethical Waste and Early-Stage Rescue

A wardrobe left beside a skip is not a waste problem. It's a problem of a system with no mechanism to intercept an item before it becomes waste.

Among every stream examined in this book, clothing, toys, and furniture occupy a distinct place. They don't “become” waste the way packaging does after a single use. In the great majority of cases, when someone actually brings one to a container, it still works, is still intact, is still wanted by somebody. This is ethical waste: a category where an item is discarded not because its useful life is exhausted, but because no infrastructure exists to catch it before that point.

1. The actual scale

The global textile figure is well documented: the fashion industry produces roughly 92 million tonnes of textile waste annually, clothing production has doubled since 2000, the average number of times a garment gets worn has roughly halved over the same period, and less than 1 percent of textile material is recycled back into new clothing (Ellen MacArthur Foundation).^[26] In Israel specifically, textile waste is not tracked as its own category in national statistics — itself a sign of how invisible this stream remains to policy, and consistent with what this book already found when it looked for a national textile-tonnage estimate and couldn't locate one (Ch.1 §5).

Furniture and bulky items — pesolet gushit under the Cleanliness Law — sit in a separate legal category. The municipality is obliged to organise their removal, but collection frequency and system vary widely between authorities. Per State Comptroller data, bulky items abandoned near containers are among the most frequent complaints logged on the national 106 municipal-services line — a direct symptom of residents having no convenient channel for transferring these items properly, the same first-mile gap this book has traced for every other material (Ch.4).

2. What actually happens to an item: four paths

Path	What it means
Early-stage rescue — the ideal case	The item is intercepted while still functional: given to someone directly, dropped at a themed organisation or hub, sold second-hand. Lifecycle extended without any processing at all.
Repair and restoration	The item is broken but repairable. A repair café, a cobbler, a seamstress, an electrician — with the service actually accessible, the item goes back into use.
Textile recycling	The item is worn out and can't be passed on. Conversion into wiping material, wadding, or filling is technically possible, but infrastructure is limited and volumes small.
Landfill or incineration — the failure case	The item ends up in a mixed container or beside a skip. For furniture, that's landfill. For clothing, it's landfill or, if the resident happens to have one nearby, a charitable organisation's container.

Charitable containers sit outside the official system entirely

Clothing containers run by organisations such as Yad Sarah, Second Hand, and Sela are widespread in Israel, but they are not part of the TAMIR system and are not counted in MoEP's or TAMIR's official recycling statistics. That means the real scale of textile rescue in Israel is structurally undercounted — not because the activity isn't happening, but because nothing collects the data. At the same time, some of this material enters the same export stream this book already examined in detail (Ch.1 §5): garments tracked by GPS from Israeli

donation bins turned up in Togo, Ghana, and Burkina Faso, not resold locally or recycled into new clothing — a documented reminder that a donation container is not, by itself, proof that an item stays in useful circulation.

3. Early-stage rescue: three levels

Early-stage rescue means intervening before an item ever reaches a container — fundamentally different from recycling, which only ever operates after the fact. Here the concern is interception while the item still has value, still needs no processing, and can move directly to whoever needs it next.

- Level 1: person to person. Apps and platforms — Yad2, Facebook Marketplace, local neighbourhood chats. Minimal infrastructure, maximum speed, but it depends entirely on the donor's own time and initiative.
- Level 2: organised drop-off points. Second-hand shops, charitable organisations, and the circularity hubs examined in the previous chapter (Ch.25) — this requires a physical point within walking distance and a genuinely regular schedule.
- Level 3: corporate buyback and extended life. A producer taking back its own product — Keter Group's collection of used plastic garden furniture for reprocessing into new products is a working Israeli example (Ch.11 §4 examined Keter's recycled-content programme in more detail). Still rare in Israel overall.

4. Re-Haifa Lab: what the experience actually taught

Re-Haifa Lab has operated since 2018 as an organisation specialising in early-stage rescue of textiles and furniture, repair, and community swap events (Ch.25 §5 covers its role as a circularity-hub model in more detail). Several years of practical work produced lessons that generalise well beyond any one organisation.

- Textiles and small items are the easiest entry point. Moving a sofa requires transport; accepting a bag of clothing requires two volunteers and five minutes. Starting with textiles is almost always the right first move.
- Furniture needs real logistics. Without an agreement with the municipality or a partner organisation on transport, it's simply not workable. The most realistic path is coordinating around scheduled bulky-waste collection days, so the hub can intercept material before the truck does.
- Repair is the highest-value service and the hardest to scale. A repair café works when it has regular, skilled practitioners showing up — enthusiastic volunteers are necessary but not sufficient on their own.
- Data matters. Without tracking item counts, categories, and each item's eventual fate, there's no way to demonstrate real impact to a municipality or a funder. Re-Haifa has maintained this tracking since 2022.
- Partnering with other NGOs multiplies reach. Passing along what the hub itself can't process — to youth clubs, care homes, social services — extends impact without adding cost.

5. What doesn't work: an honest accounting

It would be dishonest to examine this only from the angle of success stories. Several persistent barriers meaningfully limit how much early-stage rescue actually happens in Israel today.

Barrier	Why it exists, and what would help
No data in official statistics	The textile stream isn't regulated by the Packaging Law and isn't tracked by TAMIR (Ch.1 §5). Without data, there's no basis for policy — a dedicated category in MoEP's own national statistics is the minimum needed to change that.
No statutory producer obligation	The EU's revised Waste Framework Directive introduced mandatory textile EPR schemes across all member states, entering into force in October 2025, with a 30-month window to establish national schemes. ^[28] Israel has no equivalent, and

Barrier	Why it exists, and what would help
	no producer currently bears responsibility for what happens to a garment after sale.
Low quality of the rescued stream when collection is disorganised	An item left beside a container in an open bag, exposed to rain, loses value fast. Closed, clearly marked drop-off points hold their value far longer.
Furniture needs transport and storage	For an NGO without municipal support, moving and storing bulky items is often simply an unsolvable logistical problem on its own.
Export of second-hand clothing is not, by itself, a solution	Some “rescued” material from Israel enters export streams to countries with weaker regulation — not closing the loop, but relocating the same problem (Ch.1 §5 already traced exactly this pattern with GPS-tracked garments).

Practical questions for an activist

- Are there organised clothing and furniture drop-off points with a known schedule in your city, or only containers in inconvenient, unstaffed locations?
- When is bulky-waste collection scheduled in your neighbourhood, and could that become a real interception opportunity rather than a straight path to landfill?
- Does MoEP publish textile-waste data separately, or is it folded invisibly into an “other waste” category?
- Is there a repair café or workshop operating on a genuinely regular schedule in your city?

Conclusion

Clothing, toys, and furniture are ethical waste: they become rubbish primarily for lack of interception infrastructure, not because their useful life has run out. Early-stage rescue deserves priority over recycling for a simple reason — catching an item before it reaches a container is cheaper, cleaner, and more effective than processing it afterward. Israel's textile stream remains untracked in official statistics, which is a form of structural invisibility with a direct policy consequence: what isn't measured doesn't get addressed. Charitable containers, real and valuable as they are, sit outside the formal EPR system entirely, and their contribution never reaches the official figures this book has spent so many chapters teaching a reader to read carefully.

Re-Haifa's experience shows a working early-stage-rescue model at city scale, with documented, transferable lessons about logistics, data, and the limits of scaling repair. A systemic fix would need three things together: a separate, tracked textile-waste category in national statistics; statutory producer responsibility, following the direction the EU has already set with its 2025 textile EPR rules; and municipal support for organising drop-off points and the logistics — especially for furniture — that no NGO can solve entirely on its own.

The next chapter shows what happens when these individual streams stop being scattered across separate initiatives and come together in one place: a circularity hub, and the practical steps for building one (Ch.25).

Chapter 25

Before It Becomes Waste: Circularity Hubs as Local Prevention Infrastructure

Circularity is not a personal virtue. It is infrastructure. And the best moment to manage waste is before it becomes waste.

There is a point where the formal waste system this book has spent twenty-four chapters describing usually begins: a bag goes in a bin, a truck arrives, a contractor logs a route, and the material enters the chain of transfer stations, sorting facilities, landfills, recyclers, or energy-recovery plants this book has traced gate by gate. By that point, though, a great many better options have already quietly closed. A broken chair becomes bulky waste instead of a repair job — the same chair this book first met beside a bin enclosure in Chapter 1, with nowhere defined to go. Extra food becomes organic waste instead of a shared resource. Old clothes become textile waste instead of local reuse. Kitchen scraps become a methane problem instead of compost. A repairable appliance enters the residual stream simply because no affordable repair option existed nearby.

A circularity hub is the missing layer before any of that happens — a physical place and a set of services standing between everyday household life and the formal collection system this book has spent its first twenty-four chapters mapping. Its purpose isn't asking residents to be more conscious in the abstract. It's making prevention, repair, reuse, sharing, composting, and correct downstream sorting practically available, in the same way this book has argued convenience and access determine outcomes at every other gate (Ch.4).

1. Why personal awareness is not enough

Picture a resident who genuinely wants to do the right thing before throwing something away — repair a broken toaster, pass on a child's outgrown toy, share food that would otherwise spoil, compost kitchen scraps, move a usable table to someone who needs one. In most neighbourhoods, none of these actions is actually simple. There's often no nearby repair service, no organised exchange point, no food-sharing channel, no local composting option, no affordable way to move something bulky.

The toaster, the toy, the food, the table, and the scraps enter the waste stream not because the resident failed morally, but because the necessary service layer was never built. This is a problem of access, proximity, and infrastructure — not of awareness. The same logic Chapter 4 applied to first-mile packaging access applies here to an even wider set of materials: a resident knows food waste shouldn't go in the grey bin, but if the nearest composting point is a fifteen-minute walk through a park, the organics go into mixed waste anyway. This book keeps arriving at the same conclusion from different directions: we demand behaviour from people for whom we haven't built the conditions to make that behaviour easy. A circularity hub is what building those conditions actually looks like.

What the research base shows

Renee Lertzman's Environmental Melancholia (2015) documents that most people genuinely care about environmental outcomes but experience a kind of powerlessness in the face of the problem's scale — and that “be more conscious” messaging tends to intensify that feeling rather than resolve it. The OECD's 2020 work on circular cities frames the practical alternative as a “proximity and convenience” principle: the closer and easier a service is, the higher its uptake — a claim treated as obvious for retail and transport, and inexplicably ignored for waste management.

2. What a circularity hub is: seven functions in one place

A circularity hub — a neighbourhood zero-waste centre — is a multifunctional place within walking distance that brings several services together in one location. The key word is multifunctional. A single repair café or a single

drop-off point is good and insufficient on its own. A hub works as an ecosystem of services that reinforce each other.

Function	What it does
1. Early-stage rescue	Residents bring items they're discarding but feel bad about — clothing, furniture, appliances, books, toys. The hub checks, sorts, and prepares them for reuse. What's suitable stays in circulation; what isn't goes into the correct disposal stream, not onto the pavement beside a container (Ch.1 §9's chair, given somewhere to go).
2. Repair and restoration	A repair café, workshop, or partnership with local craftspeople. Fixing a toaster, a bicycle, a garment doesn't only reduce the waste stream — it builds skills and creates jobs.
3. Sharing and a library of things	A drill, a stepladder, a child's bicycle, a tent — items needed only occasionally. Borrow and return. Reduces both new-goods consumption and the burden of private storage.
4. Organics: a controlled stream	A drop-off point for organic waste, with on-site composting or transfer to anaerobic digestion — critical for the buildings without a brown bin that Chapter 23 already identified as the norm, not the exception, in Israel.
5. Verified secondary material streams	Acceptance of clean, sorted secondary materials — HDPE, PP, paper, metal — with direct transfer to a recycler, complementing rather than routing around the standard TAMIR circuit. The hub knows the downstream fate of what passes through it, in a system where Chapter 10 showed that knowledge is otherwise rare.
6. Education and information	Workshops and demonstrations, not moral instruction. Not 'sort better,' but 'here is what happens to your bottle' — the same civic-transparency instinct this book has argued for since Chapter 8.
7. MRV — monitoring, reporting, verification	The hub tracks what passes through it: tonnage by fraction, items repaired, CO ₂ avoided. This is what makes it a measurable partner for a municipality, rather than a well-meaning but invisible activity.

None of these seven functions is individually revolutionary. What's distinctive is combining them in one place, with permanent staff and a real accounting system — which is exactly what the international examples below show cities doing piece by piece, rarely all at once.

3. International analogues: a family of models, not one template

The circularity hub proposed here isn't invented from nothing. Its components already exist in different cities under different names — reuse centres, repair hubs, municipal second-hand shops, reuse shops at recycling centres, community waste hubs, circular-neighbourhood initiatives. What's less common is their integration into one neighbourhood-level prevention infrastructure with clear boundaries, municipal partnership, real data, and measurable outcomes. The international experience is best read as a family of models rather than one template to copy: each case demonstrates one function a hub needs, and the task in front of Israel is combining those functions in a form that fits local municipal structure, housing types, climate, and language.

Tallinn: municipal reuse and repair, at city scale

Tallinn is among the closest existing examples to the municipal logic proposed here. Its first circular economy centre, at Mustjõe 40 in the Lilleküla district, opened in October 2024^[22] — a reuse room for free item exchange, a furniture and wood-repair workshop, and a sewing workshop for clothing repair, run out of what used to be an ordinary waste station. A second, considerably larger centre opened in the Lasnamäe district in April 2026: an €8 million facility serving roughly 120,000 people — about a quarter of Tallinn's entire population — combining a waste station, repair workshops, a reuse room, a circular shop, classrooms, a café, and a 2.5-hectare public park built on a former illegal dumping site.^[22] The city's stated plan is one such centre in each of its eight districts, part of a formal Waste Management Plan running through 2026 and a longer “Tallinn 2035” strategy, with explicit national

targets attached: 66 percent of municipal waste collected separately by 2026, 70 percent by 2030, and landfilling held under 5 percent by 2026.^[22]

The underlying pilot that made this possible was itself modest and well-documented: a roughly €421,000 EU-funded project, run with Tallinn University of Technology, the Stockholm Environment Institute's Tallinn centre, and the NGO Uuskasutuskus (Estonia's national reuse centre), which tested reuse rooms and repair workshops at individual waste stations before the city committed to the larger, permanent centres.^[22] The lesson for Israel isn't that every neighbourhood needs an €8 million building. It's that repair and reuse can be treated as ordinary urban waste infrastructure, financed and planned the same way a transfer station is — and that starting small, at a single existing site, with a real partner organisation, is exactly how a city this size actually got there.

Munich Halle 2: a reuse centre run by the waste utility itself

Munich's Halle 2 is a municipal second-hand shop operated as a core part of the waste-prevention programme of the Munich Waste Management Corporation (AWM) — the city's own waste utility, not a separate NGO. Items collected at the city's twelve recycling centres are sorted, tested, repaired where needed, priced, and sold on-site or online, alongside repair cafés, pop-up stores, and public auctions. The scale is concrete and independently documented: roughly 1,000 tonnes of goods reused or repaired each year, about 4,300 customers a month, roughly 200,000 items sold annually, average monthly revenue near €60,000, and a permanent staff of fourteen^[62] — six in sales, eight in stock and merchandise management — alongside named partner organisations handling specific categories: a social enterprise repairing bicycles for youth employment training, a certified electronics recycler, a tool-lending library.

What makes Halle 2 useful as a model isn't that it's a shop. It's that it functions as the public interface between the municipal waste utility and residents — a place where material is rescued, skills are taught, and the city visibly demonstrates that prevention is part of its waste policy, not an addition to it. The renovation budget of roughly €1 million came directly from the same household waste fee that funds ordinary collection^[62] — prevention treated as a line item in the existing waste budget, not a grant application. For Israel, the institutional lesson matters more than the retail format: a circularity hub needs a recognised relationship with the municipality or its waste contractor — access to material, permission to intercept it before disposal, and a financial model that treats prevention as a paid-for service, not a favour.

Reuse shops at recycling centres: the UK model

In the UK, reuse shops attached to household waste recycling centres are a well-established, practical format. WRAP publishes formal guidance for local authorities, waste companies, and third-sector reuse partners on setting up or improving these services; Hertfordshire and Bristol Waste both run shops where quality household items dropped at a recycling centre are sold on for reuse rather than processed as waste.^[88]

This model solves one specific problem well: bulky and household items often become waste not because they've lost value, but because the resident has no convenient route for passing them on. A reuse shop at a recycling centre catches that material at the very last moment before disposal. Its limitation is built into its position in the chain — it sits at the edge of the system, after a resident has already decided to discard something. A neighbourhood circularity hub should move that intervention earlier: repair before disposal, sharing before purchase, food redistribution before waste, bulky-item rescue before the object ever reaches the curb.

Bratislava KOLO: reuse as a civic practice, not just material diversion

Bratislava's KOLO reuse centre grew out of informal garage-sale events into a permanent, city-supported reuse centre, which the city itself describes as a reliable scheme for collecting unwanted but still functional items, reducing waste generation, and creating an educational and creative space at the same time.^[21]

This example matters because it treats reuse as a civic practice, not only a diversion statistic. A hub that works isn't simply a storage point for old objects — it needs curation, repair, education, and public trust, so residents understand exactly what can be brought, what can't, and what happens to it afterward. Without clear rules, a reuse initiative quickly becomes what might be called a moral landfill: people bring everything they feel too guilty to throw away, whether or not it has any real value. A circularity hub needs explicit operational boundaries — reusable items,

repairable objects, clean textiles, defined food-sharing channels, documented downstream partners — not “anything that might be useful to someone, someday.”

C40: organics as the missing anchor

C40's community-centric waste-hub framework is especially relevant for the organics function this book examined on its own terms in Chapter 23. It argues that local waste hubs deliver the most benefit — to residents, the city, and the environment together — when organic waste is collected and treated close to where it's generated, producing compost, biogas, or another local benefit instead of landfill methane.^[4] C40 frames waste and materials management more broadly as a city-level climate, equity, and infrastructure question, not simply a sanitation service.

Organics is often the missing anchor in a hub model built mainly around reuse and repair. Reuse and repair reduce item waste; food-sharing reduces edible-food loss; small-scale composting or digestion reduces methane and creates a visible local resource all at once. A hub that includes organics stops being only a reuse centre — it becomes a prevention node at exactly the point where household routines, food culture, garden waste, and municipal logistics intersect. Chapter 23 already showed why local design matters so much here: climate, odour, collection frequency, and contamination make organics unforgiving of symbolic half-measures. A hub can't simply “add composting” as a gesture — it has to define scale, input rules, hygiene, responsibility, and where the output actually goes, with real municipal permission behind it.

Circular neighbourhoods: the neighbourhood as the unit of infrastructure

The Ellen MacArthur Foundation's circular-neighbourhood framing is useful less as a specific programme than as a conceptual bridge. It describes the neighbourhood as the scale at which circular practices — reuse, sharing, repair, regeneration — can be embedded into everyday life using existing spaces and social networks, rather than depending only on large industrial systems.^[26]

This is close to the central claim of this chapter. Waste prevention doesn't happen only in national legislation, producer-responsibility schemes, or large sorting facilities — all subjects this book has already covered in detail. It also happens where people actually live: in courtyards, community centres, schools, markets, housing-block WhatsApp groups, repair workshops, and local food-sharing networks. The neighbourhood isn't a soft add-on to the formal waste system. It's the scale at which prevention becomes something a resident can actually do.

What these examples prove — and what they don't

Together, these cases prove the components of a circularity hub are realistic, not utopian. Cities already run reuse centres, repair workshops, municipal second-hand shops, recycling-centre reuse points, community organics programmes, and circular-neighbourhood initiatives — real, funded, operating today. But the examples also show fragmentation as the default state: one city has a strong reuse shop and weak organics; another has repair workshops but no real measurement; a third composts well but has no bulky-item rescue at all. Many depend on grant funding or social-enterprise energy rather than being written into ordinary waste contracts and KPIs, the same instability Chapter 22 already flagged as a general problem for anything funded outside a stable contract.

That's the gap a circularity hub is meant to close. It isn't a shop, a repair café, a compost corner, a charity drop-off, or a recycling station on its own. It's a synthesis of these already-proven elements into one local prevention infrastructure — repair and reuse before disposal, food-sharing before food waste, organics separation before landfill methane, bulky-item rescue before curbside dumping, defined downstream routes for what can't be reused, education through practice rather than slogans, MRV that reports accepted, reused, repaired, rejected, and transferred material separately, and a municipal partnership specific enough that the hub never becomes a quiet substitute for public responsibility. The lesson from the international record isn't that Israel should copy Tallinn, Munich, Bratislava, or the UK model directly. It's that circularity hubs become possible exactly when prevention is treated as infrastructure, with the same seriousness this book has argued should apply to every other gate in the system.

4. Jobs and the economics of a hub

One of the most underused arguments for circularity hubs is employment. RREUSE, the European network of social enterprises working in reuse and repair, reports that its member organisations create on average 70 jobs per 1,000 tonnes of material collected for reuse^[68] — a figure that ranges from roughly 20 to 140 depending on the material handled, from textiles and furniture to electronics and toys. A joint report from Zero Waste Europe and RREUSE puts the comparison starkly: reuse policies create over 200 times more employment than landfilling or incineration for the same tonnage — directionally consistent with academic labour-multiplier studies elsewhere, which find recycling and reuse activity generating several times more direct and indirect employment than landfill disposal per tonne handled, even under more conservative methodologies.^[68]

This isn't abstract statistics. A neighbourhood hub means a coordinator organising streams, keeping records, and liaising with residents and the municipality; repair technicians — an electrician, a cobbler, a seamstress, a bicycle mechanic; sorters and intake staff; educators running workshops; and, where the hub handles verified material transfer, someone coordinating logistics to a recycler. Some of these roles are full-time, some part-time, some volunteer-supported — but all of them are local, and the money involved stays in the neighbourhood rather than flowing to a landfill operator or a WtE concessionaire located somewhere else entirely (Ch.7).

Where a hub's money actually comes from

A municipal service contract, paying for a measurable service — landfill diversion, reduced pressure on the sorting system, resident education — the same way a municipality already pays for ordinary collection (Ch.20–22), not as a grant. Sale of verified, sorted secondary material directly to recyclers, typically at a higher price than material recovered from a mixed stream. Sale of repaired goods through a shop, market stall, or online channel, with revenue offsetting operating costs. Cleanliness Fund grant support, where a hub can document real diversion from landfill (Ch.15 §8, Ch.17 §3 already examined this fund's troubled record — worth knowing before relying on it as a primary source). And social-employment programme funding for roles serving population groups those programmes are designed to support. A hub doesn't need to be profitable. It needs to be sustainable — meaning its financing is diversified enough that no single funding source can shut it down by disappearing.

5. Re-Haifa Lab: one Israeli model, not the only one

Re-Haifa Lab is not the only possible model of a circularity hub in Israel, and not necessarily the best one — but it is a real Israeli experience that can be examined, evaluated, and adapted rather than imagined from scratch. Founded in 2018 in Haifa, it works across three main areas: early-stage rescue of textiles and furniture, a repair café, and pilot organic-waste projects. Over several years, thousands of items have passed through the hub that would otherwise have ended up at landfill or beside a container.

What matters more than any single figure is what the experience actually taught.

- A physical place matters. Events “in the air” — markets, festivals — create visibility but not a habit. A permanent address and a known schedule turns into a reflex: I have an old lamp, I'll take it to Re-Haifa.
- Staff is critical. A hub doesn't run itself. Someone has to open the door every day, accept items, and answer questions. Volunteering is necessary but not sufficient — volunteers don't replace permanent staff.
- The municipality is a partner, not a sponsor. The most durable form of collaboration is a service contract, not a grant — a grant ends, a contract obliges both sides, and obliging both sides requires the hub to offer measurable KPIs in return.
- MRV isn't bureaucracy — it's survival. Without tracking tonnage, item counts, and CO₂, there's no argument to make to a municipality, a funder, or MoEP. A hub without data is invisible to the system, in exactly the sense Chapter 8 used that word.
- Don't replace municipal responsibility. When a hub works too well, a municipality can start quietly cutting a parallel service — “they have Re-Haifa now.” That's a trap: if the hub closes, residents are left without a service, and the municipality gains a ready argument for why it managed fine without one. The hub complements the system. It doesn't substitute for it.

6. What works and what doesn't, at this scale

A decade or more of European experience is long enough to start separating what genuinely scales from what sounds appealing but doesn't. A municipal contract with measurable KPIs outperforms dependence on a single grant-funder. A permanent location with a regular schedule outperforms occasional events. Paid coordination outperforms volunteers-only staffing. Integrating organics and verified secondary materials into one site creates synergy that an isolated, single-function service doesn't. Transparent MRV with a public report builds the trust a hub without any accounting system can't. Partnership with local recyclers and craftspeople outperforms trying to handle every function in-house. And orientation toward resident convenience — the same first-mile logic from Chapter 4 — consistently outperforms an orientation toward “correct behaviour.”

Reported figures from specific European pilots — Brussels' neighbourhood hub programme and Amsterdam's repair-hub network among them — describe tonnage diverted and contamination-reduction rates in the range this chapter's broader argument would predict, though the specific figures circulating for these two programmes could not be independently re-confirmed for this edition and are treated here as illustrative rather than settled. What's better established, and worth building on directly, are the fully verified cases above: Tallinn's district-by-district rollout, Munich's utility-embedded reuse shop, and RREUSE's employment data.

7. How to build a hub: from pilot to infrastructure

A hub isn't built at full scale on day one. A sustainable model moves through stages, and the minimum needed to start a pilot is smaller than it looks: a physical place — a room in a community centre, a corner of a synagogue, mosque, or church, garage space in a residential complex, no dedicated building required; one responsible person, who doesn't need to do everything but does need to open the door, take in items, and keep records; basic tracking — a spreadsheet noting date, item type, weight where possible, and outcome; one downstream partner willing to accept what the hub itself can't process; a basic municipal sign-off — not a full contract, just a letter of intent or meeting minutes, so the hub isn't operating in the shadows; and public communication — a schedule and address on the neighbourhood website, in building chat groups, on a noticeboard.

Stage	What it looks like
1. Pilot (6–12 months)	One or two functions. A fixed location, even a few hours a week. Simple tracking: count items and kilograms. Goal: prove that demand exists, to yourself and to any future partner.
2. Extended pilot (12–24 months)	Three or four functions. One coordinator, at least half-time. A first agreement with the municipality — even just on use of premises or collection of a verified stream. MRV begins: tonnage by fraction, items repaired or redistributed.
3. Sustainable model (24–36 months)	All seven functions, or close to it. A municipal service contract with KPIs. Diversified financing. An annual public report with real data. Early replication — helping a second neighbourhood launch.
4. Systemic integration	The hub becomes a recognised operator with a formal role in the chain; its data enters municipal reporting. Not the end of the journey — a change of status, from civic initiative to local infrastructure.

8. Hub and municipality: how the relationship matures

The relationship between a hub and its municipality typically moves through recognisable stages as the hub's measurability grows: invisible or seen as a hobby, with no contract and no municipal awareness; noticed but not recognised, with informal contact and occasional, non-obligatory help; recognised as a partner, under a cooperation protocol where the municipality provides premises or logistical support and the hub provides data in return; a service contractor, under a real contract with KPIs, where the municipality pays for a verified outcome — tonnes diverted, residents served, data delivered; and finally an infrastructure element, where the hub is written into the city's waste management programme and its indicators enter official municipal reporting.

The goal isn't necessarily reaching the last stage immediately. It's moving up this ladder deliberately, adding measurability and formalising the relationship as trust accumulates on both sides.

Complementarity, not substitution

The most serious trap in hub operation is becoming a quiet substitute for municipal service. It happens gradually: the hub works well, the municipality trims a parallel service, residents come to depend on the hub for something the city was always supposed to provide. This is a genuinely dangerous position, legally and politically — if the hub closes, residents are left without a service, and the municipality gains a ready-made argument that it managed without one before. The correct frame is that a hub provides services the municipal system doesn't have and won't build on its own timeline soon — repair, early-stage rescue, verified streams — while basic collection, basic sorting, and basic container servicing remain squarely the municipality's own responsibility (Ch.2 §2), which the hub doesn't take on and shouldn't be asked to. The OECD frames this precisely as a “complementarity, not substitution” principle: civil-society organisations should be contracted for additional services, not asked to quietly absorb what a public operator is legally obligated to deliver.

Conclusion

Sorting remains necessary in this book's account of the system — but it was never meant to be the starting point for a system oriented around prevention. It's the last gate, reached only after repair, reuse, sharing, organics diversion, and early-stage rescue have already done what they can (Ch.1 §9). A circularity hub is seven functions in one place, and their combination creates a synergy no single function achieves alone. It creates real, local, better-than-landfill employment. Its financial sustainability depends on diversified income — a municipal contract, material sales, social-programme funding — rather than dependence on any single grant. Its data, tracked honestly through MRV, is not paperwork but the argument itself: without it, a hub is invisible to exactly the kind of system this book has spent twenty-four chapters teaching a reader how to see.

A circularity hub complements the municipal system. It does not, and should not, substitute for it — a clear boundary of responsibility is the actual condition for a partnership that survives past its first enthusiastic year. Re-Haifa is one path through this, not the only one; every neighbourhood will adapt the model to its own scale, resources, and political context. The chapters that follow describe the practical tools for doing exactly that — data requests, civic audits, a concrete weekly action plan — and a circularity hub is where those tools find their most natural first use, because it is already the point of contact between residents and the system this whole book has been mapping.

Chapter 26

Who Can Change Israel's Waste System?

Political Ownership, Institutional Capacity, and Civic Action

After examining how waste is defined, collected, financed, sorted, recycled, exported, landfilled, and considered for energy recovery, one question remains: how can the situation be changed?

It would be easy to answer with a list of actions for residents, schools, or environmental organisations. Yet the preceding chapters suggest that Israel's waste problems are not primarily caused by a shortage of public awareness or volunteer activity. They arise from legislation, contracts, infrastructure, economic incentives, fragmented responsibilities, and political priorities.

Change therefore depends not only on what individuals are willing to do, but also on what different actors are authorised, equipped, and motivated to do.

This chapter does not offer a universal route to reform. It asks three more limited questions:

Is there a political or administrative actor that clearly owns the transition from waste disposal to waste prevention?

What can individuals, neighbourhood groups, NGOs, municipalities, and national institutions realistically contribute?

Which necessary functions still lack a clear institutional home?

The answers are provisional. Policies change, organisations gain and lose capacity, and local conditions differ. The purpose is not to prescribe one model, but to identify possible roles and their limits.

1. The missing owner of change

Israel does not lack institutions involved in waste management.

The Ministry of Environmental Protection has adopted a national strategy aimed at reducing landfill dependence, increasing recycling, and moving from a linear towards a circular economy. The strategy sets measurable targets for recycling and landfill reduction. However, although reducing municipal waste generation per person is one of its stated aims, it does not establish an equivalent measurable national target for prevention at source (Ministry of Environmental Protection, 2020; OECD, 2023).

The OECD has also noted that Israel still lacks a clear and comprehensive legislative framework for waste management, despite the adoption of the national strategy and several laws governing individual waste streams (OECD, 2023).

In January 2026, an inter-ministerial team began work on broader regulation of the waste sector. Its establishment reflected official recognition that the existing market and regulatory structure could not be addressed through another isolated amendment concerning a single stream or operator (Ministry of Environmental Protection, 2026; Knesset Research and Information Center, 2026).

These developments are significant. But policy activity is not the same as unified ownership.

Different actors approach the system from different positions. The Ministry of Environmental Protection is concerned with regulation, licensing, enforcement, treatment infrastructure, recycling and landfill reduction. The Ministry of Finance is concerned with costs, competition, market structure and public expenditure. Municipalities are responsible for collection and public-space services. Producer-responsibility bodies manage particular regulated streams. Private contractors collect, transport, sort and treat waste under contractual arrangements.

Each actor controls only part of the chain.

The gap is particularly visible in waste prevention. A ministry can establish recycling targets. A municipality can commission collection. A producer-responsibility body can finance the treatment of regulated packaging. A contractor can be paid for tonnes collected. But responsibility is much less clear when the objective is to prevent edible food, repairable furniture, clothing, household equipment or organic material from entering the waste stream at all.

Some relevant practices already exist, but they do not yet form a consistently available layer of public infrastructure.

The problem can therefore be stated cautiously:

Israel has relatively identifiable institutions for managing waste after it appears, but no equally clear institutional owner of preventing useful materials from becoming waste.

2. Is there political ownership?

A problem becomes politically owned when an identifiable political actor treats it as more than one element in a general environmental programme. Political ownership implies continuing attention, proposed instruments, resources, measurable objectives and some willingness to be held accountable for results.

This chapter does not claim that no Israeli party or politician is interested in waste. Nor does it attempt a comprehensive comparison of changing electoral programmes. The narrower observation is that no clear and durable political owner of municipal waste prevention is visible in the institutional and policy documents reviewed here.

Official strategies describe targets and instruments. Government decisions divide responsibility among ministries. Parliamentary committees discuss specific regulatory or infrastructural problems. Yet the following questions remain politically dispersed:

Who should finance local prevention and reuse infrastructure?

Should municipal contracts reward reduced residual waste rather than tonnes collected?

Who should be responsible for repair, reuse, food redistribution and bulky-item rescue?

How should waste prevention be measured?

Who should report whether prevention policies work?

How should costs be divided among producers, central government, municipalities and residents?

These are not only technical questions. They distribute costs, responsibilities and benefits differently among public institutions, producers, contractors and households.

Waste policy is also connected to broader political issues: the cost of living, municipal budgets, public health, environmental justice, land use, infrastructure, industrial regulation and relations between central and local government. This can make waste less visible as a separate political field. It can also create possible entry points.

Waste prevention may acquire greater political significance when it is linked to:

- reducing household expenditure;
- lowering municipal treatment costs;
- improving neighbourhood conditions;
- reducing food insecurity;
- limiting truck traffic and local pollution;
- creating repair and reuse employment;
- reducing pressure on communities near treatment and disposal facilities.

Waste does not necessarily become politically important because it is presented only as a waste problem. It may become harder to ignore when its relationship to money, health, public services and inequality is made visible.

3. Administrative capacity without unified direction

The absence of clear political ownership does not mean that public administration is inactive.

National institutions can propose legislation, regulate producer responsibility, establish reporting requirements, distribute funding, license facilities and change economic incentives. The inter-ministerial process launched in 2026 indicates that the state recognises the need for a broader regulatory framework (Ministry of Environmental Protection, 2026; Knesset Research and Information Center, 2026).

The Ministry of Environmental Protection's strategy provides a national direction for reducing landfill dependence and increasing circularity. At the same time, its most visible quantitative targets concern treatment outcomes. Waste prevention appears as an objective, but without a binding per-capita reduction target comparable to the recycling and landfill targets (Ministry of Environmental Protection, 2020; OECD, 2023).

Municipalities occupy a different position. They are closest to residents and to the physical places where waste appears. They influence collection contracts, sanitation schedules, bulky-waste removal, public land, schools, community centres and local communication.

Their capacity, however, varies greatly. Municipalities differ in financial strength, professional staffing, available land, political priorities and relationships with contractors. The State Comptroller has repeatedly documented both the central responsibility of local authorities for waste services and serious differences in their practical ability to provide those services (State Comptroller of Israel, 2025).

The national system of funding is itself a source of uncertainty. The Maintenance of Cleanliness Fund was established to support cleanliness, waste treatment, recycling and alternatives to landfilling. Yet the State Comptroller found low utilisation of landfill-levy revenues, large accumulated balances and weaknesses in evaluating the effectiveness of expenditure (State Comptroller of Israel, 2022).

Administrative capacity therefore exists, but it is divided and uneven:

- national government can alter rules but remains distant from everyday practice;
- municipalities can organise local services but cannot rewrite national producer-responsibility law;
- contractors can provide operations but generally follow contractual incentives;
- specialised schemes can manage defined streams but not the entire material system.

The problem is not simply that actors fail to coordinate. Different parts of the system are designed to deliver different outcomes.

4. Is there a strong civil-society actor?

Israel has an organised environmental movement. Life & Environment serves as an umbrella body for more than 150 national and local environmental organisations and works to strengthen cooperation and organisational capacity within the movement (Life & Environment, n.d.).

Some organisations have substantial legal and policy expertise. Adam Teva V'Din has worked on waste legislation, landfill closure, municipal collection standards, recycling policy and environmental litigation, and explicitly presents waste reduction and circular economy as part of its current agenda (Adam Teva V'Din, n.d.).

It would therefore be inaccurate to suggest that civil society is absent from the waste field.

The more difficult question is whether these capacities are assembled into a sustained coalition focused specifically on municipal waste prevention and local reuse infrastructure.

Environmental organisations work across many urgent fields: air pollution, climate change, water, planning, biodiversity, hazardous industries, public health and environmental justice. Waste may be one programme among many. Other organisations redistribute food or household goods without identifying themselves as waste organisations. Local groups may organise exchanges, repair activities or clean-ups without participating in national policy work.

The available evidence therefore suggests not an absence of civil society, but a distribution of functions among different organisations:

- legal action;
- policy advocacy;
- research;
- public education;
- local mobilisation;
- practical reuse or redistribution;
- environmental justice work.

These functions are valuable, but they do not automatically form one institutional system.

An NGO can define a neglected problem, test a limited approach, support a community, litigate, produce research or advocate for policy change. It cannot normally provide permanent citywide infrastructure without premises, staff, transport, insurance, stable funding and a formal relationship with public authorities.

Civil society should not be treated as a substitute for government. Its role may instead be to make institutional gaps visible, connect actors, preserve knowledge, support experimentation and translate local experience into evidence.

5. What can different levels contribute?

No level of action is irrelevant. But no single level can replace all the others.

Level	Possible contribution	What it cannot normally replace
Individual	Preventing personal waste, repair, sharing, using existing return and reuse channels, identifying problems	Public infrastructure, regulation, contracts and stable services
Neighbourhood group	Local exchange, mutual support, small pilots, trust-based coordination and identification of needs	A permanent citywide service with staff, transport, insurance and guaranteed access
NGO	Research, advocacy, litigation, coalition-building, pilot support and institutional mediation	Public authority, national legislation and secure long-term public funding
Municipality	Premises, local logistics, contracts, data, communication, schools and scaling	National producer rules, packaging standards and countrywide financial regulation
National government	Legislation, standards, EPR rules, funding systems, economic incentives and reporting obligations	Local trust, neighbourhood knowledge and everyday coordination
Political actor	Giving the issue priority, connecting reforms and accepting public accountability	Technical implementation and daily service management

This is not a hierarchy through which a small initiative naturally develops into national policy. The levels have different capacities and limitations.

A system involving several partly independent centres of action is often described as polycentric governance. Such arrangements can support experimentation and learning at different scales, but they still require mechanisms for coordination, accountability and the distribution of resources (Ostrom, 2010).

6. Individuals: necessary participation, limited power

Individuals make decisions that affect when an object becomes waste. Food can be stored, frozen, shared or discarded. Furniture can be repaired, transferred to another household or placed outside for collection. Clothing can remain in use, move through a local exchange or enter a donation stream.

These decisions matter. Prevention cannot occur without someone choosing not to discard an item.

But the realistic range of choices depends on infrastructure. Repair is easier when affordable repair services exist. Sharing is easier when people have a trusted local channel. Food redistribution is safer when rules, refrigeration, transport and responsible organisations are available. Returning materials is easier when collection points are accessible and reliable.

The individual can reduce part of their own waste stream, participate in an existing system, document a recurring problem, contact a responsible institution or support a political demand.

The individual cannot independently create the conditions under which an entire population can act in the same way.

Households are one point at which materials become unwanted. They are not the point at which packaging is designed, collection contracts are written, producer obligations are established or treatment facilities are selected.

Individual responsibility exists, but it is bounded by institutional design.

7. Neighbourhood groups: proximity, trust and fragility

Neighbourhood groups have capacities that national ministries and large contractors cannot easily reproduce. They know the local population and may recognise needs that remain invisible in administrative data.

A local group may notice that elderly residents cannot transport reusable furniture, that one community centre has unused space, that food is regularly left over after events, or that residents would use a tool-sharing system if someone coordinated it.

At this scale, a group may organise:

- a local exchange channel;
- a reuse shelf;
- occasional repair activities;
- food-sharing arrangements;
- shared use of tools or equipment;
- coordination around bulky household items;
- a small gardening or organics project;
- observation of a recurring local problem.

These functions do not need to be combined into a full circularity hub. One modest and useful practice may be more appropriate than an ambitious centre that cannot be maintained.

Research on grassroots sustainability initiatives suggests that their strengths lie in local knowledge, experimentation and the ability to develop solutions fitted to a particular place. The same literature emphasises their limited power, dependence on volunteer labour and constrained access to resources (Seyfang and Smith, 2007; Middlemiss and Parrish, 2010).

Neighbourhood action can identify a need and test whether a practice is useful. It should not be expected to replace a public service indefinitely.

A functioning local initiative may demonstrate that public infrastructure is possible. Its survival should not depend permanently on the unpaid work of one or two organisers.

8. NGOs: several roles, not one model

An NGO may contribute in different ways depending on its mission, expertise, legal status, funding and organisational capacity.

One organisation may act mainly as an advocate. Another may specialise in litigation. A research-oriented body may collect data or evaluate a pilot. A community organisation may support neighbourhood groups. A larger service organisation may operate a reuse, food-redistribution or repair programme.

These roles should not be treated as interchangeable.

Not every NGO working on waste needs to rent a building, employ drivers, collect materials or negotiate a municipal service contract. Physical operations involve responsibilities related to staff, transport, storage, safety, insurance, accounting and long-term funding.

An NGO can also act as an intermediary: connecting a neighbourhood group with a municipality, translating local experience into a pilot proposal, bringing together technical and social knowledge, or ensuring that residents are involved in project design. Research on sustainability transitions identifies such intermediary functions as potentially important in connecting innovations, institutions and users (Kivimaa et al., 2019).

Where an NGO becomes a service provider, formal agreements and measurable responsibilities may be necessary. But service provision is one possible role, not a universal organisational destination.

The most credible role for an NGO is the one that corresponds to its actual capacity.

9. Municipalities: where a practice may become infrastructure

A municipality can provide what most informal groups cannot: continuity.

It can allocate space, incorporate a programme into a community facility, commission transport, establish requirements in tenders, coordinate with schools, publish local information and maintain a service after a particular volunteer has left.

This does not mean that every municipality should establish the same kind of circularity hub. It means that prevention becomes public infrastructure only when some institution accepts continuing responsibility for it.

Local initiatives can help a municipality understand demand and reduce the risks of experimentation. A pilot may show whether residents use a repair point, which items are exchanged, how much transport support is required, or why a food-sharing arrangement fails.

A successful pilot, however, is not automatic proof that immediate expansion is appropriate. It may depend on an unusually committed organiser, temporarily available premises or a socially cohesive group. Scaling requires understanding which resources and relationships made the original initiative possible.

The municipal role is therefore neither to absorb every civic initiative nor to leave prevention entirely to volunteers. It is to decide which functions merit public support, what form that support should take and how responsibility should be divided.

10. What requires national action?

Some changes cannot be achieved locally, however successful a neighbourhood or municipal project may be.

Only national institutions can establish common rules for:

- producer responsibility;
- packaging design and labelling;
- national reporting standards;
- economic incentives and landfill levies;
- minimum service requirements;

- the use of dedicated environmental funds;
- treatment infrastructure;
- the relationship between prevention, recycling, energy recovery and landfill.

National policy is also necessary to limit territorial inequality. If prevention infrastructure depends entirely on municipal wealth, residents of stronger municipalities may gain access to repair, reuse, source separation and local organics options, while weaker municipalities remain focused on residual waste and basic removal.

At the same time, regulation written at the national level cannot determine every local arrangement. It cannot by itself create trust in a multilingual neighbourhood, identify a reliable community partner or explain why one service is accessible and another is not.

National institutions can create enabling conditions. Local institutions must translate them into workable arrangements.

11. The role of political actors

Political actors do not need to operate collection vehicles or manage repair centres. Their role is to turn a dispersed administrative issue into a public commitment.

They can demand that waste prevention be budgeted rather than merely encouraged, that targets address residual-waste reduction rather than only recycling, and that responsibility for implementation be identifiable.

Political ownership becomes especially important when institutional interests conflict. Producers, municipalities, ministries and contractors may favour different distributions of responsibility and cost. Questions such as who should finance reuse infrastructure, how far producer responsibility should extend, or whether public funds should support prevention or treatment infrastructure cannot be resolved by technical coordination alone.

A credible political programme would not need to promise an immediate zero-waste transition. It could begin with narrower commitments:

- define prevention as a measurable public objective;
- identify responsibility for coordinating it;
- finance a limited number of evaluated municipal pilots;
- establish transparent reporting;
- distinguish prevention from collection presented as prevention;
- publish failures and limitations as well as successes.

Such measures would not solve the system, but they would make responsibility more visible.

12. No level can change the system alone

The familiar distinction between change "from above" and change "from below" is inadequate.

Individual action without infrastructure is difficult to sustain. Neighbourhood projects without organisational support may disappear when their organisers leave. NGOs without administrative partners cannot easily establish permanent services. Municipalities cannot redesign packaging or national EPR systems. National policies without local knowledge may create formal programmes that residents cannot use.

The different levels can be complementary:

Individuals make everyday prevention possible. Neighbourhood groups contribute proximity and trust. NGOs connect experience with research, advocacy and organisational support. Municipalities provide continuity, logistics and scale. National institutions establish legal and financial conditions. Political actors turn outcomes into matters of public responsibility.

This is an analytical model, not a description of a coalition that already operates effectively.

In practice, one or more links will often be absent. A neighbourhood may have volunteers but no premises. A municipality may have premises but no suitable coordinating organisation. An NGO may possess expertise but lack a local base. A ministry may finance treatment infrastructure but not prevention.

The first question is therefore not necessarily:

What new project should be started?

It may be:

Which necessary function is currently missing, and which actor could realistically accept responsibility for it?

Conclusion: from isolated action to public responsibility

Israel's waste system is not without policies, institutions, professional knowledge or civic activity. The difficulty is that these capacities are fragmented, while the transition from disposal to prevention requires them to operate across institutional boundaries.

No resident, neighbourhood group, NGO, municipality, ministry or political party can perform every necessary function.

Individuals and small groups should not be expected to compensate indefinitely for missing public infrastructure. NGOs should not be assumed to possess the staff, premises, transport and financial capacity required to operate municipal services. Municipalities cannot solve problems created by national product regulation and financing systems. National institutions cannot design workable local practices without community participation.

The most important unresolved question is therefore not whether residents care enough or whether another awareness campaign is required.

It is:

Who will accept political and administrative responsibility for ensuring that prevention, repair, reuse, food redistribution and local organics management become ordinary parts of municipal life rather than exceptional acts of private commitment?

This chapter does not claim to know which institution will provide such ownership, or whether one institution should provide it alone. It suggests that the absence of clear ownership is itself part of the waste problem.

Local initiatives matter, but not because they can replace government. They matter because they can reveal unmet needs, preserve materials that the formal system loses, test limited alternatives and show what kind of public infrastructure is still missing.

The movement from cleanup to system design begins when these gaps are no longer treated only as matters of private behaviour or volunteer responsibility, but as questions of public policy.

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Bela Nikitina / Re-Haifa Movement
Haifa, July 2026

CONCLUSION: WHAT SYSTEM DESIGN REQUIRES

1. What This Book Has Established

1. Waste is not one stream.

What gets called “waste” is split across different legal categories, collection channels, operators, and reporting systems. A plastic bottle, a construction pallet, and a broken chair are all waste in the everyday sense, but they belong to entirely different regulatory and statistical worlds, and their fate cannot be read off a single number.

2. Collection is not recycling.

A container is only the first, or an intermediate, point in a chain. Material has to be tracked all the way to actual recycling, reuse, or final disposal before a claim of success is credible; anything less names an intention, not an outcome.

3. Percentages require gates and denominators.

Every figure in this book has needed a denominator, a year, a system boundary, a statistical gate, and a source before it could be trusted. The finding that only around 6 percent of household packaging reaches a designated collection container, for example, is not evidence of a simple failure of household motivation. It marks a failure at an early capture gate, before sorting and recycling can even begin — a distinction that disappears the moment a percentage is quoted without saying what it is a percentage of.

4. Incentives shape outcomes.

Contracts, tariffs, EPR fees, and the way costs are distributed explain the behaviour of collectors, sorters, and producers more reliably than appeals to responsibility. A system that pays for tonnes collected will optimise for tonnes collected, whatever its stated environmental goals.

5. End-of-pipe choices redistribute burdens.

Landfill, RDF, export, and waste-to-energy are not neutral technical endpoints. Each one commits the system to a particular long-term infrastructure and moves the environmental burden to a particular place and a particular set of residents, often for decades.

6. Prevention requires an institutional home.

Repair, reuse, redistribution, local organics, and bulky-item rescue cannot be sustained indefinitely as the sum of individual habits and volunteer effort. They become durable only when some institution accepts continuing responsibility for financing, accessibility, and operation — the question Chapter 26 leaves open rather than resolved.

2. What Remains Unresolved

Some questions this book has not answered, and does not claim to:

- A single, final recycling rate cannot be established across all material streams; each has its own denominator, and several remain genuinely contested.
- Which version of the 2026 reform, if any, will be adopted remains unknown at the time of writing.
- Whether circularity hubs are a generalisable model for Israel, rather than a promising but still local practice, has not been demonstrated.
- Who should finance and operate that kind of infrastructure at scale has not been decided by anyone with the authority to decide it.
- No political or administrative actor currently holds clear, durable ownership of the transition from waste disposal to waste prevention.

- Municipal capacity to take on any of this varies enormously, and nothing in current policy addresses that inequality directly.
- A successful local pilot does not, by itself, prove that its results will scale.

3. From Local Practice to Public Responsibility

Individuals can participate in prevention. Neighbourhood groups can organise proximity and trust. NGOs can contribute research, advocacy, mediation, or service delivery where their capacity allows. Municipalities can provide continuity, logistics, and scale. National institutions can change legal and financial conditions. Political actors can make the outcome a matter of public responsibility. None of these levels can substitute for all the others.

The question this book keeps returning to is therefore not whether change should come from above or below. It is which function belongs at which level, who has actually accepted responsibility for it, and which necessary function remains unassigned.

4. A Final Word

Israel's waste problem is not simply that too much material is discarded or too little is recycled. It is that responsibility is repeatedly lost between legal categories, collection channels, contractors, producers, municipalities, and national institutions. Materials disappear at statistical gates; costs move between actors; environmental burdens move between places.

Waste does not disappear when it leaves the household. Its costs are transferred to workers, municipal budgets, host communities, land, air, and sometimes other countries. Any claim of successful recovery should therefore be examined not only by tonnage, but also by labour conditions, geography, downstream fate, and the burden left behind.

This book has argued that system design begins by making those movements visible. It requires knowing what is being measured, who is responsible, what the contract rewards, where the material goes, and what burden remains after the reported success.

Prevention adds one further question: who is responsible before an object becomes waste?

Individuals and local initiatives can answer part of that question in practice. They can preserve useful materials, reveal unmet needs, and test limited alternatives. But they cannot be expected to create permanent public infrastructure through private commitment alone. An initiative becomes infrastructure only when responsibility for its continuity, accessibility, financing, and operation is clearly assigned.

No single actor currently appears able to redesign the whole system. That is not an argument for inaction. It is a reason to distinguish more carefully between participation and responsibility, between experimentation and public service, and between a promising local practice and an institution capable of sustaining it.

The movement from cleanup to system design therefore does not end with a list of actions for the conscientious resident. It begins with a public question:

Which useful materials could be prevented from becoming waste, what conditions would make that possible, and who should be responsible for creating those conditions?

This book does not claim to have a complete answer. Its more limited purpose has been to make those questions visible, name the actors involved, and show where responsibility still disappears.

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